

2017-2019

Internship Interview Experience



Prepared by :
Alumni Cell –PGDBA 2020
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Roles Offered : a) Financial Research Analyst
b) Consumer , Asset Backed Securities , Energy & Investor Relations

Job Description :

a) Resourceful candidates to intern with our Financial Research (FinRes) group at DESIS based out of Hyderabad. The Financial Research group works closely with the front office to provide high quality and in-depth support to the investment research team and for the decision-based processes of the firm's global proprietary trading strategies. If hired, your role will involve working closely with teams within the FinRes group on data-driven analysis and statistical modeling to generate research/analysis that will assist the trading desks with investment decisions. The ideal candidate should have a keen interest in statistical analysis and modeling along with excellent communication skills.

Internship Stipend Offer : Rs 1,25,000 (Cash) , Rs 1,40,000 (Effective)

Shortlist Steps : a) Resume Shortlist –All Candidates shortlisted.
b) Technical & Business Interview (Skype Interview) (Round 1(Elimination) and Round 2)
c) HR Round

Duration : Maximum Duration was 110 minutes.

Technical Round

Technical Round Questions (Puneeth Gandla)

- a) Explain the load forecasting problem ? With respect to data? location? household/industrial? Any trend/seasonality?
 - a).1 Why yearly, weekly and daily seasonality? Explanations? How are multiple seasonality identified?
- b) What would you do if there are very large number of variables? What are the cases where PCA doesn't work?
- c) Given Railway Ticket data with names, berth etc., How to generate money out of it?

Technical Round Questions : (Manoj Mana)

a) Tell me about yourself ?

b) Gave me a hypothetical use case, "Imagine that all the cars in the country have GPS attached and you have all the data generated by each GPS device, How to generate revenue using this data?"

b).1 Asked me to think from investors point of view ?

c) Amazon project resume

c). 1 Why did I use deep learning techniques , is it because they give better accuracy or some other reason to which I responded used deep learning out of curiosity.

d) Work experience at Reliance.

d).1 How to forecast all the planned and unplanned shutdowns in the country? What techniques would you use? What extra variables will you try in improving the forecasting accuracy?

Round 2:

e) Asked if I have any questions?

He explained 4 different roles(teams) available for the internship and preference is asked. No technical questions were asked.

Technical Round Questions : (Ankita Jha)

a) You are given prescription data for the past five years (Name of the hospital, doctor, patient details, illness and medicines prescribed). Derive revenue for the company from this data. ?

a).1 A lot of follow up questions happened on this, went on for about 20 mins.

b) You're given marks of 30 students for three subjects. Each subject has a missing mark for one student. How should you impute it?

- c) There are two Indian firms, one manufactures durable goods (television, washing machine etc) and the other manufactures laptops, phones, tablets etc. You need to forecast their sales for the next quarter. For which company would you use linear regression and for which time series forecasting?
- d) Explain my internship project at Cummins and then asked when should you not use moving average imputation for time series? (I had mentioned moving average as a technique used in the project).
- e) DE Shaw mostly picks up projects related to time series and regression and asks you questions pertaining to them.

Technical Round Questions : (Mayank More)

- a) Given mobile identifier and location data, Create revenue for D.E. Shaw. ?
- b) Questions on project on your CV.
- c) Two Envelope Puzzle.
- d) Question on Residual v/s Leverage

Technical Round Questions : (Tajendra Mehta)

- a) Tell us about yourself.
- b) Case - You have WhatsApp data for all Indian users since they have signed up, i.e. conversations, images and videos shared. Use this data to generate money for D E Shaw.
- c) Why is ARIMA model not suitable for forecasting visitor count of a restaurant (from a Project that I had done) ?
- d) Puzzle - Two envelopes Problem - You are given two indistinguishable envelopes, each containing money, one contains twice as much as the other. Having chosen an envelope, but before inspecting it, you are given the chance to switch envelopes. Should you switch?
- e) Do you want to ask us anything?

Round 2

They first explained the work of all the verticals that they were hiring for.

- a) Explain how do you fit for this role.
- b) Case - Suppose you have to recommend most important investors, what will be your approach.
- c) What is outlier and leverage and what are their interpretation?

Technical Round Questions : (Shaswat Roy)

- a) You have been given all social media data, how would you generate revenue for DE Shaw?
- b) In your work experience you have done dash boarding, Explain the same? How did it help the clients?
 - b) 1. How were you able to pinpoint the rise in sales of a drug to the effort of a medical sales representative ?
(Based on work ex, but in layman terms they asked me difference between causality and correlation)?
- c) Explain a time series project (volatility in Indian financial markets)?
- d) You have 9 red socks and 11 blue socks in a dark room, how many minimum no. of socks you need to draw to guarantee getting a pair?
 - d)1.How many minimum no. of socks you need to draw to guarantee getting a red pair?
 - d)2.How many minimum no. of socks you need to draw to guarantee getting a blue pair?
 - d)3.How many minimum no. of socks you need to draw to guarantee getting a red and blue sock?
- e) Difference between outliers and leverage points ?
- f) What would you do differently in your work ex given all you have studied here in PGDBA?

HR Round

HR Round Questions (Puneeth Gandla)

- a) Is Hyderabad ok with you?
- b) What sports do you play? etc.

HR Round Questions (Manoj Mana)

- a) Asked me about my other shortlists and my preferences.?
- b) Asked my experience as a placement team member?
- c) My reason for preferring DE shaw over other companies?

HR Round Questions (Ankita Jha)

- a) Is work life balance important to you?
- b) Why do you want to join this company?
- c) What are you looking for in long term from this job?

HR Round Questions (Mayank More)

- a) Why D.E.Shaw?
- b) Why not Amazon?

HR Round Questions (Tajendra Mehta)

- a) Why would you be a good fit for this role?
- b) Asked about a project from my work ex which the interviewer felt was relevant to the profile.

HR Round Questions (Shaswat Roy)

- a) Which role do you want?

Other Technical & HR Round Questions

- a) Round 1- Guestimate, Project on Time Series.
- b) Case study on generating revenue from airline ticket data ?
- c) Forecasting planned and unplanned maintenance of a refinery and validating result ?
- d) Imagine you have the CCTV footage of all the shopping malls. How would you make investment decisions out of this ? Post this there were some easy questions on time series and regression (outliers, leverage etc.)
- e) Case - import and export data of country for past 25 years is given including prices and merchant details. Find investment opportunity for an investor.
- f) One question about project(Time series)
- g) One question about Work Experience(Effects on market prices by significant market player, how to leverage machine learning in this)
- h) Case: You are an investment firm. How will you use the transaction data of a country of last 5 years to make money?
- i) Discussion on project. What was the most interesting and insightful thing you learnt from this. Applications of Fourier transform.

- j) Applications of PCA.
- k) How do you decide when to use linear regression and when time series models like ARIMA for forecasting?
- l) You have two firms- one is a consumer goods firm and other is a technology firm. In which case will you use linear regression and where time series models for forecasting?
- m) How will you decide potential investors for your firms and the products they will be interested to invest in? (they were looking for a model based approach)
- n) You are a school teacher. You have marks for 30 students for 3 subjects. Some values are missing. How will you impute them?
- o) How much risk averse/loving are you? Give examples.
- p) Discussion on extra curricular activities?
- q) Discussion on PORs?
- r) Do you mind living in Hyderabad?
- s) Case: How to choose to invest given 20 emerging countries list and discuss few macro economic factors to consider like GDP & growth rate & inflation?
 - s)1. Top 3 GDP growth rate countries?
 - s)2. Real vs nominal interest rates, interest rates relation with foreign exchange?
- t) How to deal multiple seasonality in a time series?
- u) What is Fourier transform ?
- v) Given Video Traffic Data , how can you make investments that would be helpful for DE Shaw?
- w) Case on how to find the potential investors for a particular strategy ?
- x) Given the PGDBA exp what would you have done differently during your work exp?
- y) Investment decision using WhatsApp chat/audio/ video data ?
- z) Investment decision using GPS data of people

- aa) How would you find if a FMCG product is doing good.
- ab) Why manhole cover is circle and not rectangle?
- ac) You have Minimum temperature, maximum temperature and rainfall data of Kolkata for last 35 years, How will you generate business out of it for DE Shaw?
- ad) Questions on Time series project - Stock value prediction ?
- ae) Generate revenue for DE Shaw from clickstream data (data from various ISPs with timestamp of when a page was visited). Explain a project.
- af) Puzzle: Two envelope problem and mean imputation.
- ag) Case study on how to get business out of a hypothetical dataset. Focus on the broad picture for this without getting into details (Example: Given the telephone/mobile bills data of each individual in India, how will you get revenue out of it? Please exclude any operational ideas and focus on from trading or investment purposes).
- ah) Expect a few questions on very basic concepts of regression, missing value handling, PCA etc. (Example: Given a data set how will you impute missing values: answer they look for is KNN).
- ai) Also, there may be basic puzzles asked. (Example: You are given two letters with x and $2x$ amount in it. You picked up one and looked at the amount and now you are given a chance to switch. Will you switch the letter or not). Communication skills also play a factor for this round.
- aj) Discussion on how to measure the economic loss during the recent Kerala flood ?
- ak) Questions on Regression - multicollinearity- how to check for it and what to do when two variables have high multicollinearity?
- al) What to do when R^2 is 0.2 on a model with only one explanatory and a response variable ?
- am) Case questions like how will you generate revenue for DE Shaw given transaction data.
- an) Case how will you generate revenue for DE Shaw given Receipt of Medicines sold in India?
- ao) Guesstimate question- How many pen will be sold in Kolkata on a given day?

- ap) Case Study : How can you generate revenue for D.E Shaw, given the data about all the manufactures cars, their past owners, insurance paid etc.
- aq) Asked about the click prediction project I had done ?
- ar) What is the difference between statistical modelling and econometrics modelling?
- as) How will you model losses due to Kerala flood? Rest questions were based on my econometric projects ?

Courses & Skills Suggested

- a) Selected Aspects of Predictive modelling.
- b) Regression and Time Series Forecasting/ Modelling.
- c) Computation for Data Sciences.
- d) Macro economics, Corporate Finance.
- e) Linear regression, Time series.
- f) ISI-CDS, Inference, Probability, DBMS
- h) IIT: RTSM, FADML
- i) Finance, Business Economics and Strategy
- j) Machine Learning



ACADEMIC QUALIFICATIONS

Year	Degree/Board	Institution/University	Performance
2019*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-
2012	B. Tech (Mechanical Engineering)	IIT Madras	7.87/10
2008	Class XII, BIE – AP	Sri Sai Vikas Junior College, Vijayawada	92.1%
2006	Class X, BSE – AP	Dr. KKR's Gowtham Concept School	90.0%

ACADEMIC AWARDS

Awards & Achievements	- Ranked All India 8 in DATA TALES Machine Learning Competition by Great Lakes, in Public Leaderboard '18 - Secured All India Rank 862 (99.73 percentile) in IIT Joint Entrance Exam (IITJEE) out of 320,000 applicants '08 - Achieved 99.70 percentile in All India Engineering Entrance Exam (AIEEE) out of 850,000 students '08 - Ranked All India 3rd in Mathematics Olympiad by The Association of Mathematics Teachers of India '04 - Received prestigious Prathibha award from Hon. Chief Minister for securing District 2nd rank in Board exams '04
Certification	Certified Part 1 Financial Risk Manager Candidate by Global Association of Risk Professionals (ID 226538) '13

WORK EXPERIENCE

58 Months

Cummins Technologies India Pvt. Ltd.	Senior Engineer (Applied Mechanics)	Pune (Jul '12-May'17)
Part of the Centre of Excellence team that delivers Analysis Led Design solutions to Cummins entities located at Brazil, China, India, UK & the US , through Computer Simulations and Experimental Testing		

Executive Achievements	Promoted twice over the hierarchical structure and received the best performance rating for five straight years - Received Managing Director Award twice for 'Customer Support Excellence' and 'Delivering Superior Results' - Credited with Six-Sigma Green Belt, through successful execution of 3 6S projects for developing technical processes
Roles & Responsibilities	- Designed and validated Power Generator-set and its components for maximal efficiency, minimal vibrations and noise - Administered as India SPOC for structural soft validation of biggest ever 3.5MW Power Generator set by Cummins - Performed commendably in 3 challenging roles across 2 Business Units with cross-functional and cross-cultural teams
Key Technical Initiatives	- Played pivotal role in CE Audit and Certification of Multi-Million dollar Experimental Test Centre to facilitate R&D - Pioneered design, validation and establishment of multiple technical simulation and testing processes from scratch - Developed MATLAB based tools to automate data pre-processing and analysis, reducing turnaround time by ~60%

Int'l Publication	• International conference publication on Low-frequency Noise Modelling at NOISECON, Michigan, USA [Link] '17	
Cummins Global Analytics Centre	Data Science Intern	Pune (Dec'17)

Missing Data Handling	- Developed R models for effective imputation of missing values using imputeTS, MICE and missForest packages - Evaluated these ML packages against imputation time and accuracy, across different datasets and % of missingness
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DATA SCIENCE PROJECTS

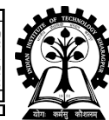
Sentiment Mining	- Used Amazon Electronics reviews & ratings(1.7M) dataset, to build a sentiment classifier that converts text into ratings - Employed this classifier to predict the customers' perception of new product launches, using tweet or comments - Modelled the ratings by building a Bag of Words model using factorization machines, to deal with high sparsity - Captured context of reviews using Sequence LSTMs embedded with word2vec, to further improve accuracy by 14%
Revenue Maximization	- Predicted propensity of policy renewal for 80k+customers using their demographics, income and renewal history Calibrated the classifiers using Isotonic Regression, to arrive at the true probabilities instead of probability 'scores' - Optimized individual agent's incentive to maximize the policy renewal propensity, and boosted net revenue by ~10%
Travel times prediction using GPS data	- Predicted route-based travel times for a city by capturing historical spatial & temporal dependencies, given start time Engineered cyclic temporal features, geodesic distance and 2D spatial grid; and detected outliers using these features Stacked tree based RandomForest & XGBoost models, and non-tree based SVR model to minimize MAPE scores
Information Retrieval (IR) System	- Constructed a tf-idf based IR system to retrieve ranked list of docs for a given query, from a corpus of ~6k+ docs - Implemented extractive summarization of documents using graph-based degree-centrality & text-rank algorithms - Expanded and updated queries using word2vec and Rocchio algorithm respectively, to improve F1 score by ~15%
Load Demand Forecasting	- Forecasted weekly electricity demand of households using past 5 year data, to avoid blackouts & promote sustainability - Identified trend and seasonality by decomposing the time-series and confirmed stationarity using Dicky Fuller test - Employed ACF & PACF plots to identify the orders and fit an ARMA model to forecast demand with 13% MAPE
Market Basket Analysis	Recommend 2 next best items to go into the cart, given user purchase history and existing items in the current cart - Analyzed dataset with 3mn online orders of 50k+ different products with information pertaining to ~2L customers Engineered product, user, time and user-product interaction based features for effective product recommendation

EXTRACURRICULARS & PORs

Activities	Spearheaded team of 15, that bagged 1st place in Poster Presentation at the Cummins India Conference '14 Elected as hostel Secretary and a member of Student Hostel Council at Indian Institute of Technology Madras '12 - Successfully finished 525ft high bungee jump, 27ft deep scuba diving and motored to KhardungLa pass (18,400ft)
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TOOLS & LANGUAGES : R, Python, MATLAB, MySQL and MS Office

SELECTED COURSES : Computing for Data Sciences, Business Data Mining, Information Retrieval and Decision & Games



ACADEMIC QUALIFICATIONS

Year	Degree/Board	University/Institution	%/CGPA
2019*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-
2014	B.Tech (Mechanical Engineering)	IIT Bombay	8.55/10
2010	Class XII, BIE-AP	Narayana Junior College, Hyderabad	97.50%
2008	Class X, BSE-AP	Trinity High School, Mancheril	91.00%

DATA SCIENCE COMPETITIONS

World Data Science Game 2018 (World Rank #9) (India Rank #1)	- Only team to represent India in the World Data Science Game finals, Paris sponsored by BNP Paribas - Predicted customer interest to buy/sell a bond in the coming week leveraging past trading & market data Clustered bonds & Users, engineered 200+ features, reduced feature space dimension to 80 using Boruta Out-performed BNP baseline AUC by ensembling multiple base models from LightGBM, Xgboost, RF
CrowdAnalytx.com (World rank #2) (Won \$750 cash prize)	Second prize winner of the “Predict Airline Delay in US” challenge with 650+ teams across the globe - Engineered features, performed likelihood encoding of huge categorical variables on 8.5 million entries - Built a GBM model using h2o.ai API in R and investigated potential reasons for Airline delay [Github Link]
Miscellaneous Achievements	Winner of the American Express-“Data & Beyond” challenge to target customers maximizing returns - One of Top 6 Finalists & 2nd in final LB among 9600+ in Machine Learning Challenge by Société Générale Hired as a Business Analyst in an Analytics firm, winning a hiring hackathon on Analytics Vidhya

WORK EXPERIENCE

(35 months)

Senior Business Analyst Hansa Customer Equity Pvt Ltd *Mumbai (Nov '15 – June '17)*

Handled Customer Analytics for a “major Life Insurer”, a “Financial Services conglomerate” & a “major DTH provider”

Achievements	- Awarded as “Best newcomer of the quarter” & “Proactive employee”, within one month of joining - Got promoted within a year with highest performance rating in the Analytics department of 40 employees
Cross-Sell Models a) Insurance customers b) Retail Traders	- Identified high propensity customers to cross-sell using Logistic Regression & recommended next best policy - Strategized campaigns using model scores to generate monthly revenue impact of INR 29 million - Built a Random Forest model to recommend 1 of 5 financial products to retail trading customer base
Lead Scoring Model	- Built a model to Identify top 50k of 160 million consumer loan leads to sell insurance using credit history data - Improved “Test vs Control” appointment rate by 300% by running a pilot campaign on top 50k leads
Churn Models	- Predicted next 30 & 90 day churns using current market conditions, past transactional data of traders - Reduced churn rate of equity & derivative traders by 4% by strategizing activities based on model output
Fraud detection (General Insurance)	- Enhanced existing rule-based fraud detector using bag-of-words model on textual Insurance claims data - Reduced costly false positives by doubling the precision as compared to existing rule-based model
Customer Profiling	Carried out income, behavioural segmentation for a 17 million base leveraging CIBIL data additionally

Inventory Engineer Reliance Industries Limited *Mumbai (Jul '14 – Nov '15)*

Central Materials Management	- Designed process flow for ‘Dynamic Inventory Modelling’ of mechanical maintenance material - Involved in planning, resource allocation & safe execution of shutdown activities, during training period
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DATA SCIENCE PROJECTS

Electricity Load demand forecasting	- Forecasted hourly load demand for next one month using 5-year data to plan power availability in advance - Adjusted for multiple seasonality to reach stationarity; Analyzed ACF/PACF plots to fit ARMA model
Ratings Prediction for Twitter Reviews (Electronic products)	Objective: To assign probable ratings (1-5) to live twitter reviews during electronic product launch - Leveraged 1.7 Million Amazon electronic product review-ratings to train a FM model & a LSTM model - Demonstrated both models using 50 sample tweets on electronics with 80% & 92% 1-step accuracies
Real-time face recogniser	- Built a real-time face recogniser on webcam feed using OpenCV and Keras libraries in Python - Achieved a validation accuracy of 99% using a 2-layer CNN model to classify four different faces
Deep learning on Medical Images	- Developed a tool to detect one of 14 different thorax diseases given an X-ray as an image (jpeg format) - Implemented transfer learning using two different pre-trained weights of VGG-16, ResNet-50 architecture - Handled high class-imbalance partly by augmenting data and partly by tuning assigned class-weights

ACADEMIC ACHIEVEMENTS & POSITIONS OF RESPONSIBILITY

Academic Achievements	- Secured an All India Rank of 758 in IITJEE-2010. Ranked among top 10 students of the PGDBA batch - Recipient of MHRD’s Central Sector Scheme Scholarship for being in top 1% in XIIth board (2010) - Recipient of IIT Bombay Heritage Fund Scholarship (awarded to only 5% of 800+ students) (2011-14)
PGDBA Placement Representative	- Formulated Placement Charter & key policies; facilitating industry interaction, promoting PGDBA brand - Achieved a 26.5+% YOY increment in avg. & median compensation in the 2016-18 final placements
Social Activities	Founding Member of IIT Bombay’s first ever student driven Social Festival “Abhyuday” [Link] (2014) - Organized lectures, workshops, NGO exhibition (30+ NGOs) to spread social awareness among students - Mentored low-income youth at risk as a part of “Mentors of Mumbai” program Antarang Foundation
Other Interests	Trekking Successfully completed 65 kms trek from Lonavala to Bhimashankar in 38 hrs (2014)

ACADEMIC QUALIFICATIONS

Year	Degree/Board	University/Institution	Performance
2019*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-
2014	B.Tech (Hons.) – Electronics and Communication	LNM Institute of Information Technology, Jaipur	9.98/10 (Rank 1)
2010	Class XII, ICSE	City Montessori School, Lucknow	95.50% (Rank 1)
2008	Class X, ICSE	St. Mary's Convent Inter College, Lucknow	95.71%

SCHOLASTIC ACHIEVEMENTS

*Institute Rank 3 – PGDBA 2017-19**

Academic Achievements	- Received a cash scholarship from ISI, Kolkata for excellent academic performance in PGDBA semester-I '17 - Secured Director's Gold medal for scoring highest CPI in the university, with a perfect 10 SPI in 7/8 semesters '14 - Scored 730/800 in GMAT, 116/120 (99th percentile overall, 30/30 in writing) in TOEFL and 326/340 in GRE '16 - Ranked 4/179 teams, national finalists in Data and Beyond event sponsored by American Express and IIM-B '17 - Placed in the top 7% to qualify for the grand finale of Smart India Hackathon organized by GOI '18 - Featured in the top 0.6% globally among over 1.25L+ data science community members on Analytics Vidhya '18
Publications	Co-authored a research paper on blood pressure measurement published at IEEE Technical symposium, 2014 (Link)

DATA SCIENCE INTERNSHIP

Cummins Global Analytics Center

December 2017

Missing Value Analysis and Imputation	- Analyzed univariate and multivariate time series data from sensors to understand causes and impact of missing values - Performed comparative analysis of imputation algorithms like Interpolation, Kalman Smoothing and Moving Average - Implemented ImputeTS, MICE and Amelia packages in an efficient manner and visualized the imputation results
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WORK EXPERIENCE

36 Months

Interactive Web Developer

Sapient Consulting

June 2014 – June 2017

Marketing Campaign	Led a team of 3 to develop a website for a Fortune 500 tyre manufacturer in 35+ locales with a load of 300 hits/sec - Integrated the website with Adobe Market Analytics tool to identify high-value customers using multi-channel data
Product Dashboard	- Designed a product dashboard for a financial investment firm, as a central location for all product related data - Addressed issues of data accessibility, visibility and collaboration in the firm while saving ~20 man-hours per day - Implemented the dashboard as a fully responsive single-page web application powered with AngularJS and AdobeCQ
Achievements	Global semi-finalist (top 10) for Sapient Impact Awards: “Rookie of the year” category among 2000 nominees - Trained, managed and up-skilled 50 new hires on front-end technologies and mentored them on project opportunities - Received a “You rock” and “You make a difference” award for exemplary contribution to a project launch

DATA SCIENCE PROJECTS

Youtube-8M Video challenge	- Assigned video-level labels to 31GB data released by Google, relevant in automatic caption and subtitles generation - Trained a Logistic Regression model, using TensorFlow and Google Cloud to achieve a GAP score of 72% - Improved GAP score to 76% using Mixture of Experts model with a dense layer of 25K units and PReLU activation
Leads conversion Prediction - McKinsey	- Identified potential customers with a high conversion probability using demographic and income data of last 3 months - Engineered new credit related features such as debt-to-income ratio, surplus income, EMI principal and annual loan - Leveraged the h2o platform to train RF, GBM, ANN and XGBoost models and achieved an AUC score of 84%
Opinion Fact Classification - Microsoft Bing	- Classified 600+ news articles into facts and opinions subcategories, useful in revenue modelling for online publishers - Extracted new features using topic modelling via Latent Dirichlet Allocation and community detection algorithms - Ensembled SVM, LGBM, GRU and bi-directional LSTM models to achieve a final classification accuracy of 74%
Trade Propensity Prediction - BNP Paribas	- Predicted probability of customer-bond interaction in coming week using trading and market data of past two years - Generated synthetic data imitating test data and ensembled logistic regression, tree-based models for an AUC of 62% - Applied Collaborative Filtering and LightFM models to extract additional features, improving the AUC to 68%
Electricity Load Forecasting	- Predicted electricity load demand in Indian states using multi-seasonal ARIMA model, leveraging the TBATS package - Reduced MAPE from 12% to 8% using wind and solar data and implemented an LSTM model for a MAPE of 3% - Visualized results with an interactive web dashboard using R-Shiny for a real-time analysis of load demand
Investors Recommendation	- Analyzed a start-up dataset on time, domain and region levels to predict top 5 investors likely to invest in a start-up - Clustered investors using KMeans post applying dimensionality reduction via PCA for a between/within SS of 97.2%

POSITIONS OF RESPONSIBILITY AND EXTRA CURRICULAR ACTIVITIES

Core Team, XTSummit	- Organized XTSummit, the front-end development community's annual event <i>Sapient, 2015, 2016</i> - Managed a footfall of over 200 participants from across India's four office locations, for two consecutive years
Placement Coordinator	- Conducted campus placement activities and secured offers for 200 students of the outgoing class <i>LNMIIT, 2012</i> - Invited new companies, organized training sessions and handled all on-floor activities during recruitment drives
Student Senate Member	- Resolved issues faced by the student community by bringing them to a common platform and made policy changes - Linked students and administration on academic matters, as part of academic council-undergraduate committee
Extracurricular Achievements	- Won silver medal at the swimming event held during the inter-section sports meet, 2017, at IIM Calcutta - Mentored underprivileged students for Sapient's CSR initiative Prayaas, organizing annual days and sports meets

TOOLS AND LANGUAGES: R, Python, Google Cloud, SQL, JIRA, Git, Sourcetree, Java, Javascript, HTML

ACADEMIC QUALIFICATIONS			
Year	Degree/Board	University/Institution	%/CGPA
2017-19*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-
2012-16	B.Tech (Computer Science & Engineering)	NIT Nagpur (VNIT)	8.22/10
2011-12	Class XII, HSC	Pace Junior Science College, Mumbai	86.50%
2009-10	Class X, ICSE	Vibgyor High, Mumbai	92.57%
AWARDS AND ACHIEVEMENTS			
Institute Rank 1 – PGDBA 2017-19*			
Scholastic Achievements & Publications	- Recipient of merit-based scholarship for excellent academic performance at ISI Kolkata ‘17 - Co-authored Ivey published case study – Pulse candy: Sustaining the Brand Differentiation [Link]; Featured in Harvard Business Review and currently a part of the Marketing pedagogy at IIM C ‘18		
Data Science Competitions	Ranked 1st in American Express’ Data & Beyond – an analytics event at IIM Bangalore ‘17 Ranked 1st in ISI Kolkata Stat Wars, a Data Science Hackathon among 706 participants worldwide ‘18 Ranked 2nd at the national level Consultant Hunt event, Sumantran organized at IIT Kharagpur ‘18 Finalist (top 7% teams) at Smart India Hackathon organized by Government of India ‘18 - Awarded bronze medal (top 6% among 3967 teams) in Kaggle’s click fraud detection challenge ‘18		
DATA SCIENCE INTERNSHIP		Cummins	December ‘17
Manufacturing Defect Detection	- Assessed under-sampling/over-sampling techniques to deal with a class imbalance ratio of 100 - Employed autoencoders as an anomaly detection technique to detect minority classes - Improved the AUC of the base models by 10% and presented the findings using R Markdown report		
WORK EXPERIENCE			10 months
Software Engineer		MAQ Software	July ‘16-March ‘17
Roles and Responsibilities	- Designed and developed an ETL framework for client requirement analysis in SSIS and SQL Server - Built OLAP cubes and developed PowerBI dashboards for real-time analysis and tracking of data - Responsible for end-to-end development and deployment of code/processes to production server - Developed a tool to check for discrepancies in upstream data sources and monitor index usage		
Awards and Achievements	- Captured ~900 source failures through adoption of the tool in 40 projects across the organization Microsoft Certified Solutions Associate (MCSA): SQL 2016 Database Development		
Database Developer		Media.net (Directi)	April ‘17-June ‘17
Roles and Responsibilities	- Designed processes involving Hive scripts and scooping data from Hadoop clusters to databases - Responsible for architecting, developing and maintaining database-centric applications		
ANALYTICS PROJECTS			
AdTracking Fraud Detection Challenge	- Predicted if a click will result in an app download by implementing FM in Python on Google Cloud - Explored ~200M clicks data using Tableau; Engineered features to identify click fraud characteristics - Developed LightGBM models to achieve AUC of 98%, improving the firm’s existing baseline model		
Prediction of Electric Load Demand	- Developed LSTM models to predict hourly load demand for the next week using ~5 years of data - Utilized TBATS package in R to incorporate the multi-seasonal short-term electric load time series - Achieved a MAPE of 3.1% through LSTM by considering weather variables such as temperature		
Prediction of Operational Losses	- Predicted the expected quarterly operational losses for next 1 year for 3 business divisions of a bank - Estimated the loss distribution from severity and frequency of past losses using Monte-Carlo methods		
Workforce Demand and Optimization	- Modelled employee demand for departments of a retail store using XGBoost via Poisson Regression - Created an automated process generating daily workforce schedule using Linear Optimization		
Time-Series Analysis of Cryptocurrencies	- Analyzed the trends and volatility in Bitcoin, Ethereum and Ripple closing price daily data - Gauged the suitability of modelling log-returns of cryptocurrencies using ARIMA and GARCH		
Family Identification of Bacterial Genome	- Used pairwise alignment and hierarchical clustering to construct a profile Hidden Markov Model - Computed HMM’s emission and transition probabilities to classify bacterial genome into 2 families		
POSITIONS OF RESPONSIBILITY AND EXTRA CURRICULARS			
PGDBA ‘17-19	Represented the batch at PGDBA’s first Industry Advisory Meeting; among 4 students of the batch		
President, E-Cell VNIT, 2015-16	- Led a 13-member team to successfully plan and execute events throughout the academic session - Organized a pro-show by The Viral Fever (TVF) and YouTube sensation Shirley Setia		
Public Relations Head, Consortium ‘15	- Envisioned and organized Nagpur’s first Startup Conclave featuring 10 startups and 3 investors - Secured sponsorship to meet the budget of ₹ 0.6 million for the event with a footfall of ~7000		
Sports	Winner of Gold Medal in Roller Hockey, State Level; Ranked 2nd in Inter-section Pool at IIM C		
Extra-Curricular Achievements	Winner of Startup Weekend powered by Google for Entrepreneurs, Nagpur, March 2015 - Represented School at State Level, ranked 2nd in Inter-School in MaRRS International Spelling Bee		

ACADEMIC QUALIFICATIONS

Year	Degree/Board	University/Institution	%/CGPA
2019*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-
2015	B. Tech. (Material Science and Engineering)	IIT Kanpur	8.1/10
2010	Class XII	CBSE	92.20%
2008	Class X	CBSE	93.50%

AWARDS AND ACHIEVEMENTS

International Competitions	Silver Medal (Top 4% in ~4k) in Global Kaggle, 'TalkingData AdTracking Fraud Detection Challenge' - Achieved World Rank 18 among 450+ teams in Capsim Global Business Strategy Challenge, Spring 2018 AIR 5 in 'Data Science Game', an International hackathon sponsored by BNP Paribas, AXA and Microsoft
National Competitions	Awarded internship by Société Générale for ranking among Top 20/9600 participants in hackathon - Shortlisted by P&G out of 1600+ teams to present prototype of store Inventory Recommendation system - Ranked 1/418 (Probability & Statistics) and 1/304 (Ensemble Learning) nationally in Analytics Vidhya quiz - Certificate of Excellence and Rank 10 among 350+ teams in 'Stat Wars', national hackathon by ISI Kolkata
Co-curricular Achievements	AIR 2431 in IIT-JEE'11 among 4.8 lac students; AIR 2058 (state rank 186) in AIEEE'11 among 11 lac students AIR 651 overall and AIR 24 in Chemistry in National Level Science Talent Search Examination in 2008 State Rank 3 in International Informatics Olympiad organized by Computer Literary Foundation in 2007 - Achieved AIR 231 and State Rank 51 among 20,000 candidates in National Cyber Olympiad held in 2006

WORK EXPERIENCE (21 months)

Business Analyst	Grail Research	(Aug '15 - Jun '17)
Achievements and Initiatives	- Ranked among top 5 out of 400+ employees and Rated 5/5 in two consecutive annual evaluation cycles 3 times recipient of the best performance award; Delivered 10 hrs of training on SPSS to 50+ employees - Conceptualized firm website increasing visibility of offerings leading to incremental revenue of 5% (\$1M)	
Cost Optimization	- Single-handedly managed \$2.5M+ Asia Pacific region business of a \$700Bn+ Global technology giant - Executed cost management and forecasting modules; engaged independently with Head of Asia Payroll - Implemented ARIMA model in R to forecast employee headcount; Saved \$40K, 1.6% of annual account budget - Reduced error by 10% and saved 180 man-hrs yearly by automating the daily dashboard updation task	
Opportunity Assessment	- Spearheaded development of market sizing model for 2 industry verticals of \$10.2Bn data storage firm - Analysed 500 online surveys and 30 interviews using SPSS; developed strategy to migrate to digital records	
Business Development	- Helped in securing \$0.6M revenue for firm by converting 4 potential clients in collaboration with CSOs Converted one-time engagement with \$162bn client into permanent account, additional revenue of \$150k	

DATA SCIENCE PROJECTS

Click Through Rate Prediction	- Predicted deal probability for online classified ads based on textual, image & historical data for similar ads - Used TF-IDF and Word2Vec for vector representation of text in the ad and clustered sellers using LDA - Extracted image features by Transfer Learning using VGG-16 and tuned LightGBM to achieve 0.22 RMSE
Restaurant Visitor Forecasting	Forecasted visitor headcount for 829 restaurants using their past one-year reservation and visit data - Modelled seasonal ARIMA, LSTM and generated 300+ features by rolling window to capture seasonality - Achieved better performance using boosted tree model with RMLSE 0.5 in comparison to 0.8 with LSTM
Bond Trading Prediction (BNP Paribas)	- Identified corporate bonds a customer will trade in coming week using customer trading data of two years - Generated synthetic data, extracted customer-bond features and achieved AUC 0.63 using Random Forest - Identified similar users using cosine-similarity and improved AUC to 0.71 by modelling with XGBoost
Recommender System (P&G)	Recommended products with highest probability of purchase for 6000+ stores based on product ratings - Implemented collaborative filtering by Singular Value Decomposition to get normalized RMSE of 0.22 - Modelled non-linear latent factors with neural network based variational auto-encoder in Python Keras
Fraud Detection in Mobile Ad Clicks	- Classified if click on a mobile ad is fraudulent given device attributes using a dataset of 240 million rows - Leveraged random under sampling to handle class imbalance, engineered features based on time of click - Generalized model by rank averaging Gradient Boosting and Factorization Machine models (AUC: 0.98)
Stock Market Volatility	- Performed comparative analysis of SENSEX and NIFTY returns to study impact of global financial crisis - Analyzed stationarity using ADF test, ARIMA order with ACF & PACF plots and volatility with GARCH

EXTRA CURRICULAR ACHIEVEMENTS

Sports	Won badminton tournament for 2 years consecutively among 32 teams at Grail's sports tournament - Won Silver Medal (1st Runner-up) in inter-section badminton tournament, 2017 hosted by IIM Calcutta
Events and Social Work	- Organized annual sports competition & Christmas Celebration at Grail Research; 100 people participated Volunteer at Social Mass Awareness Campaign against Cancer and AIDS by Caring Souls Foundation

TOOLS AND LANGUAGES: Python, R, SQL, VBA, MS Excel, SPSS, Visualization in Tableau

ELECTIVES: Pricing and Revenue Optimization, Corporate Finance, Supply Chain Analytics, Information Retrieval



ACADEMIC QUALIFICATIONS			
Year	Degree/Board	Institute/University	% / CGPA
2019	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-
2014	B. Tech (Mechanical & Automotive Engineering)	Delhi Technological University (DTU)	7.5/10
2010	Class XII (CBSE)	The Mother’s International School, Delhi	87%
2008	Class X (CBSE)	The Mother’s International School, Delhi	88.8%
Areas of Interest: Linear Algebra, Complex Networks (applications in sport analytics)			
AWARDS AND ACHIEVEMENTS			
Institute Rank 10 – PGDBA 2017-19*			
Academic Distinctions & Publications	• Recipient of Bronze medal in the national level Manav Sthali Inter School Mathematics Talent Test ‘10 • Among 184 out of ~1.65 lakh applicants to qualify the UPSC conducted SCRA written exam ‘10 • Placed amongst Top 1.2% in IITJEE ‘11; Secured 99.51 percentile in QA & DI, 99.04 total percentile CAT’13 • Awarded prestigious Junior Science Talent Scholarship by Science Branch of Education, Delhi ‘06 • Published research paper on design and fabrication of a Micro Air Vehicle in IJMPE Nov ‘13 issue [paper]		
Sports Achievements	• Represented Delhi in U-14 table tennis at the 51st National School Games held at Vishakhapatnam ‘06 • Received monthly scholarship from Sports Authority of India from 2007-14 for excellence in table tennis • Conferred with Emerging Player award in table tennis at IIM Calcutta Inter-Section Sports Meet ‘17 • Won an aggregate of 4 Gold, 1 Silver and 2 Bronze medals in sports of table tennis and football at tournaments of Inter-IIMs, Inter-Sections, IIT KGP GC and other inter-varsity competitions		
WORK EXPERIENCE (26 Months)			
Decision Analytics Associate		ZS Associates	Gurgaon (Apr ‘15 - Jun’17)
Incentive Compensation Design	• Designed motivational compensation programs for salesforce of a US pharmaceutical giant, across 9 BUs • Created IC plans for 2,500+ salesforce to drive changes in rep behavior, in line with brand strategy • Designed and implemented IC plan for 2 key drug launches across cardiovascular and diabetes space • Incorporated latest news of formulary wins and losses in setting goals for sales reps, ensuring fairness		
Data Visualization	• Observed geographical distribution of sales uptake by constructing heat maps with ZS’ proprietary tool • Analyzed dependency relationships between sales and its potential indicators using correlation plots • Built an executive summary dashboard for VPs to highlight budget spend on sales force and its impact		
Modelling & Simulation	• Predicted sales targets for reps using regression model, while checking for fairness through simulations • Estimated payout sensitivity associated with new product launch by performing Monte Carlo simulation		
Impact	• Reduced turnaround time of operations by ~2 days by populating deck from excel workbook using VBA • Led sessions on effective communication and Monte Carlo simulations for more than 20-member team		
Recognitions	• Secured first position amongst ~ 40 participants in the Annual Awards Design Workshop held in 2016 • Won the Project Champions Award 2016 out of more than 100 project teams for operational excellence		
DATA SCIENCE PROJECTS			
Early Detection of Follower Loss	• Analyzed factors behind Twitterati’s follower loss with data of 9.3M users at two different time points • Created user-user network based on topic similarity of tweets and computed various centrality measures • Applied Louvain algorithm to detect communities and performed word content analysis of topics • Augmented content-based features with graph features in SVM model obtaining an F1 score of 0.8		
Product Recommendation	• Predicted propensity of a customer to buy a certain item based on a sequence of click and buy events • Engineered user, item and time-based features from data of ~9M users, ~53K items & 339 item categories • Combined Factorization Machines output with Collaborative Filtering to provide recommendations		
Volatility in Indian Market	• Performed comparative analysis of SENSEX & NIFTY returns to study impact of global financial crisis’08 • Analyzed stationarity using ADF, ARIMA order with ACF and PACF plots and volatility with GARCH		
Stay Alert Ford Challenge	• Built a classifier to predict a driver’s alertness using physiological, environmental and vehicular features • Investigated feature importance using Lasso Regression and found physiological variables as insignificant • Achieved better AUC with Logistic Regression compared to SVM, Random Forest, Naïve Bayes models		
Image Masking Challenge	• Removed studio background from car images taken at 16 different angles using CNN based architecture • Stacked multiple models at different resolutions capturing local and global effects to get F1 score of 0.995		
EXTRACURRICULAR & POSITION OF RESPONSIBILITY			
Content Writing	• Authored ‘meet the team’ section of a ZS monthly newsletter with a reach of ~150, including clients • Collaborated with PlaceComm on content creation of internship placement report & placement brochure		
Mess Manager	• Managed wealth of ~₹6 lakh in 1st semester at ISI and made food arrangements for 2 celebratory events		
TOOLS: R, Python, SQL, Microsoft Excel, VBA ELECTIVES: Supply Chain Analytics, Product Analytics, Corporate Finance			

ACADEMIC QUALIFICATIONS			
Year	Degree/Board	University/Institution	%/CGPA
2019*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-
2017	CFA Level 1	CFA Institute, VA, USA	Qualified
2016	B.Tech (Mechanical Engineering)	Nirma University, Ahmedabad	8.31/10
2012	Class XII, GSEB	Fatima Convent High School	89.00%
2010	Class X, GSEB	Fatima Convent High School	91.69%
AWARDS AND ACHIEVEMENTS			
INSTITUTE RANK - 6 PGDBA 2017-19*			
International/ National Competitions	- Representing India (AIR 1, World Rank 9) in Data Science Game, Paris sponsored by BNP Paribas '18 - Ranked 11th worldwide among 706 participants in Stat Wars hackathon organized by ISI Kolkata '18 - Finalist (top 7%) in Smart India Hackathon by GOI as a part of PM Modi's Digital India Initiative '18 - Finished in the top 5% among 32000+ participants in Predict the Criminals hackathon '18		
Scholastic Achievements	- Scored 99.96 percentile in Quantitative Ability section and overall 99.52 percentile among 200K in CAT '16 - Awarded CR Mehta scholarship for achieving 100% marks in Mathematics in SSC examination '10 - Secured 149th position in National Science Olympiad (NSO) among 100K participating students '08		
Certification	MapR Certified Spark Developer (MCSD) - Machine Learning with Apache Spark v1.6 '18		
DATA SCIENCE COMPETITIONS			
Data Science Game 2018 (BNP Paribas)	- Recommended corporate bonds to customers which they are most likely to buy or sell in next week - Engineered features from transactional and market data; performed feature selection using Boruta - Ensembled XGBoost and LightGBM base models outperforming BNP Paribas baseline AUC by 0.21%		
Smart India Hackathon 2018 (GOI)	- Forecasted region-wise hourly load demand using LSTM for next one week and achieved 3.1% MAPE - Captured multi-seasonality with TBATS library using past 5 years hourly load and weather data - Designed interactive web dashboard for GOI using Shiny in R to monitor load availability in real time		
Stat Wars 2018 (Bank in Uzbekistan)	- Predicted the expected quarterly operational losses for next 1 year for 3 business divisions of the bank - Estimated the loss distribution from severity and frequency of past losses using Monte-Carlo methods		
WORK EXPERIENCE			12 months
Production Engineer		Delta Industries Pvt. Ltd	Jun '16 - May '17
Awards	- Awarded "Innovator of the year" among 250+ employees for reducing lead time of deliverables by 4 weeks - Received letter of appreciation for continual improvement in quality and reliability of product design		
Responsibilities	Led a 20 member team responsible for executing and monitoring Volvo-Eicher project worth 10 million INR - Single-handedly deployed 3D Printing infrastructure worth 3.5 million INR in existing production line		
Impacts	- Increased revenue by 25 million INR by eliminating tooling costs and reducing material costs by ~5% - Integrated Cura software to existing design processes for better job scheduling, increasing OEE by 6%		
ANALYTICS PROJECTS			
Portfolio Optimization	- Constructed diversified portfolio of US stocks in S&P 500, generating market neutral return of 4.6% - Identified co-integrated pairs of stocks based on 19 fundamental factors using PCA and K-means - Optimized trading parameters using Hidden Markov on 5 years daily data sourced from Bloomberg		
Product Recommendation from clickstream	- Predicted the most likely item to be bought based on clicks of users and recommended similar items - Leveraged undersampling to handle 33M clicks of 9.3M e-commerce users with ~1% click conversion - Ensembled FM and GBDT to predict buy propensity and collaborative filtering to recommend items		
News Summarization	- Built a system that retrieves news for given query from a cluster of news articles and summarizes them - Used TF-IDF and Word2vec for retrieval (0.52 F1 score); degree centrality method for summarization - Evaluated generated summary against human reference summary achieving 57% matching bigrams		
Modeling of cryptocurrencies	- Analyzed trends in prices and volatility of Bitcoin, Ethereum and Ripple using past two years daily data - Gauged suitability of modeling log returns of cryptocurrencies using ARIMA and GARCH techniques		
Venture Analytics	- Predicted whether a start-up is successfully funded by analyzing data of 3 million Kickstarter projects - Achieved 88% accuracy using Random Forest generating insights for both entrepreneurs and investors		
POSITIONS OF RESPONSIBILITY			
Placement Representative	- Conducted final placements of 2nd batch of PGDBA achieving 27% increase in average salary offered - Formulated the placement charter, framed placement policies and organized industry interactions		
Regional Head, NGO Basal	Led a team of 12 to conduct teaching sessions for 78 under-privileged children in Dahej, Gujarat - Organized 5 weekend vocational training workshops for 165 unemployed youth as part of 'Skill India' - Managed inventory of products worth 2.5 million INR sanctioned by State Government of Gujarat		
ELECTIVES: Information Retrieval, Corporate Finance, Pricing and Revenue Optimization, Supply Chain Analytics			
LANGUAGES/TOOLS : R, Python, SQL, IBM-SPSS Modeler, Spark, Tableau, Shiny, Google Cloud, Bloomberg			



Roles Offered : a) Capacity Optimization Analyst
b) Business Intelligence Engineer

Job Description :

Capacity Optimization Analyst

The successful candidate will work in lock-step with multiple business teams, CS Operations, Global Outsourcing, Program Management and Finance leadership. He/she will:

- Develop forecasting/Capacity Planning models that captures the dynamics of retail and digital e-commerce business
- Develop large scale optimization and predictive models using AI/ML tools/technique
- Promote process improvement and standardization of processes across all sites in the network
- Lead critical projects to improve planning and forecasting efficiency for global operations.
- Cost and service optimization for customer service organization spread across 20+ countries worldwide.
- Evangelize customer service improvements across WW Customer Service Operations and gain buy-in from senior leadership on proposed changes
- Improve performance to plan by identifying, measuring and managing key metrics related to customer service
- Coordinate with internal and outsourcing network operation teams to meet business service levels

(<https://www.amazon.jobs/en/jobs/706255/business-intelligence-engineer-intern>)

Business Intelligence Engineer *

- Monitor existent metrics, analyzing data and partnering with internal teams to identify process, system improvement opportunities and come up with forecasting/BI solutions
- Take a proactive approach to defining new metrics, reports, and dashboards.

Drive business decisions by analyzing of demand and supply side metrics, designing and developing dashboards and reports

- Extensive stakeholder management of data suppliers and data consumers worldwide who operate out of diverse geographies.

- Improve data flow between multiple systems and provide self-service tools to the customers

- Perform database management of the team, develop frameworks to process very large datasets

- Support reporting modules, design and maintain data architecture for statistical research and development of forecasting systems.

- Maintain data integrity and data quality of the team inputs and outputs.

- Participate in global capacity planning and customer service initiatives and project roll outs to cater to growing business needs

(<https://www.amazon.jobs/en/jobs/706189/capacity-optimization-analyst-intern>)

* JD of BIE and work might differ.

Internship Stipend Offer : Rs 1,00,000

Shortlist Steps : a) Resume Shortlist

b) Technical

c) HR Round

Duration : 2 Round of interviews Each 1 Hour.

Capacity Optimization Analyst

Technical Round Questions (Arjun Gupta)

- a) Different types of JOINS (SQL)
- b) Expectation Maximization, K- means (These were part of a project that I did)
- c) Case Study: How do we calculate the number of people interested in buying Amazon Kindle ?
(Considering its launch is due next week)

Technical Round Questions (Shubhank Bhandarkar)

Projects and Competitions by Shubhank

Projects: Recommendation System, Time series and NLP

Competitions: BrainWaves, Capsim, Kaggle and Analytics Vidya

Project based questions:

- a) Which project did you like the most? Explain it. (Explained recommendation system in detail)
- b) Cross questions on techniques used in project:
- c) Explain how K-Means works (on whiteboard). Algorithm, Limitations (can't handle categorical data). Other methods for clustering which overcome limitations of K-Means.
- d) What is PCA? How it works?

Analytics questions:

- a) Explain R-Squared in Layman terms ?
- b) What is p-value? How is hypothesis testing done?

- c) Rate yourself on scale of 1-10 in SQL and R. Asked some R questions: What does library function do in R? Which packages did you use?
- d) Which visualization tools you have worked on? I said Spotfire. He was looking for Tableau so didn't get cross question here

Guesstimate:

- a) Total number of cars in Kolkata

Work Experience:

- a) Brief explanation about the role in Fractal?
- b) Any project for which you were recognized?

General:

- a) What do you think amazon does? It's business model.

Technical Round Questions (Abhi Jain)

- a) Work Ex related question
- b) Case Study based on the work done by Amazon(first the scenario was explained by the interviewee)
- c) Necessary condition for joining two tables in SQL. Give an example of Self Join

Technical Round Questions (Deepanshi Seth)

Projects and Competitions by Deepanshi

Projects: Recommender System, Object Detection and Classification, Anomaly Detection, Sales Forecasting, Opinion Spam Detection, Predicting Startup Acquisition

Competitions: Some competitions on Kaggle, AV and Hackerearth

a) Case question: You have to plan capacity for one week for Amazon customer support center for calls regarding the Amazon Prime Video accounts. Also during that week there would be promotional activities going on for the same.

Estimated number of existing customers are 500.

b) Give an example of self join in SQL

c) Brief discussion about work experience- focus on one of the projects and the tools and technologies used ?

d) What is more important: process or data?

Technical Round Questions (Namit Agarwal)

Projects and Competitions by Namit Agarwal

Projects: a) Hotel Recommendation Systems b) Store Sales Prediction c) Product Image Classification d) Stock Price Forecasting e) Document Summarization f) Entity Representation

a). Tell me About yourself - both interviews started with this.

b) Work Ex - A lot of questions revolved around what I did during my work. They also asked the rationale behind my decisions and approach. Discussion on numbers mentioned on the resume. Specific role in all the projects,

c) Cases - They gave me situations and asked me what type of features/data will you collect. To predict number of persons required for the customer care for a product

Then discussion on multiple cases - If the product is new, If it is a festive season ?

- d) Guesstimates
- e) MySQL - interviewer asked me to rate myself on SQL and gave me two questions. Both were on Join queries.
- f). Projects - Store Sales Prediction - What you did in this project? Description of data? Missing data imputation?
Important variables? Why time series? Why Regression and XGboost?
- g) Difference between Python and R and what would you choose?
- h) Skills you learned in pgdba and at your work ?

Technical Round Questions (Siddharth Das)

- a) CV based interview with focus on work ex and projects done ?
- b) What was your typical day in the organization? Roles, responsibilities etc.?
- c) Couple of questions related to the achievements during the job ?
- d) Explain your favorite project. I choose time series project as it was what the role was for basically ?
 - d)1. What all time series models can be applied given a scenario etc ?
- e) Explained the role in and out and asked how I found it interesting and what all challenges I could face.
I related it to a competition I took part in and told the challenges faced and a naive approach to forecasting and scheduling as well as the actual approach used in firms.
- f) Some scenario based questions related to the three profiles they have within capacity optimization and which one would suit me ?

Business Intelligence Engineer

Technical Round Questions (Korah T Malayil)

Projects and Competitions by Korah T Malayil

Projects: Yelp Image Classification ,Yelp online review usefulness prediction ,MovieLens movie recommendation engine, Instacart Market Basket analysis

1st Round

- a) Explain the process of fitting a Regression model.
- b) When and why do we use Regularization?
- c) How does the closed form solution of linear regression change when we add regularization?
- d) Explain the working of decision trees? (From splitting criteria to bias-variance)
- e) What are the advantages of using Random Forest? Explain how it reduces variance
- f) Explain the difference between bagging and boosting models

2nd Round (On Python interpreter)

- a) Write a program to find the kth largest number out of 100 billion numbers, taking complexity into consideration.
(Answer: Use a map-reduce cluster, and write the program accordingly)

Technical Round Questions (Sambeet Tiady)

Projects and Competitions by Sambeet Tiady

Projects : Advertising, Demand forecasting, Text classification, etc. (Focused on feature engg. and Deep learning, tried to cover as many tools and techniques as possible in the projects.

Competitions - Kharagpur semester I was mostly focused on competitions.

ROUND 1 -

a) Word embeddings:

- a)1. How did you train word embeddings? What was the input and the output?
- a)2. What was the architecture of the network used?
- a)3. What are the different ways you can get word embeddings and when would you use what?

b) Compare LSTM and SVM (this was a text classification project):

- b)1. Why LSTM worked better than SVM? Why not bi-directional LSTMs? What features did you use for SVM?
- b)2. What kernel did you use? Give a case where a non-linear kernel works while a linear kernel doesn't (XOR) ?
- b)3. Briefly describe the optimization for SVM ?

ROUND 2 -

a) Brief intro.

b) Rate yourself on R and Python. Was asked simple syntaxes/code on R and python for 20-30 mins following that.

- b)1. How to remove variables? How to access any particular row? Removing rows and columns?
- b)2. Changing column names, converting text to upper case?

c) Was asked to explain a project from work experience mentioned in the resume.

d) Explain another from pgdba projects.

- d)1. Difference between Random Forest, GBM, xgboost and lightgbm.

e) Was given a case:

Kindle is launched first time in India. It is different from other products because it has technical issues as well as logistics issues. The customer service associates who handle technical issues need to be trained separately than logistic ones.

e)1.How would you estimate the number of associates for each type? What data would you look at?

e)2.During Diwali sale week, customer service calls jump 5x from normal weeks. How would you manage workforce in such a case? Suggest something innovative.(I thought in terms of cost, training, variable pay and outsourcing)

HR Round

HR Round Questions (Arjun Gupta)

- a) Work Experience
- b) Qualities of a team leader
- c) My strengths/ weaknesses as perceived by others
- d) Placement Representative Experience and learnings
- e) What do I bring to the table
- f) Why Amazon

HR Round Questions (Shubhank Bhandarkar)

Note: Amazon focuses a lot on its 14 leadership principals and will test you on these. It is good to have basic idea about them. Don't try to fake yourself as per these principals. Be yourself and clearly justify your opinion.

Interviewer: Head of World Wide Capacity Planning Team

- a) He started with: Who interviewed your technical round? I didn't know his name. (Advice: Always try to know your interviewer's name)
- b) Asked about personal background. About family, hometown, school & college
- c) Why Capacity Optimization role? Was it your first preference?
- d) Strengths and weakness as perceived by others
- e) Asked if I would be willing to work if no role consisted of ML work but only basic Analytics (Be clear about your preference here)
- f) Any experience where you influenced 10-15 senior management with your results
- g) Any experience (in office or other) where you built something very simple which had huge impact and was recognized for it
- h) Various questions behavioral questions such as what would you choose between: Growth and stability? Why?
- i) Working in team vs working alone? etc

HR Round Questions (Abhi Jain)

This was more challenging than technical round. It mainly revolved around if I am fit for the organization.

HR Round Questions (Deepanshi Seth)

- a) Explain what you know about the role. (asked to use white board)
- b) Short term and long term career goals
- c) What do you lack in yourself?
- d) What is more important: process or data?
- e) Where do you see yourself: complete technical role, business role or project management role?
- f) Do you want to be flexible in life or go for a niche career?

HR Round Questions (Namit Agarwal)

- a) Tell me About yourself?
- b) Why Amazon? - Revenue? Number of employees?
- c) Why this role?
- d) Long term and short term goal?
- e) Questions on Work experience? Skills learned ,challenges faced?
- f) Why CDOT? (Work-ex - Research Engineer ,Centre for Development of Telematics (CDOT))
- g) Asked me to rate myself on innovation and then asked me what I did innovative in my work?
- h) Strength and weakness? Cross questions on them
- i) What do you do for self development?
- j) Questions on different roles of Amazon and situations in them?
- k) What would you do If you get a lucrative offer from some other company?
- l) Would you work for a leader or an organization?
- m) Would you like to work in a team or lead the team?

HR Round Questions (Siddharth Das)

This was a long interview which basically was to check the fit with the culture of the organization and the role. Lots of very well phrased and tricky questions. The interviewer was the site Head of WWCP(Looking for very fast answers)

- a) Questions on sig sigma certification that I just completed in May this year as he had experience in six sigma implementation.
- b) What do you prefer Data or the process?
 - b)1. How do you go about when you get any data?

- c) What is your top priority Learning, stability, growth or pay?
- d) What is your strategy in an era when a lot of firms are failing?
- e) Why PGDBA?
- f) Why Amazon?
- g) Strength? Weaknesses (Said not to use clichés answers and say something unique- I guess this was the point where he got impressed by my answer)
- h) Apart from academics what do you do and how does that add value to my career/life?
- i) Again a few questions related to the role and how I see myself fit?
- j) Some scenario based questions (how to convince a manager when he is against you/ project management questions)

No HR Round for BIE selected people

Other Technical & HR Round Questions

- 1) Started with a project on Amazon data on my Resume (Manoj Mana)
 - 1.1) Explain Factorization Machines?
 - 1.2) Dimensions of the word2vec that we used for LSTM model(He skipped further questions on LSTM as I said I am not that confident)
- 2) Derive expression for Beta matrix in linear regression.
 - 2.1) Intuitive understanding of the hat (projection) matrix.
 - 2.2) Write the dimensions of all the matrices involved in previous derivation?
 - 2.3) What to do if number of regressors are higher than the number of data points (explained PCA)
 - 2.4) Write down all the steps while performing PCA

2.5) Link PCA to SVD

2.6) What to do if some of the betas are very close to zero (Lasso)

2.7) What to do if some of the beta is very large (Ridge)

2.8) Why did we use L2 loss in OLS & what other loss functions can be used (advantages/ disadvantages)?

3) Started with simple SQL queries, different joins. (joins, rank, group by, extract weekday from date, supply etc)

4) Asked me whether I am more comfortable with R or Python.

4.1) Asked me to write codes for a lot of cases in R

Having a good understanding of the ML techniques and hands on experience on R or Python can easily land you up in Amazon

5) Theoretical concepts of ML with special focus on a project.

6) Programming in Python (Arrays, Strings, Binary Trees)

7) In depth Discussion on NLP project (on approach, working of LSTM model, model parameters, packages used, loss function, evaluation metric)

8) Coding based on Strings

9) Difference between univariate and multivariate time series data?

10) Difference between gbm, xgboost and light gbm ?

11) Explain the most difficult project from your CV ?

12) Explain mixture of experts model?

13) Explain latent dirichlet allocation?

14) Why is feature engineering important?

15) Simple coding questions from R and SQL.

16) Write a code to implement K-means clustering in Python.(No choice of language given, had to be done in Python),

- 17) What is the complexity of K-means clustering, Is the solution of k means clustering always optimal?
- 17)1. Assume K means clustering is done in 2 dimensions, how will you do it using dynamic programming?
- 17)2. Write down the optimal sub-structure?
- 18) SVM v/s Logistic, Whether any one of them/both are susceptible to outliers? Intuition behind SVM?
- 19) MLE equation for Logistic?
- 20) Bias Variance Tradeoff, Regularization,
- 21) How does Lasso work, difference between Lasso and Ridge?
- 22) Explain how Lasso does feature selection (The interviewer was looking for a Math oriented answer)
- 23) Difference between Random Forest, XGBoost, LightGBM
- 24) Asked if I was aware about the detailed working of XGBoost/Adaboost and then explain?
- 25) Different types of ensembling approaches possible?
- 26) Question about FM, FFM
- 27) Pseudo codes for some algo's in the second round - Cormen may help in this case - If correct answer is given, the interviewer will then go for efficiency and urge the code to be written in $O(n)$ - data structures (hashing, arrays), Dynamic Programming
- 27)1. The algos were given an array of n elements of stock prices of every day, find the optimum gain (When should we buy and then sell), then asked for a solution in $O(n)$.
- 28) Find k th largest/smallest element in array - sorting, then use of heaps, and then some cross questions on that ?
- 29) Given an array of n elements, find 2 elements such that their sum is s - Gave 2 solutions, one with hash maps, then extend the algo so as to find k elements such that their sum is s (Recursion and DP)
- 30) Explain DSG problem and approach.
- 31). logistic regression vs xgboost
- 32) tf-idf

- 33) Logistic loss function and how to solve ?
- 34) Code entire logistic regression from scratch ?
- 35) Code entire K-means from scratch ?
- 36) Print all substrings for a given string ?
- 37) Many questions on XGBoost and its working,
- 38) LSTM, Comparison between Random forest and XGBoost,
- 39) Discussion on projects
- 40) Coding task - 1)code to randomize a given list,
2) code that samples from data randomly
3) Time and space complexity of each code
- 41) Tell me something which is not in your resume.
- 42) Rate yourself in R and Python on scale of 1-10
- 43) What will you use for text analytics btw R and Python. Why?
- 44) Tell me few libraries in which you have used in Python. what are the libraries available for NLP.
- 45) Time series Project: Explain ARIMA, write mathematical equation of ARIMA, how will you explain MA smoothing and ETS smoothing to a layman.
- 46) Case: How will you predict the number of calls/emails to the customer support during the sale period. What all predictors will be needed.
- 47) Asked to explain one of my projects.
- 48) Why PGDBA?
- 49) Suppose you have developed a forecasting tool by investing a lot of your time and the finance and HR team says that your tool is underestimating and they deny to use your tool, how will you handle such a situation.
- 50) Tell me about your work-ex.

- 51) Given the knowledge that you have gained during PGDBA, what is something that you would have done differently in any of the project during work-ex.
- 52) Which tool are you comfortable with, R/Python?
- 53) Given an opportunity to work in purely technical role vs purely managerial role vs a mix of both, which one will you choose and why?
- 54) Where do you see yourself in 5 years?
- 55) SQL question based on partition by, various types of joins,
- 56) A graph without labels was given along with diagram which depicted clustering algorithm without any labeling- identify the clustering algorithm and what the graph depicted?
- 57) Explain R2 in layman terms?
- 58) Explain K means clustering?
- 59) Asked a few things about Lean sigma black belt(As I had in my resume) (Ashish Das)
- 60) How do you approach Kaggle competitions(As I had few in my resume) (Ashish Das)
- 61) What strategies would you take to reduce the call wait of the customer(in queue) using data?
- 62) HR Round was kind of a psychometric interview where there were rapid fire questions and they were checking whether you are honest. Also they checked your stability and how motivated are you to join the company ?

Courses & Skills Suggested

Arjun Gupta

In my interview, CDS and DBMS were the highlights but it was the combined knowledge gained in various courses that played the role in hindsight. I'll briefly share my thoughts about the courses in each semester.

Courses & Skills Suggested

Arjun Gupta

In my interview, **CDS and DBMS** were the highlights but it was the combined knowledge gained in various courses that played the role in hindsight. I'll briefly share my thoughts about the courses in each semester.

ISI lays down the foundation. **Inference, DBMS, CDS** are very important, for the learnings here would make your journey smoother. ****Get your hands on coding during this semester****

During IIT semester, make your machine learning concepts stronger. I found **FADML, Time Series, DSL Lab, Supply Chain Analytics** really useful.

The courses at IIM are really important because they give you a perceptive of how to use the knowledge you acquire in the last one year to the real world problems. **Selected Aspects of Predictive Modelling, Pricing Revenue and Optimization, Strategic Management** were really useful.

Apart from the curriculum that, I believe, everyone has to follow religiously, I will list down 4 things that I feel are paramount.

- 1) Code. Start coding, even if you don't like it. You'll do fine after a year of regular coding. A lot of companies have coding tests as a way to shortlist candidates.
- 2) Your resume. Start working on your resume right now. Companies shortlist basis your resume. Participate in competitions, hackathons. For the interview, you should be thorough with each and every work that you write in your resume.
- 3) Confidence: I feel the key to crack interviews, apart from your technical knowledge, is confidence. Making interview as a discussion/ interaction than a very formal interview helped me.
- 4) Complacency: Stop being complacent, if you are. PGDBA bubble is growing fast, make sure it doesn't burst.

Shubhank Bhandra

- a) Inference,
- b) FDBMS (SQL),
- c) IR for NLP & Programming (R & Python)
- d) CDS (Machine Learning Algorithms),
- e) Cases and Guesstimates
- f) IIT - Information Retrieval
- g) IIM - Courses which requires solving cases
- h) Be thorough with all the points in your CV and be confident.
- i) Programming is the basic requirement and companies assume we all are good at it so try to master at least one of the programming language (R or Python).
- j) Try to focus on business application

Abhi Jain

Prepare your Resume thoroughly. Focus on the "Why" question for everything - why this technique, why not some other technique, why this evaluation metric, etc

Deepanshi Seth

- a) Database management
- b) CV is very important. Be thorough with whatever you have written
- c) Both the interviews were more of a conversation. They were checking how clear we are with respect to what we want to do in life. So consistency in answers is very important as they asked same questions in different forms.

Namit Agarwal

- a) Technical - Full understanding of all points mentioned in your CV,
- b) Clear Understanding of the projects in PGDBA and in work
- c) HR - Self Awareness
- d) Amazon tests you on their 14 Leadership Principles and see if you are a cultural fit
- e) CDS, DBMS, Inference, RTSM, FADML, Strategic Management
- f) Maintain your CG. Also do as many projects or competitions as you can.

Siddharth Das

- a) Time series, CDS, Practical scenario based Competitions, Spontaneity
- b) Predictive Modelling(IIM), CDS(ISI),Project management(IIM)

Sambeet Tiady

Apart from course work these definitely helped

- a) Deep Learning (Andrew Ng)
- b) Questions were asked on simple syntaxes in R and python.
- c) Computation for Data Science
- d) Inference
- e) Prepare CV and projects well. Focus on machine learning and statistics.
Everything else is secondary for revision. Follow a structure when asked to explain projects.

Korah T Malayil

- a) CDS (ISI)
- b) FADML (IIT)
- c) All Projects
- d) Machine Learning Theory
- e) Programming skill
- f) Inference (ISI)
- g) Predictive Modelling (IIM)
- h) While learning basic Machine Learning theory, read about the algorithms on Analytics Vidhya, Medium etc.
You'll gain a better understanding of the nuances of the techniques by this.

ACADEMIC QUALIFICATIONS			
Year	Degree/Board	University/Institution	%/CGPA
2019*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	--
2014	B.E (Electronics and Electrical Communications)	Punjab Engineering College, Chandigarh	7.30/10
2010	Class XII, CBSE	Mount Carmel, Chandigarh	91.0%
2008	Class X, CBSE	DAV Centenary Public School, Mandi	92.8%
ACHIEVEMENTS			
Academic Achievements	- Received Bronze medal in Kaggle TalkingData Ad Fraud Detection Challenge; Among Top 6% teams Top 1.7% in IITJEE '10; Top 0.4% in AIEEE '10; 99.61 percentile in Quant, 99.53 percentile in DI in CAT '16 - Won 1st prize among 400+ students in a Mathematics event in VYOM, annual technical fest of the college School Topper among 100+ students in Maths, Class 10; Awarded Achievement Certificate by DAVCMC		
Sports Achievements	National Badminton Player; Represented Himachal Pradesh twice in National Badminton Tournaments National Semi Finalists in Badminton in CBSE Inter-School Sports Competitions held in Jharkhand - Secured podium finishes (1 Gold, 1 Silver, 1 Bronze) from 2010-14 in PEC Open Badminton Tournament		
POSITIONS OF RESPONSIBILITY			
Placement Representative	- Formulated Placement Charter & key policies; facilitating industry interaction, promoting PGDBA brand - Conducted Placements of 2nd PGDBA Batch of 51 students successfully; 26.7% rise in avg. compensation		
Sports	Captained Badminton Team of 6+ people at PEC; Won 1 Silver, 1 Bronze in Inter-Varsity tournaments - Elected as Sports Coordinator for PGDBA 2017-19 Batch for Intersection Tournaments at IIM Calcutta		
WORK EXPERIENCE (32 months)			
Business Analyst		Evalueserve (Consumer Insights and Data Analytics)	<i>Gurgaon (Sep '15 – May '17)</i>
Client Engagement	- Handled end-to-end process transition and new work migration followed by process re-engineering - Managed capacity planning- scheduling, prioritizing and resource mapping for 50+ ad hoc requests - Identified requirements and drafted execution plans by coordinating with various key stakeholders		
Automation & Process Improvement	- Achieved business efficiency by automation of reporting processes in VBA, saving 200+ hours annually - Conducted over 20 training sessions in process improvement and automation for a team of 5 members - Streamlined induction process for new members by designing & documenting 10+ operational procedures		
Business Analyst		Evalueserve (Corporate and Professional Services)	<i>Gurgaon (Sep '14 – Sep '15)</i>
Automotive Telematics Market Analysis	- Helped a Fortune 500 telecom client gain revenues of USD 25M, generating USD 150K revenue for EVS - Carried out the project end-to-end from developing project proposal plan to final solution delivery - Built forecasting models to estimate the addressable market size for 6 different European geographies - Conducted interviews with SMEs to identify key trends and growth factors in the telematics industry		
Identifying Target for Acquisition	- Leveraged databases such as Avention, Hoovers, Kable to perform M&A, SWOT and financial analysis - Built score-carding framework; Created interactive dashboard to compare set of candidate companies - Received “Pat On The Back” quarterly award for delivering exceptional performance for this project		
DATA SCIENCE PROJECTS			
Ad Fraud Detection Challenge	- Predicted the probability of a display ad click being non-fraudulent by developing a model in Python - Explored ~200M clicks data using Tableau and leveraged Under-sampling to reduce the dataset size - Engineered click popularity and time-based features to improve the ROC-AUC score from 0.95 to 0.98 - Stacked LightGBM and Neural Networks using Soft Voting to achieve a final ROC-AUC score of 0.982		
Pipe Breakage Prediction	- Developed an algorithm to forewarn authorities of impending water main failures of ~190k pipelines - Implemented Expectation Maximization in R to cluster 78k pipes based on the time up to first breakage - Performed feature selection using Boruta and trained a Random Forest model with an accuracy of 80% - Applied Box-Cox transformation and predicted the time gap before successive breakages using XGBoost		
Store Sales Prediction	- Built a model to Forecast sales for 1,115 Rossmann drug stores using store, promotion & competitor data - Determined order of ARIMA model using ACF and PACF plots; Checked stationarity using ADF Test - Implemented ARIMA and TSLM to forecast daily sales for the next 6 weeks; Achieved a MAPE of 0.15		
Interaction on Wikipedia	- Identified potential vandalism by analyzing editor interactions using a set of 9 different Wikipedia pages - Performed Temporal analysis of page edit history to study trends of registered and unregistered editors - Created Editor-Editor & Bipartite networks; Detected communities using Louvain & k-clique algorithm		
Product Classification	- Built models to correctly classify 60,000 products with 93 features in 9 anonymous product categories - Performed a comparative analysis of RF, NN, XGBoost achieving improved CV accuracy of 82% with RF		
EXTRACURRICULAR ACTIVITIES			
Social Endeavours	- Among the 60 Youths selected from all over country to participate in Himalayan Yuva Prerna Yatra - Volunteered for Bhumi, a non-profit organization; tutored 5 students in elementary Mathematics - Organized Nakshatra, annual cultural festival of Bhumi, with 650+ participants from 45+ shelter homes		



ACADEMIC QUALIFICATION

Year	Degree/Examination	Board/University	%/CGPA
2019*	Post Graduate Diploma in Business Analytics	ISI Kolkata, IIM Calcutta, IIT Kharagpur	-
2015	B. Tech., Electrical and Electronics Engineering	NIT, Nagpur (VNIT)	7.96/10
2011	Class XII	Maharashtra Board HSC	89.17%
2009	Class X	Maharashtra Board SSC	89.69%

AWARDS AND ACHIEVEMENTS

International Competitions	World Rank 18 among 450+ teams in 'Capsim Global Business strategy simulation' Spring challenge '18 Top 15% on Kaggle leaderboard in 'Avito Demand Prediction' and 'Toxic Comment Classification' challenge '18	
National Competitions	- Offered performance-based intern by Société Générale; top 20 finalist among 9600 participants in hackathon '18 Shortlisted by P&G among 1600+ teams to present prototype of neural network-based recommender system '18 - Among top 6 finalist in 'Conquestum', a strategy formulation case competition organized by SPJIMR '17 - Awarded Certificate of Excellence with Rank 10 among 353 teams in 'Stat-Wars' by ISI Kolkata '18 Finalist (Rank 7) among 190 teams in 'Default Prediction challenge' organized by ChiSquare Analytics '18 Ranked 12 among 261 teams in 'House Price Prediction challenge' by Great Lakes on Analytics Vidya '18	
Scholastic Achievements	State Topper among 16 lakh students in Class 10 Mathematics board exam; scored 150/150 '09 - Achieved 99.98 percentile in Maharashtra state High School Scholarship Exam by MSCE '06	

WORK EXPERIENCE

21 months

Business Consultant	Fractal Analytics, Mumbai	Aug 2015 – May 2017
Roles and Responsibilities	- Monitored brand performance and household penetration of 3 major categories for Fortune 50 FMCG client - Devised 10+ Market Mix Models (MMMs) delivering insights on pricing, promo and distribution strategies Automation Expert: Mentored cross-functional teams; Shouldered 2 cross-category projects simultaneously	
Category Driver Analysis	- Modelled 3 MMMs with 3-year data using Linear Regression; achieved 84% R-Squared and 3.5% MAPE - Identified drivers and drainers providing insights on campaign effectiveness; forecasted sales for 6 months	
Performance Dashboard	- Designed one-stop tool to track sales offset from forecasted target for 3 major brands in 6 APAC Countries - Embedded MMMs to forecast targets and provided real-time validation; automated by ETL from ~10 data sources	
Impacts and Achievements	- Exceeded goal benchmark by 23% in FY15-16; achieved NPS of 100% in FY16-17; QOS of 98% in all quarters - Conceptualized SKU initiative performance tool; adopted by clients for all SKUs to formalize business strategies - Enhanced delivery time efficiency by 62%, saving 104 man-hours/month by automating from ~10 data sources	
Professional Awards	- Received 'P&G On-Spot Award' for Spotfire tool providing 'Ease in data visualization' across hierarchy - Awarded 'Possible Excellence in Operations' and 'My Intent into Business Insights' for two projects	

DATA SCIENCE PROJECTS

Product Recommender System	- Predicted the list of items a user will add to cart in next order using 3 million past orders of 200K users - Segmented users by K-Means with PCA and generated association rules by market basket's Apriori algorithm - Engineered 40+ features capturing user and item buying behavior and applied stacked boosting classifiers - Improved recommendations by 5% using Expectation Maximization algorithm and achieved 0.39 F-1 score
Online Ad Demand Prediction	- Predicted likelihood of online ad sale based on its text description, image and 1.5 million similar historical sale - Extracted text features with LDA, TF-IDF; characterized images by blurriness, VGG-16's transfer learning - Ensembled LightGBM models to achieve RMSE of 0.22, 11% reduction over benchmark (Kaggle top 15%)
Visitors Forecasting	- Forecasted 6 weeks number of visitors for 829 restaurants using their past 1 year's reservation and visit data - Formulated seasonal ARIMA and LSTM; Generated 300+ features by rolling window to capture seasonality - Improved results by 38% using Boosting models compared to LSTM and achieved RMSLE score of 0.5
Word Ambiguity Detection	- Implemented a deep learning model to predict the actual context of an ambiguous word in an article - Generated training set by scraping POS tags of 40k ambiguous words from Wikipedia against their actual usage - Constructed Word2Vec similar to GloVe using stacked denoising Auto-Encoders and cosine similarity
Start-up funding prediction	- Predicted probability of start-up getting funded on Kickstarter with 88% accuracy using Random Forest - Identified key features and provided investment insights on country, goal and duration for project's success
Volatility in financial market	- Performed comparative analysis of SENSEX and NIFTY returns to study impact of global financial crisis - Analysed stationarity using ADF test, ARMA order with ACF and PACF plots and volatility using GARCH

POSITION OF RESPONSIBILITY

Student Mentor	- Assisted 15 first year students through their onboarding under SMP@VNIT program during 2013-15 - Received Certificate of Appreciation from Director for providing academic and non-academic counselling
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Tools: Python, R, VBA, SQL, KNIME, Excel | **Databases:** Nielsen, Kantar, GFK | **Visualization:** TIBCO Spotfire, Tableau

Electives: Information Retrieval, Corporate Finance, Pricing and Revenue Optimization, Supply Chain Analytics

ACADEMIC QUALIFICATIONS

Year	Degree/Board	University/Institution	%/CGPA
2019*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-
2015	B.E.(Honours) - Electronics and Instrumentation	BITS Pilani	6.95/10
2011	Class XII, CBSE	Delhi Public School, Bhilai	84.80%
2009	Class X, CBSE	Delhi Public School, Bhilai	90.50%

WORK EXPERIENCE (23 months)

Media iQ Digital Ltd **Campaign Analyst** **Bangalore (July '15 – May '17)**

Single-handedly managed **250+** Programmatic Ad-campaigns across multiple business domains to increase clients' RoI

Roles and Responsibilities	- Developed Linear & Log-Linear Regression Models to estimate client budgets & convert upcoming deals - Created SQL based brand performance dashboard of a Fortune 500 company - used by senior executives - Collaborated with product teams to deliver User Segmentation Tool - utilized for targeting the users - Optimized campaigns for Fortune 500 companies towards KPIs: click-through-rate and conversion-rate
Impact	- Delivered 70+ actionable Brand & User Insights for companies from user's cookie and third-party data - Managed £500K to £1.5Million of cumulative budgets every month across 10+ campaigns - Consistently generated 25% margin on budgets as revenue for company while ensuring client retention

AWARDS AND ACHIEVEMENTS

Data Science Competitions	AIR 27 among 9600 participants in Societe Generale FinTech hackathon; Offered summer internship '17 - Shortlisted by P&G to present prototype of Recommender System in National Innovation Challenge '18 AIR 5, WorldRank 47 in International Data Science Game on Bond Recommendation by BNP Paribas '18 Top 6 finalist at Business Case Study Competition – Conquestum, organized by SPJIMR, Mumbai '17 AIR 10 out of 877 participants in online round of Credit Risk Default Challenge by NMIMS '18 AIR 14 out of 261 teams in online round of Real Estate Price Prediction Challenge by Analytics Vidhya '18
Publication: Business Case	- Authored case "Deodorant Wars in India" - Analyzed deodorant industry (1995-2016) with focus on Axe - Submitted in John Molson MBA International Case Competition, 2019 among entries from 19+ countries
Scholastic Achievements	- Scored 99.46 percentile in CAT 2015 - 99.47 percentile in Quant and 99.38 percentile in Data Interpretation - Awarded CBSE Merit Certificate for being among top 0.1% in Xth board at national level in Science Scholar Gold Medallist at DPS Bhilai for excellent academic performance for 9 consecutive years ('02-'10)

DATA SCIENCE PROJECTS

Recommender System	- Developed latent factor based Collaborative Filtering Models to recommend movies to 6000+ viewers - Applied SVD model and optimized with Stochastic Gradient Descent to get Normalized RMSE of 0.22 - Implemented Neural Network based Variational Autoencoders to model non-linear latent factors
Object Detection & Image Classification	- Validated P&G product visibility norms in retail stores by detecting products in shelf images - Implemented Faster RCNN model in Keras by leveraging transfer learning from ImageNet dataset - Determined bounding box coordinates and classified objects with Mean Average Precision of 54.11%
Time Series: Customer Forecasting	- Predicted footfall for 829 restaurants in upcoming month using 1-year historical reservation & visit data - Leveraged SARIMA, uni & multivariate LSTM models to achieve RMSLE score of 0.59 on best model - Improved the score by 18% through LightGBM trained on rolling window features to model seasonality
NLP: Document Summarization	- Identified Highlights from News articles using graph-based approaches – text rank and degree centrality - Leveraged Bag-of-words technique to construct tf-idf vectors & cosine similarity matrix for sentences - Optimized Text Rank model with Power Method & evaluated summaries on Rouge-1,2, Rouge-L scores
Venture Analytics	- Identified key influential factors affecting Start-up Acquisition using company profiles from CrunchBase - Performed missing data imputation, feature engineering and built Random Forest model in R - Improved score with boosting model XGBoost; evaluated by Kappa score to account for class imbalance

EXTRACURRICULAR ACTIVITIES

Positions of Responsibility	Coordinator, Arts Club BITS Pilani - Led a team of 50 students to organize Art work for college festivals Public Relations Team, PGDBA: increased social media reach by 40% & wrote articles for press releases
Social Work	- Raised ₹18 Lakhs through Community Fund Raising events as a Volunteer at Make a Difference (MAD) - Initiated collaboration between Media iQ Digital and MAD as a part of company's CSR initiative
Sports	Silver Medalist in Inter-Section Badminton tournament at IIM Calcutta (2017)

TOOLS & TECHNIQUES: Python, R, SQL, C, MS Excel, AppNexus, Google Double Click Bid Manager



ACADEMIC QUALIFICATIONS

Year	Degree/Board	University/Institution	%/CGPA
2019*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	--
2016	B.Tech (Electrical Engineering)	Delhi Technological University (DTU)	78.22%
2012	Class XII, CBSE	Delhi Public School, Haridwar	95.80%
2010	Class X, CBSE	Delhi Public School, Haridwar	10/10

AWARDS & ACHIEVEMENTS

Academic Awards	- Conferred NTSE Scholarship (2008); School topper in Class X (2010); City topper in Class XII (2012) - Scored 750 (98 percentile) in GMAT (2016); 99.6 percentile in AIEEE 2012 among 1.1 Million+ applicants - Department Rank 2 in B.Tech 2nd year(2014); Received scholarship from Delhi Technological University Scholar Gold medalist at DPS Haridwar for excellent academic performance for 6 consecutive years
Data Science Competitions	Silver medalist (Top 5%) in Kaggle's Data Science Bowl on Nuclei Detection in images challenge '18 - Shortlisted by P&G to present prototype of Recommender System in Click Predict Innovate challenge '18 AIR 4, WorldRank 40 in International Data Science Game on Bond Recommendation by BNP Paribas '18 AIR 33 among 9600 participants in hackathon on Predicting Returns of Portfolios by Societe Generale '17 AIR 1 among 261 teams in online round of Real Estate Price Prediction by Analytics Vidhya '17 AIR 6 among 179 teams in online round of Data & Beyond by American Express & IIM Bangalore '17
Certification	Successfully completed 70-461 Querying Microsoft SQL Server 2012/14 conducted by Microsoft '16

WORK EXPERIENCE

Thorogood Associates	Business Intelligence Consultant	10 months Bengaluru (Aug '16 - May '17)
Major Contributions	- Upgraded data collection & reporting system of a Fortune 500 FMCG company used across 60 countries - Migrated & modified Data Warehouse, ETL packages and Tableau Dashboards from 2008 to 2012 server - Automated & integrated the Brand Position dimension of products into the reporting system of client - Eased the data collection process through automated E-mail generation using pre-loaded templates	
Stack used	SQL Server, Microsoft SSIS for ETL, ASP.NET and Microsoft Excel for reporting	

DATA SCIENCE PROJECTS

Recommender System	- Developed latent factor based Collaborative Filtering models to predict movie ratings for 6000+ viewers - Applied SVD model and optimized with Stochastic Gradient Descent to get Normalised RMSE of 0.22 - Implemented Neural Network based Variational Autoencoders to model non-linear latent factors
Object Detection & Classification	- Validated P&G product visibility norms in retail stores by detecting products in shelf images - Implemented Faster RCNN model in Keras by leveraging transfer learning from ImageNet dataset - Determined bounding box coordinates and classified the objects with mean average precision of 54.11%
Walmart Sales Forecasting	- Forecasted weekly sales for multiple departments across 45 Walmart stores over a horizon of 39 weeks Imputed data using Kalman Smoothing and fitted ARIMA models on store-department combinations - Further reduced MSE by 19% by preprocessing data with SVD & fitting exponential smoothing models
Anomaly Detection: WannaCry	- Identified presence of Ransomware in computer using timeseries data of hardware performance counters - Applied PCA (Recall-12%) and LSTM based variational autoencoders (Recall-84%) to detect anomalies - Converted data to frequency domain and captured 98% anomalies with the LSTM based model
Predicting Startup Acquisition	- Identified key influential features affecting Start-up acquisition using company profiles from CrunchBase - Trained XGBoost model on financial, factual and managerial features to achieve a recall score of 80.5% - Improved recall to 82% by integrating topical features obtained by applying LDA on news articles
Opinion Spam Detection (Unsupervised)	- Identified potential spammers from Yelp Reviews dataset using Frequent Itemset Mining - Performed feature engineering to identify textual, behavioural & temporal indicators of group spam - Implemented Expectation Maximisation algorithm to rank potential spammers on their spam index

EXTRA-CURRICULAR ACTIVITIES

Positions of Responsibility	Public Relations Team, PGDBA- increased social media reach by 40% and wrote articles for press releases Executive Committee Member, IEEE DTU- Organized Annual Technical Fest in years 2013 & 2014
Achievements	- Completed Senior Diploma in Kathak from Prayag Sangeet Samiti, Allahabad - Member, Street Play Society of DTU- won several competitions at Inter-University level - Represented DPS Haridwar at Inter-DPS Debate Competition held in Gwalior (2011)

TOOLS & LANGUAGES: R, Python, C, SQL, Microsoft SSIS, Microsoft Excel, Tableau



ACADEMIC QUALIFICATIONS

Year	Degree/Board	Institution/University	%/CGPA
2019*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	–
2013	B. Tech (Electronics and Communication Engineering)	NIT, Allahabad	8.96/10 (Rank 5)
2009	Class XII (CBSE)	Delhi Public School, Ghaziabad	91.80 %
2007	Class X (CBSE)	Delhi Public School, Ghaziabad	88.80 %

AWARDS AND POSITIONS OF RESPONSIBILITY

Institute Rank 9 – PGDBA 2017-19*

Scholastic Achievements	- Ranked 5 among 130 students in ECE department at NIT Allahabad, receiving Merit Scholarship for all 4 years - Awarded Scholar Badge in 8 consecutive years for outstanding academic performance in school from 1999-2007 - Secured 99.61 percentile in LR-DI and overall 99.14 percentile in CAT 2016; 51/51 in GMAT Quantitative Aptitude
Analytics Competitions	- Awarded 2nd Runner up and a cash prize of ₹10k in Case Study Challenge by IRSC and GoI at IIT BHU, 2018 Top 3% in Felicity by DE Shaw (among 1500+ teams) and Data Supremacy (1000+teams) on Analytics Vidhya, 2018 Top 13% in Image Classification of Furniture and Home Goods (iMaterialist Challenge at FGVC5) on Kaggle, 2018
President, Student Association PGDBA	- Maintained synergy between student body and administration of 3 institutes through active stakeholder management - Pioneered the formation of the magazine, alumni and PR teams to enhance brand value and conducted 10+ events - Entrusted with all negotiations on behalf of PGDBA student body; managing wealth of ₹1.4 million over the year - Formulated Constitution of student body, defining governance framework of student activities and responsibilities
Event Head, Technical Fest	- Led a team of 40 students for organizing all electronics and electrical events of the technical fest with a footfall of 10k Conceived and executed a novel “War of Currents” event in a budget of ₹50k, attracting participation from 100 teams
Other PORs	Held the post of Hostel President and part of the core team for the cultural fest, knowledge fest and farewell event

WORK EXPERIENCE

46 Months

Research Engineer	Centre for Development of Telematics (CDOT)	Delhi (Aug '13 -May'17)
Awarded with “Best Employee” award for the year 2016		
Cloud Computing	- Improved server efficiency to 70% by constructing a firm-wide distributed architecture system with add-on dashboard - Developed a load balancing cloud application for heavy network load scenarios, automatically migrating the instances - Monitored KPIs of cloud instances using NoSQL Database to create triggers for early detection of network overload - Achieved a 17.3 % reduction in response time and latency by developing an algorithm to auto-scale the cloud resources	
Video Conferencing Application	- Developed an IMS compliant voice (VoIP) and videoconferencing app for Linux, Windows and Android platforms - Designed and implemented AES encryption-based solution in signaling for secure communication in defense systems Spearheaded the product roll-out to ~8 project teams and managed all downstream activities in the product pipeline	
Leadership	Coordinated 15+ National and International level Government programs that were held at DOT and CDOT	

DATA SCIENCE PROJECTS

Hotel Recommendation	- Built a personalized hotel recommendation system utilizing 1000+ users' ratings, profile similarity & their credibility Scraped ratings of 13000+ hotels from Trip Advisor using Selenium package and combined it with adjusted user bias - Segmented users based on Euclidean distance and applied collaborative filtering & SVD to recommend top 5 hotels
Store Sales Prediction	- Predicted 6 weeks of daily sales for 1k+ drug stores using 2+ years data, enabling creation of effective staff schedules - Performed missing data imputation, feature engineering and built time series model to achieve RMSPE of 0.28 - Improved the sale prediction with an ensemble of Glmnet, Random Forest and XGBoost to reduce RMSPE by 65%
Product Image Classification	- Developed a tool to automatically recognize images of furniture and home goods from a 200k dataset with 128 classes - Classified products using CNN & transfer learning with ResNet backbone post augmentation and feature extraction - Achieved an accuracy of 86% by leveraging hyperparameter tuning, annealing and differential learning rate in 3 cycles
Stock Price Forecasting	- Forecasted the stock price of a bank for next 6 months using 9 years of data, accounting for the global financial crisis - Determined ARIMA order using ACF and PACF plots and validated stationarity with augmented Dickey–Fuller test - Performed a comparative analysis of different time series models ARIMA, ETS, Holt-Winters & TBATS span in R
Learning Entity Representation Using NLP	- Developed a deep learning model to identify the context of a given entity in case of a disambiguation using Python - Created knowledge base by scraping POS tags from Wikipedia content for ~40k entities against their actual usage - Constructed Word2Vec similar to GloVe using stacked denoising Auto-Encoders and cosine similarity
Document Summarization	- Summarized 125 documents in 250 words using Text Rank, Degree Centrality, TF-IDF and cosine similarity - Evaluated generated summary against human reference summary using Precision, Recall, F1 score & Rouge N metrics

EXTRACURRICULAR ACHIEVEMENTS

Social Initiative	Member of Community Development Cell , imparting quality education to orphans and underprivileged children
Sports	Part of IIMC Cricket Team ; Winner in CDOT Cricket League and 1 st Runner up in CDOT Football league 2016
Quiz	1 st Runner up in the Quiz competition, organized in CDOT'16 and 2 nd Runner up in Quiz competition in college '13
Cultural	Won 1st Prize in Street Play, the dramatics event of college cultural festival for 2 consecutive years, 2011 and 2012 - Member of Dramatics and Dance club of CDOT and performed for 2 years in annual day celebration, 2013 and 2016

ELECTIVES: Decision & Games, Corporate Finance,DBMS, Pricing & Revenue Optimization, Supply Chain Analytics, Product Analytics

ACADEMIC QUALIFICATIONS			
Year	Degree/Board	University/Institution	%/CGPA
2019*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-
2014	B. Tech (Mechanical Engineering)	NIT Jamshedpur	8.55/10
2009	Class XII, CBSE	Delhi Public School, Ranchi	90.60%
2007	Class X, ICSE	St. Xavier's School, Ranchi	91.28%
AWARDS AND ACHIEVEMENTS			
International/ National Competitions	▪ Bronze Medallist among 4551 teams in 'Toxic Comment Classification Challenge' organized by Kaggle '18 ▪ Ranked 1st out of 150 teams in Data Play, a Forecasting challenge sponsored by KPMG held at IIM Calcutta '18 ▪ Ranked 2nd out of 179 teams in 'Data and Beyond' sponsored by American Express held at IIM Bangalore '17 ▪ 1st Runner up in Sumantran, national level star Consultant hunt event organised by IIT Kharagpur '18 ▪ Ranked 7th nationally and 57th globally in Data Science Game organised by BNP Paribas and Kaggle '18 ▪ World rank 26 out of 750 teams in 'AO: Predicting how points end in tennis' hackathon by Tennis Australia '18 ▪ Represented India & Ranked 2nd nationally at Shell Eco-Marathon, vehicle design event held in Malaysia '12		
Achievements	▪ Awarded Black Belt Certification on Lean Six Sigma methodology by KPMG '18 ▪ Granted INSA-NASI Summer Research Fellowship by Indian Academy of Sciences; 2000+ applicants '13 ▪ Presented Paper at 11th International ISHMT-ASME conference on Heat & Mass Transfer, IIT Kharagpur '13		
WORK EXPERIENCE		27 MONTHS	
Piping Engineer	Samsung Heavy Industries, India		Nov '14 - Feb '17
Roles and Responsibilities	▪ Part of Special Task Force for re-designing world's largest semi-submersible oil facility, project worth \$34 bn ▪ Expertise on 3D plant design software PDMS and GSCAD; Responsible for imparting trainings to employees ▪ Worked as 3D Administrator to generate quality reports; Ensured 99.99 % quality by efficient CAD execution		
Initiatives	▪ Developed an in-house multidisciplinary clash management tool and increased the clash check efficiency by 30% ▪ Proposed a 3-Level WBS based on weightage to disciplines in line with quantum of work & planned man-hours ▪ Introduced user friendly Status Control, which reduced user effort by ~50% compared to the existing interface ▪ Devised a tool using Programmable Machine Language for efficient Material Take off extraction from PDMS		
DATA SCIENCE COMPETITIONS			
Toxic Comment Classification Challenge	▪ Identified negative sentiments in comments from Wikipedia Talk page edits to facilitate healthy online discussions ▪ Leveraged NLP techniques to represent comments using N-grams, TF-IDF and GLOVE word embeddings ▪ Developed Bi-Directional LSTM architecture to achieve an AUC score of 0.98 on private leader board in kaggle		
Product Returns in E-Commerce	▪ Predicted return rates of orders for 2 months using Poisson regression model, for an Indian fashion retailer ▪ Performed customer segmentation through K-means clustering and feature selection using Boruta package in R		
DATA SCIENCE PROJECTS			
Music Recommendation System	▪ Built a recommender system for KKBox dataset of 3 lac+ users and 3 Cr+ songs to enhance user engagement ▪ Implemented Collaborative Filtering using Jaccard similarity to predict the probability of user re-listening a song ▪ Captured user-song interactions through SVD and user diversity through entropy and probability based features ▪ Applied ensemble of XGBoost and LightGBM using rank averaging; achieved AUC score of 0.69 on test data		
Social Media Analytics	▪ Identified reasons for follower loss using web based crawls of tweets of 9.3 mn users at two different time points ▪ Created user-user network based on topical similarity of tweets; applied Louvain algorithm to detect communities ▪ Built a SVM model using graph & content based features to predict early follower loss, obtained F1 score of 0.8		
Sales Forecasting	▪ Forecasted 6 weeks of daily sales for 1,115 Rossman drug stores to enhance effectiveness of staff scheduling ▪ Analysed seasonality by STL decomposition, stationarity by ADF test and ARMA orders by ACF-PACF plots ▪ Implemented ARIMA, TSLM and Prophet models to achieve best score of RMSPE as 0.13 from TSLM		
Credit Risk Modelling	▪ Predicted loan repayment ability of applicants using previous application status, POS and credit bureau data ▪ Handled class imbalance using Oversampling technique and imputed missing values using KNN method ▪ Applied Logistic Regression and Random Forest to achieve an AUC score of 0.72 using engineered features		
POSITIONS OF RESPONSIBILITIES AND EXTRA-CURRICULAR ACTIVITIES			
Vehicle Design Team ('12-'14)	▪ Led 5-member team in designing and fabrication of ultra energy efficient vehicle; fund raising of INR 5 lacs ▪ Secured 9th rank in Electric Solar Vehicle Championship, Punjab'14; built an in-house eco friendly vehicle [link]		
Sports Achievements	▪ Semi Finalist out of 40 teams in Red Bull sponsored Neymar Jr. Futsal tournament held at IIT Kharagpur '18 ▪ 1st Runner up in ARMA-RUN, a 5 km marathon conducted by Armageddon, IIM Calcutta (time~27 mins) '18 ▪ 1st Runner up in Table Tennis tournament representing PGDBA in Intersection held at IIM Calcutta '17 ▪ Represented college twice in football at Inter-NIT Sports Meet & Table Tennis at Spardha, IIT BHU '12-'14		
Extra-Curricular	▪ Member of editorial team at Samsung Heavy, involved in release of annual and monthly edition of magazine '16 ▪ Ranked 1st in CNC Programming event in the national technical fest, OJASS, held at NIT Jamshedpur '11		
TOOLS : R, Python, SQL, MiniTab, Tableau ELECTIVES : Supply Chain, Complex Networks, Pricing & Revenue Optimization			



ACADEMIC QUALIFICATIONS

Year	Degree/Examination	Board/University	%/CGPA
2019	Post Graduate Diploma in Business Analytics	ISI Kolkata, IIM Calcutta, IIT Kharagpur	-
2016	M.Tech (Dual Degree)-Biochemical Engineering	IIT (BHU) Varanasi	8.05/10
2011	Class XII	CBSE	92%
2009	Class X	ICSE	93.57%

AWARDS AND ACHIEVEMENTS

Scholastic Achievements	- Selected by Justus-Liebig University, Germany for internship to work on Animal Tissue Engineering AIR 35 in GATE 2015 out 8000+ candidates; AIR 5713 in IIT-JEE 2011 out of 4,80,000+ candidates
Professional Achievements	Award of Excellence for work on Annual Revenue Forecasting Model, 2017 - MSD Pharmaceuticals - Successfully completed the KPMG Lean Six Sigma Green Belt Certification training program, 2015
Analytics Competitions	AIR 13 - Out of 1018 participants - The Data Supremacy - All India Student DataFest Hackathon 2018 AIR 19/WR 84 - Capsim Challenge 2018; Formulized Cross-functional Strategy of Million-dollar firm AIR 10 - Data & Beyond 2017 by American Express & IIM Bangalore; Credit/Debit card marketing

WORK EXPERIENCE/INTERNSHIPS

11 Months

Aspect Ratio	Analyst	July 2016 - June 2017
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- Played a key role in the **Development Team** of a Revenue forecasting patient flow model for an immuno-oncology drug
- Responsible for the development of an **annual forecasting model**; **Achieved 96% accuracy** under steady state conditions
- Incorporated** utility of having multiple indications in a single model for **MSD Pharmaceuticals (Fortune 100)** company)
- Developed** the **tool** for replicating market assumptions among models of different versions for the ease of stakeholders.

Accubits Technologies

Data Science Intern

December 2017

- Built **classification models** to predict fashion trends that will be successful in 2018; Utilized last 10 year's data of the client
- Performed Data cleaning, Cross-validation and Imputed missing data points using **K-NN**; Ensembled **XGBoost** and **GBM**
- Developed the **Rest APIs** for the classification models using **Django** and successfully tested the model output on the API

DATA SCIENCE PROJECTS

Yelp Restaurant Photo Classification Computer Vision	- Built a multi-class image classification model to suggest tags for users when uploading photos to app - Utilized 2,34,000+ crowd-sourced images of 2000 restaurants and tagged them with multiple labels - Extracted features by Transfer-learning pre-trained architectures- Resnet50, InceptionV3 & VGG16 - Leveraged Google Cloud services to run Deep learning models; Achieved a mean F1-score of 60.38%
Review Relevance Prediction NLP	- Modelled text data, consisting of 5mn reviews; Rated usefulness using Natural Language Processing - Implemented Degree-centrality and Term-Document Matrix techniques; Achieved NRMSE of 33.2% Improved result by 12% using Bi-directional LSTM on Word2Vec word embeddings of the reviews
MovieLens-26M Hybrid Recommender System	- Built Recommendation engine on 26mn reviews of 270k users & 45k movies by Collaborative Filtering - Used Restricted Boltzmann Machines network & Matrix Factorization on Keras for Latent Factors - Integrated weighted scores of both approaches; Solved cold start problem by Content-based filtering
Instacart - Market Basket Analysis	- Performed Market Basket Analysis of 32mn orders; Used Apriori algorithm for Association Rules Generated 20+ features capturing user, item and user-item buying behaviour by Feature engineering - Achieved F-1 score of 0.39 using stacked Boosting classifiers & Expectation maximization algorithms
Spatio-Temporal Prediction of Travel Times	- Developed Travel time forecasting model of Cabs between GPS coordinates of pick-up & drop points - Used data of Cab travel times from a Cab aggregator app for trips within London between 2014 - 2017 - Utilized an ensemble of Random Forest and LightGBM algorithms to build the ETA prediction model
Retail Sales Forecasting	- Developed Time-series model to forecast Online Sales; Segmented market on the basis of profitability - Applied ARIMA and Holt-Winters exponential smoothing algorithms; Obtained MAPE value 12.44%
Chatbot Design	- Built a chit-chat chatbot on DialogFlow; Deployed bot on Slack using Heroku app through Webhook Incorporated functionality of responding to weather queries for places via the OpenWeatherMap API

POSITIONS OF RESPONSIBILITY

Placement Coordinator	Coordinated placement session 2015-16 of IIT (BHU) Varanasi, along with team of 20 representatives Organized placement preparation training sessions for 2014-2015 batch; Witnessed 100% placements
Publicity Head Spardha 2014	- Led a 25-member team of junior executives; Generated profit: INR 5,00,000+ via effective marketing Spearheaded the Publicity campaign; 20% increase in participation; Increase of 7000 followers on FB
Event Coordinator Spardha 2013	Core Festival Team member of Annual National Sports Fest of IIT (BHU) with footfall of over 30000 Managed event ceremonies with budget of INR 30+ lakhs with 1600 participants; Over 90+ colleges

TECHNICAL SKILLS AND COURSEWORK

- Programming Languages:** Python, R, SQL, VBA, C++; **Analytical Tools:** Tableau, Microsoft Excel, Spotfire, SPSS, Minitab 17
- Electives:** Information Retrieval, Supply Chain Analytics, Product Analytics, Corporate Finance, Computational Finance

ACADEMIC QUALIFICATIONS			
Year	Degree/Board	University/Institution	%/CGPA
2019*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-
2014	B.Tech + M.Tech (IDD), Metallurgical Engineering	IIT (BHU), Varanasi	7.37/10
2008	Class XII, CBSE	Central Academy, Kota, Rajasthan	82.6%
2006	Class X, CBSE	Kendriya Vidyalaya No. 2, Balangir, Odisha	92.0%
ACADEMIC AWARDS AND COMPETITIONS			
Institute Rank 4 - PGDBA 2017-19*			
Data Science Competitions	- Ranked 1st in Stat Wars by ISI Kolkata, Data Science Hackathon at Integration, 2018 out of 353 teams - Won Bronze medal (Top 6%) in Kaggle’s TalkingData Fraud Detection challenge among 3967 teams - Ranked 2nd in ‘Sumantran’ 2018, a consultant hunt case competition organized by IIT Kharagpur - Ranked 3rd among 1018 participants in DataHack Supremacy, 2018 organized by Analytics Vidhya - Ranked 54th (Top 1%) among 7000+ participants in HackerEarth’s Deep Learning Challenge #2, 2018 - Finished in the Top 5 out of 179 teams in ‘Data & Beyond’, 2017 organized by IIM Bangalore - Finalist (Top 7% teams) at Smart India Hackathon, 2018 organized by Government of India		
Academic Awards	- Recipient of merit-based scholarship for excellent academic performance at ISI Kolkata - Awarded Amul Shakti Vidya Shree for academic excellence in CBSE Class 10th Board Examinations - Secured AIR-5483 in IIT-JEE, 2009 and AIR-893 in NSTSE, 2003 by Unified Council		
WORK EXPERIENCE (36 months)			
Data Scientist	iQuanti Pvt. Ltd.	Bangalore (June '14 – June '17)	
Services - Digital Marketing			
Local Search Competitor Analysis	- Used Logistic Regression to predict likelihood of a business listing appearing in Local Search results - Created two key features in title-query similarity and distance from business clusters within a city - Helped in renewal of the project, bringing in additional USD 120,000 revenue for the organisation		
Website Optimization and Reporting	- Identified bottlenecks in User Flow funnels of the client’s website, applying Markov Chains in SQL - Increased the conversion rate by 50% and annual revenue by USD 2 Million through A/B testing - Designed & automated daily, weekly & monthly reports in Excel, saving 140 man-hours per month		
Products - Search Engine Optimisation (SEO) Tool			
Search Engine Rank Prediction Algorithm	- Utilized content, backlinks, technical aspects and social presence of a webpage to create features - Extracted topics through LDA from ~4MM Wikipedia articles to capture webpage-query similarity - Improved the evaluation accuracy by 60%, stacking Logistic Regression, GBM, RF & NN in Python - Implemented the module into the product, acting as the Product Owner for the Engineering Team		
Click-Through Rate (CTR) Prediction	- Engineered features utilising brand names and interaction with rich snippets in the search results - Modelled CTR curve using Linear Regression (R-square 85%) for forecasting traffic to the website		
Professional Achievements	- Received highest rating across the organisation of 5/5, during performance appraisal in 2015-16 - Awarded annual ‘Innovation’ award and ‘Employee of the Quarter’ award during 2015-17 - Conducted training sessions on R & Python as a member of the Leadership Development Program		
DATA SCIENCE PROJECTS			
Detecting Fraudulent Behaviour in Display Ad Clicks	- Leveraged Random Undersampling to reduce dataset size and tackle a class imbalance ratio of 400 - Engineered features based on popularity and time of clicks to improve AUC score from 0.95 to 0.98 - Ensembled LightGBM and Factorization Machines by Soft Voting to get a final AUC score of 0.982		
Scoring Classified Ads on Ability to Generate Demand	- Represented title and description of the product in the Ad as vectors through TF-IDF and Word2Vec - Characterized product images using blurriness, whiteness and Transfer Learning utilizing VGG-16 - Created features based on user’s ad-posting behavior and tuned LightGBM to achieve RMSE of 0.22		
Forecasting Demand for a Retail Store	- Forecasted hourly demand at all departments of a retail store through XGBoost and ARIMA models - Generated employee schedule with Linear Optimization formulating employee and shift constraints		
Visualising IPL Data	- Scraped websites using Selenium in Python and aggregated ball-by-ball data for 10 IPL seasons - Designed two interactive web applications to visualize the data with Shiny & R; Links: App 1 App 2		
Removing Photo Studio Background	- Applied Augmentation techniques to pre-process images of 400 vehicles clicked at 16 distinct angles - Trained a CNN-based architecture, U-Net with Adam optimizer in Keras to obtain F1 score of 99.5%		
Predicting Drunk Texting using Tweets	- Applied SVM and LSTM models to identify drunk tweets with an AUC of 0.92 and 0.95 respectively - Identified interest-based communities and their traits using Louvain community detection algorithm		
POSITIONS OF RESPONSIBILITY			
PGDBA 2017-19	Represented the 2017-19 PGDBA Batch at the PGDBA Industry Council Meet, 2018 in IIM Calcutta		
Event Coordinator, Technex, IIT-BHU	- Led a 20-member team for Clutch 2014, a national level gaming event involving 200+ participants - Established budget with INR 1 lakh prize money and supervised publicity on social media platforms		
ELECTIVES: Decision and Game Theory, Complex Networks, Supply Chain Analytics, Corporate Finance, Venture Analytics			



Roles Offered :

- a) LEADIT (Leadership Excellence in Analytics, Data Sciences and IT). Focused on managerial profile in data science.

About this role:

You'll be successful in the LEADIT Rotational Development role if you are an analytical thinker with a desire for personal development and growth into a leadership role within a few years. You have a passion for learning and applying advanced data algorithms, mathematical concepts and leveraging technology to scale up solutions. You apply facts and logic to solve tough analytical problems, work well in teams and communicate clearly and effectively. Upon successful completion of this program, associates have the opportunity to join Gartner after demonstrating a solid understanding of the organization and the roles and industries we serve, as well as problem solving and leadership skills.

Internship Stipend Offer : Rs 125000

Shortlist Steps :

- a) Online test (87 minutes) (Included Test on LR ,Comprehension ,Python Coding and SQL)
 - b) Case Study (60 minutes)
 - c) Technical interview then HR Interview
- Mode of interview: In Person and Video

Duration : Maximum Duration was 30 minutes.

What you'll do:

- Apply Natural Language Processing to improve our ability to synthesize unstructured information and predictive modelling to develop intelligence to recommend the “Next Best Action” for the client
- Surface actionable insights from large volumes of data by leveraging statistical concepts; automating wherever possible to allow for scale
- Use Machine Learning to develop self-correcting algorithms to improve our recommendations for Services teams to increase client retention
- Work with the product owner to define product roadmap, build systems and tools, and actively review development priorities in collaboration with engineering, and business teams to deliver measurable impact.
- Build pipelines and scalable data models to allow for consolidation of large data sets and perform analysis
- Design, implement unit, integration testing and build web-based applications and services
- Communicate to business & service leaders recommended actions based on findings

Technical Round

Technical Round Questions (Shankar Kumar)

Projects & Competitions :

a) AV Competitions, Projects on Kaggle datasets

Questions:

- 1) Semi Supervised learning with applications
- 2) Questions on project
- 3) Handling class imbalance
- 4) Regularisation.

Technical Round Questions (Savita Gond)

Projects & Competitions :

- a) Image classification of home goods and furniture (Kaggle rank 53rd)
- b) Recommendation System for trip advisor.
- c) Sentiment analysis of movie reviews.
- d) Sales forecasting.
- e) Chest Xray classification(65th rank , hackerearth)

Questions:

- 1) They asked about two projects and some technical details of it.
- 2) One Puzzle

Technical Round Questions (Himanshu Gupta)

Questions:

- 1) Technical - CV based question mostly on projects.
- 2) Questions on ROC curve
- 3) Confusion matrix,
- 4) Sentiment analysis,
- 5) Boosting and bagging

Technical Round Questions (Puneet Vashisht)

Questions:

- 1) Questions on how to check and deal with multi-collinearity,
- 2) What is semi-supervised learning ? And example of the same that was used in practice?
- 3) What is the problem of data imbalance? And what are the way around for that ?
- 4) What are the similarity measures used for categorical data?

HR Round

HR Round Questions (Shankar Kumar)

- 1) Strengths & Weaknesses
- 2) Any one project with significance business impact ?

HR Round Questions (Savita Gond)

- 1) Challenging situation you have handled and how you turn around the situation?
- 2) Explain any project with business application of it?

HR Round

HR Round Questions (Himanshu Gupta)

- 1) Weakness, strengths, etc
- 2) Are you a team player or individual player ?
- 3) Difference b/w leader and follower?
- 4) Learnings from each project ?

HR Round Questions (Puneeth Vashishth)

- 1) Was asked to give a 30 seconds elevator pitch to the panel ?
- 2) What would you choose between Individual growth and being a team leader ?
- 3) Explain the business impact of any one of the projects ? And what was your learnings from that ?
- 4) What are your short term and long term goals ?

Other Technical & HR Round Questions

Questions:

- 1) HR round - They asked about the motivation to join the firm?
- 2) Explain one project (Ravi)
- 3) The second was rapid fire technical round which covered most topics from ISLR.
- 4) Guestimate-" no. of people traveling from your home town to Calcutta in a day" SQL queries- "mostly join question"
- 5) CV- "Time series", "Sentiment Analysis"
- 6) Linear Regression basics
- 7) General HR questions- "One thing you regret in your life", "biggest weakness"

Courses & Skills Suggested

- 1) Mainly CDS ML models
- 2) Hypothesis testing from inference
- 3) Ability to explain clearly a concept.
- 4) Python programming and SQL.
- 5) Time series
- 6) Regression analysis
- 7) SQL
- 8) Information retrieval
- 9) CDA, Regression time series, IR, DBMS
- 10) CDS, FADML, Inference

Suggestions:

- 1) Be thorough with the basics of ISLR. Companies ask about projects and their details .
- 2) Revise your basics for data science and give importance to HR questions as well
- 3) Practical knowledge of all the methods that you use and be well versed with projects.
Also, practice coding and be well equipped with the syntax. Many companies have started to have a coding round.

Resumes of Selected Candidate Attached.

ACADEMIC QUALIFICATIONS

Year	Degree/Board	University/Institution	%/CGPA
2017-19*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-
2010-14	B.E. (Mechanical)	Army Institute of Technology(University of Pune)	71.03%
2010	Class XII, CBSE	Army School Devlali	93.2%
2008	Class X, CBSE	Army School Devlali	93.4%

WORK EXPERIENCE (35 months)

Software Engineer (Developer)	Dassault Systèmes	July '14 – June'17
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Worked on Computer Aided Design softwares **Catia**, **Solidworks** and Product Data Management software **Enovia**

Achievements	- Conferred Best Innovation Award among 1,300+ employees for developing a new web application - Received Kudos Achievement Award for timely completion and superior delivery of critical projects
Major Contributions	- Delivered 10+ new functionalities related to object and workflow process management in the product - Developed a new Web Application for storing and managing bugs related to Language Translation - Rectified 200+ major software bugs in Enovia SmarTeam and ensured smooth delivery to the clients - Reduced product dependency on third party softwares; Efficiently handled critical migration projects - Managed client queries & delivered knowledge sharing sessions on product to the team of 40+ peers

DATA SCIENCE PROJECTS

Credit Risk Modelling	- Identified high default risk customers from data of ~1.5 lac loan applicants using Logistic Regression Oversampled defaulters and undersampled non defaulters using SMOTE and achieved AUC of 0.85 - Leveraged hierarchical clustering to identify features differentiating defaulters from non-defaulters
Image Classification	- Classified restaurants from their images into multiple classes in a multi-label, multi-instance dataset - Extracted features using pre-trained CNN architectures - Inception V3, VGG16 and ResNet 50 - Trained CNN on Kaggle problem of 2.34 lacs+ Yelp images with 2k restaurants having multiple labels
Movie Recommender	- Developed a Hybrid Movie Recommender System with an added ability to mitigate cold start issue - Trained the model on Movie Lens dataset of 26 million ratings given by 2.7 lac users to 45k movies - Built model on content based and collaborative filtering using matrix factorization SVD and RBM
YouTube Comments Classification	- Developed a classifier to bucket web-scraped Youtube comments in 12 different sentiment categories - Built comments graph, detected communities with Louvain algorithm & computed centrality metrics - Performed a temporal analysis of sentiment persistence in a sequence of comments for different users - Built an ensemble model of Random Forest, Logistic Regression & Long Short Term Memory network
Time Series Modelling	- Predicted sales and demand for a Retail Giant for the next 6 months using Holt-Winters Forecasting - Identified profitable market segments based on maximum expected profit & minimum profit variance - Utilized ACF, PACF plots to identify order of ARMA model and checked stationarity using ADF test - Applied ARIMA, Linear Modelling to compare performance, analysed sales & demand behaviour
Ransomware Detection	- Built a Wannacry detector on dataset of hardware performance counters from infected & normal PCs - Leveraged the Sliding Window technique and built the model using sequential LSTM auto-encoder - Used 3 sigma limits of reconstruction error to identify anomalous wannacry ransomware behaviour

AWARDS & ACTIVITIES

Competitions	- AIR 19/WR 84 in Capsim Challenge, devised cross functional strategy for multi-million dollar firm - Ranked 36 out of 1018 in 'The Data Supremacy' challenge to predict trainee job search probability - Secured rank 42 out of 608 participants in Student Data Fest 2018 Skill test – Python for Data Science - Achieved top 5 percentile in McKinsey Online Hackathon to predict probability of customer defaulting on insurance premium payment and formulate an incentive plan for the policy agents
Academic Awards	- Awarded National level Merit scholarship from AWES in Higher Secondary Education CBSE exams - Obtained First Rank in school out of 240 students in Higher Secondary Education CBSE board exams
Certifications	Completed two Certification courses from DataCamp Data Science Toolbox, Databases in Python
Online Courses	- Completed Coursera: Machine Learning by Andrew Ng and Deep Learning by Andrew Ng - Completed EdX: The Analytics Edge in R by MIT, Speech Recognition Systems by Microsoft

TOOLS & LANGUAGES : R, Python, HTML, JavaScript, ASP.NET, C#, MySQL, MS Excel, Map Reduce, Tableau

ELECTIVES : Computational Finance, Corporate Finance, Decision and Games, Supply Chain Analytics

ACADEMIC QUALIFICATIONS

Year	Degree/Board	University/Institution	%/CGPA
2018*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-
2014	B.Tech (Electrical Engineering)	IIT- BHU(Varanasi)	7.8/10
2009	Class XII, CBSE	Delhi Public School , Vindhyannagar	78.8%
2007	Class X, CBSE	Kendriya Vidyalaya Shaktinagar	84.8%

WORK EXPERIENCE (33 MONTHS)

Assistant Manager	Coal India Limited (CIL)	(Aug '14 – June '17)
Energy Resource Forecasting	Predicted the consumption of energy resources (electric power & diesel) using Regression modelling methods (Linear, Polynomial, Exponential) with eight years of historic data of consumption Analysed R-square values for 12 different models for energy consumption in model selection process	
Impacts & Contribution	Implemented Unique Identification Marking of inventories (12000 items) for ERP tender Economic Profits: Estimated cost, prepared document and e-tendered the Rapid Loading System valuing ~23 crore and obtained per annum savings of ~ 20,000 crore - Successfully designed and estimated the cost of the Solar Power Plant ,Ministry of Coal Project Certified for Managerial Awareness Programme at Indian Institute of Coal Management , CIL	

AWARDS & ACTIVITIES

Analytics Competitions	Ranked 53rd in Automatic Product Recognition Challenge (iMaterialist) hosted on Kaggle Ranked 65th in the Chest X-Ray Image Classification Competition hosted on Hackerearth Ranked 3rd and headed the team of 4 in the Case study Challenge of IRSC, IIT Delhi Ranked 439 out of ~16000 participants in Criminal Prediction Challenge hosted on Hackerearth	'18 '18 '18 '18
Awards and Achievements	Won 1st Prize in Leadership Development Cell's Competition "Conflictous'5" held at IIM Calcutta Awarded for Non-technical Innovation at Corporate level in Coal India Limited for CSR initiations	'17 '15

DATA SCIENCE PROJECTS

Image classification of furniture and home goods	- Automated the recognition of 128 classes of furniture and home goods using CNN architecture - Fine-tuned Resnet -50 with Hyper Parameter Tuning and Discriminative Learning Rate Annealing Attained accuracy of 86 % by using Data Augmentation and Regularization with Dropout
Personalized Hotel Recommendation System	- Built a personalized hotel Recommendation System for Trip Advisor using collaborative filtering - Scraped data from Trip Advisor and coalesced with adjusted user bias to predict user's rating matrix - Implemented SVD and used euclidean distance measure to recommend top 5 hotels to user
Stock Value Prediction	- Analysed Canara Bank's historical Stock Prices of 10 years by different forecasting methods - Predicted 6 months of Stock Price Returns using ARIMA with AIC 14721 and RMSE 9.60 - Compared above model with ETS (Exponential Trend Smoothing) , Holt-Winter and TBATS model
Sentimental Analysis of movie reviews	Modelled sentiments of movie reviews in 4 classes by employing Logistic regression (accuracy 56%) - Implemented Deep Learning techniques LSTM & GRU to train sentiments using Word Embedding - Attained accuracy of 68% for bidirectional GRU and LSTM utilized crawl-300d-2M embedding matrix
Node Embedding of heterogeneous graph	Built latent feature matrix for 8 features of karate club network of friendship by Deepwalk Algorithm - Synthetically designed citation-collaboration network of 1000 nodes by paper ids and author names - Implemented LINE algorithm and analyzed the feature with cosine and euclidean distance measures
Rossmann Sales Prediction	- Forecasted 6 weeks of daily sales for 1,115 stores of Rossmann located across Germany - Performed feature engineering and missing data imputation and achieved RMSPE 0.14 by XGBoost - Improved sales prediction using Ensemble of XGBoost and Glmnet trained models (RMSPE 0.11)

EXTRA CURRICULARS

POR	- Represented 2017-19 batch of PGDBA at the intersection sports meet held at IIM Calcutta - Appointed as Tutor to teach mathematics to unprivileged students of 11th and 12th standard in CIL - Selected for dedicated team of 7 to give Career Counselling to school going students in CIL
Sports & Technical Festival	- Won Silver medal in sports festival Spardha-2012 of IIT-BHU in Basketball Tournament - Participated in Basketball event of UDGHOSH'11 the annual sports festival of IIT KANPUR - Participated in an All India Robotics Competition TECHNEX-11 & 12, Technical Festival of IIT-BHU

TECHNICAL SKILLS AND RELEVANT COURSEWORK

Tools/Languages : R, Python, SQL, Visualization in Tableau, MS Excel, Hadoop, Map Reduce , MongoDB

Electives : Complex Network , Product Analytics, Supply Chain Analytics , Pricing and Revenue Optimization

Certified Courses : Search Engine and Text Retrieval by **Coursera**

ACADEMIC QUALIFICATIONS

Course/Examination	Institution/University	Year	Performance
Post Graduate Diploma in Business Analytics	IIM Calcutta/ISI Kolkata/IIT Kharagpur	2019*	-
Financial Risk Manager (FRM) Level 1	GARP ID - 592217	2017	Qualified (1 st attempt)
B.Tech (Hons.) – Electrical Engineering	IIT Kharagpur	2014	9.04/10 (Rank 3)
Senior Secondary School Certificate, CBSE	Mount Carmel School, New Delhi	2010	90.8%
High School Certificate, CBSE	Kendriya Vidyalaya No. 2, Pathankot	2008	91.6%

AWARDS AND ACHIEVEMENTS

<i>Institute Rank 2 – PGDBA 2017-19*</i>			
Academic Awards and Publications	- Received a merit scholarship from ISI Kolkata for excellent academic performance in PGDBA first semester ‘17 - Ranked 3rd out of 90 students in Electrical engineering, IIT Kharagpur; Recipient of MCM scholarship for 4 years ‘14 - Published a paper on blood pressure measurement at IEEE 28th International Conference on VLSI Design [Link] ‘15		
Data Science Competitions	Won Silver medal (Top 5% out of 3967) in Kaggle “Talking Data Ad-Tracking Fraud Detection” challenge ‘18 - Featured in top 20 teams out of 9,600 to qualify for the final round of Brain Waves, Société Générale hackathon ‘18 1st Runner up at national level event, Sumantran - The Hunt for Start Consultant, organized by IIT Kharagpur ‘18 - Awarded with a cash prize for securing a rank in top 1% in Stat Wars hackathon, organized by ISI Kolkata ‘18 - Ranked in top 7% of 17K teams to qualify for the grand finale of Smart India Hackathon, organized by GOI ‘18		

WORK EXPERIENCE

			36 Months
DATA SCIENTIST	Easyrewardz Software Services Pvt. Ltd.	Gurugram (Jun’14 – Jun’17)	
<i>Awarded with “Best Manager” and “Best Employee” awards for the year 2016 and 2015 respectively</i>			
Retail Analytics	- Predicted customers’ propensity to buy in the near future using logistic regression and achieved an AUC of 77% - Conceptualized win-back campaigns retaining 1.5% of the potential lapsed with an ROI of more than 5X - Clustered stores using K-means for same store growth analysis to identify reasons for differing performances - Predicted sales and bills of multiple brands by time series analysis and drove campaigns to increase sales by 3-5%		
Campaign Management Tool	- Developed a campaign tool using PHP and MySQL to design, execute and track segment-specific campaigns - Enabled managers to automate campaigns a month in advance and target customers from 1000 different segments - Unified online, offline campaigns via Omni-Channel communication including SMS, Email and push notifications		
Business Intelligence Tool	Led a team of 5 to develop an automated reporting panel for 50+ brands with an average load of 500 hits per sec - Created a Unified Customer Database in MySQL by aggregating data from 10 disparate sources via an ETL script - Incorporated the panel with campaign, segmentation, tier and cohort analysis saving ~30 man-hours per day		
Notification Engine	- Designed database structure in Cassandra and developed Spark jobs to push notification within 10ms of page load - Increased the number of customers signing up by 40% and the average time spent on the website by 2.5 times		

DATA SCIENCE INTERNSHIP

Cummins Global Analytics Center		Dec’17
Detecting Manufacturing Defects	- Assessed problems of class imbalance and identified methodologies to rectify it on a dataset with an imbalance of 1% - Applied Autoencoder as an anomaly detection model post employing SMOTE and ENN sampling techniques - Reduced the features space using PCA and applied K-means clustering to improve AUC from 0.84 to 0.86	

DATA SCIENCE PROJECTS

Bond Trade Prediction (BNP Paribas)	- Predicted probability of customer-bond interaction in coming week using trading and market data of past two years - Generated negative samples and ensembled Random Forest and LightGBM models to achieve an AUC of 0.62 - Extracted additional features using Collaborative Filtering and LightFM models to improve the AUC to 0.68
Multi-Label Classification (YouTube 8M)	- Assigned video-level labels to 31 GB data released by Google, relevant in automatic caption and subtitles generation - Leveraged TensorFlow and Google Cloud to implement logistic regression and achieved a GAP score of 0.72 - Boosted the GAP score to 0.76 with Mixture of Experts model and a dense layer using PRELU activation function
Opinion-Fact Classification (Microsoft)	- Classified 600+ news articles into facts and opinion subcategories useful in revenue modelling for online publishers - Extracted features from topic modelling using Latent Dirichlet Allocation and community detection algorithms - Ensembled GRU, Bi-directional LSTM and SVM models with n-gram features to improve the accuracy to 0.74
Electric Power Load Forecasting	- Forecasted electricity load demand of Indian states using multi-seasonal ARIMA model leveraging TBATS package - Reduced MAPE from 12% to 8% using wind and solar data and implemented an LSTM model with MAPE of 3% - Visualized results by creating an interactive web dashboard using R-Shiny for real-time analysis of load demand
Investors Clustering	Recommended top 5 potential investors to start-ups by scoring investors on region, time and domain attributes - Reduced dimensions of the data using PCA and applied KProto clustering to attain Between/Within SS of 97%

EXTRACURRICULAR ACHIEVEMENTS

Sports, Hockey Captain (IIT)	- Captained Bronze winning team of 20 for Hockey competition in Sports General Championship, 2013 and 2014 - Recipient of Hall Color and Order of Merit awards in hockey for being the most active player of 2014 batch
Technology and Cultural (IIT KGP)	- Won a Silver and Bronze medal in Product design competition, Technology General Championship, 2013 and 2014 - Led a team of 10 and won Silver in Case Study competition and Open Soft Coding competition, 2012 and 2013 - Won 2 Gold, 2 Silver medals in Illumination and Rangoli under Cultural General Championship, 2011 and 2012



Roles Offered :

a) Advanced Consulting / Intern Advisor /Analytics Consulting

Job Description:

- a) Rigorously and logically identify issues covering entire client problem to understand business objectives.
- b) Generate key hypotheses and independently structures work at the stream or small project level.
- c) Identify creative and useful additional analyses required and suggest new paradigms for recommendations.
- d) Create effective, impactful and quality assured storylines and slides at stream or small project level.
- e) Analyze large volumes of transaction and customer data to generate insights and actionable recommendations to drive business growth.
- f) Synthesize analyses into clear, sound recommendations; takes responsibility for structuring and writing reports and client ready presentations from a work stream or project level.
- g) Apply knowledge of metrics, measurements, and benchmarking to complex and demanding solutions across multiple industry verticals.
- h) Collect and synthesize feedback from clients, project delivery analysts, and sales teams for new solutions or product enhancements. Provide design specifications, functional and technical requirements for development of new products and solutions.
- i) Generates key hypotheses and independently structures work at the work stream or project level
- j) Build behavioral scorecards for target marketing and risk management solutions using econometric and statistical modeling techniques

- h) Create highly predictive models using segmentation and regression techniques to drive profits
- i) Reviews analytics end-products to ensure accuracy, quality and timeliness.
- j) Develops effective working relationships with global teams and business partners
- k) Proactively seeks new knowledge and structures project work to facilitate the capture of Intellectual Capital with minimal oversight

Internship Stipend Offer : Rs 125000

Shortlist Steps :

- a) Aptitude Test
- b) 3 Rounds of Interview (2 Technical Round , 1 HR Round)

Duration : Maximum Duration was 50 minutes.

Technical Round

Technical Round Questions (Piyush Choudhary)

Projects & Competitions :

Kaggle - Carvana Image Masking Challenge, Immaterialist Product Classification.

Questions :

- a) Projects (Recommender Systems , Natural Language Processing.
- b) Guestimate : How many Innovas can be parked in IIM Campus?

Technical Round Questions (Pratesh Jhari)

Questions :

Technical round 1: 20 mins

- a) Tell me about yourself ?
- b) What does Mastercard do?
- c) What is your best project? Briefly explain it?
- d) Ask me something about my Airbus Idea competition i won in Btech ?
- e) Gave a case to solve
- f) Prepare a strategy for ICICI bank in launching a new credit card in the market. What data you want?

Technical Round 2 : 25 mins

It was more like a HR round

- a) Why PGDBA?
- b) Why Analytics (He asked me 2-3 times till he was convinced)
- c) What is your biggest achievement in life?
- d) Why you want to join Mastercard? Since you do not have any prior analytics related workex ?
- e) Gave me a case to solve

Imagine you are hired as a consultant by Mastercard HR team. Prepare a strategy for us so that Mastercard becomes the top preferred company of PGDBA. Prepare 2 slides for this.

HR Round

HR Round Questions (Piyush Choudhary)

- a) What was the reason for leaving job?
- b) What was the motivation behind joining PGDBA ?

HR Round Questions (Pratesh Jhari)

- a) Why did you leave Indian Oil? Was that a good decision?
- b) What are your hobbies?
- c) Then asked me something about global politics, world wars etc?

Other Technical & HR Round Questions

- a) Technical questions on Regression, trees ?
- b) Linear Regression, Data Imputation Techniques, Inference, NLP

Courses & Skills Suggested

- a) Thorough knowledge of the projects done.
- b) Strategic management (IIM)
- c) Inference (ISI)
- d) Machine learning (IIT)
- e) SSD
- f) Probability
- g) DSL

Resumes of Selected Candidates Attached.

ACADEMIC QUALIFICATIONS			
Year	Degree/Board	University/Institution	%/CGPA
2019*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-
2016	B.E. (Computer Science Engineering)	Birla Institute of Technology, Mesra	7.57/10
2012	Class XII, CBSE	Army Public School, Delhi Cantt.	82.40%
2010	Class X, CBSE	Army Public School, Ahmednagar	89.30%
ACADEMIC AWARDS AND EXTRA CURRICULAR ACTIVITIES			
Academic Achievements	▪ Secured 10th Rank in iQ'oniQ, a Machine Learning competition organized by QLA club of NMIMS '18 ▪ Ranked 18th in Data Tales Beyond Infinity, Great Lakes Institute of Management on Analytics Vidhya '18 ▪ Awarded 2nd Runner Up and a cash prize in Case Study Competition by IRSC and GOI at IIT-BHU '18 ▪ Awarded Certificate of Merit for securing Full Marks in Mathematics in Class 10th Board Exams '10 ▪ Attained 20th National rank in Capsim Business Simulation game among teams from 100+ B-schools '18		
Music	▪ Performed in front of 1000+ audience as Lead guitarist and vocalist of the Official band of IIM-C '18 ▪ Awarded 1st Prize in Battle of Bands, Pune and 2nd Prize in college cultural fest band competition '15		
Sports	▪ 1st Runner Up in Badminton Championship at IIM Calcutta Inter-section Sports Tournament '17 ▪ Finished 2nd in Women's Pride Marathon and 3rd in two consecutive Puma Urban Marathons '15-16 ▪ Awarded Silver Medal in 100m breast stroke and Bronze medal in 100m freestyle in swimming '17		
WORK EXPERIENCE			
Software Developer		Headstrong Capital Markets	Noida (July '16 – Jan '17)
Received "Award of Recognition" twice in 6 months			
Search and Analytics Engine	▪ Implemented a real-time analytics platform to monitor error logs at a rate of 10,000 logs per second ▪ Identified regular expressions and used Ruby string manipulation to structure multiple log files ▪ Integrated ElasticSearch, logstash & Kibana for parsing, storage, exploration and visualization of the logs		
Trade Management Platform	▪ Worked on Calypso, a Java based platform for trade management on IRS and FOREX financial products ▪ Developed customizations to meet the requirements of one of the leading CCP Clearing House globally ▪ Analysed performance of calypso modules using Jprofiler and provided technical guidance for efficiency		
DATA SCIENCE PROJECTS			
Kaggle's Image Masking Challenge	▪ Developed a pixel-wise classifier to mask background by automatically identifying boundaries of a car ▪ Pre-processed data using Augmentation and trained CNN based on encoder-decoder architecture (U-Net) ▪ Stacked multiple models implemented at different resolutions to capture local as well as global effects ▪ Applied Erosion and Dilation techniques for Post-Processing to obtain test Dice Coefficient of 0.995		
Document Summarization and Retrieval	▪ Implemented GREP, Index and ElasticSearch based systems for Boolean query processing of documents ▪ Compared systems using precision, recall & used single-pass in-memory indexing for index construction ▪ Performed Document Summarization using graph-based methods, Degree Centrality and TextRank ▪ Evaluated generated summary against human reference summary using ROUGE-N & ROUGE-L metrics ▪ Improved precision by 10.7% using Pseudo Relevance feedback and Query expansion using GloVe		
Stock Prices Forecasting	▪ Analyzed Canara Bank's historical stock prices of nine years using multiple forecasting techniques ▪ Determined ARIMA order using ACF and PACF plots and tested for Stationarity using ADF test ▪ Implemented ARIMA, ETS, Holt-winters and TBATS models and evaluated the forecast using MAPE		
Startup Merger & Acquisition Prediction	▪ Predicted status of start-ups using XGBoost and Random Forest models trained on data of 60k companies ▪ Imputed missing data and engineered new features using company's age, funding and industry sector ▪ Determined important features essential for streamlining the investment decision making process		
Product Image Classification	▪ Developed a classifier for automatic product recognition using ~200K furniture and home goods images ▪ Used Data augmentation techniques to handle ~40GB of noisy and imbalanced dataset with 128 classes ▪ Achieved an accuracy of 86% using CNN and transfer learning through various pre-trained architectures ▪ Trained VGG-16, ResNet-34 and ResNet-50 using hyper parameter tuning and differential learning rate		
Recommender System for Movies	▪ Built content and collaborative filtering based Recommender System using 1MM ratings of movies ▪ Performed matrix factorization utilizing Singular Value Decomposition to tackle highly sparse data		
UNDER-GRADUATION PROJECTS			
Analytics Project	▪ Performed Sentiment Analysis of 400K Amazon product reviews using Naïve Bayes and Bag-of-Words ▪ Pre-processed data using tokenization, stop-words removal and lemmatization to get accuracy of 84%		
Technical Projects	▪ Developed a human-computer interaction MATLAB application to control devices using hand gestures ▪ Developed a mobile controlled counting device to automate the indoor lighting for energy conservation		
POSITION OF RESPONSIBILITY			
PGDBA Public Relations Team	▪ Facilitated a smooth admission process for aspirants and wrote articles for the press releases of PGDBA ▪ Enhanced presence of PGDBA course by 40% in terms of users following the course on social media		
TOOLS AND LANGUAGES : R, Python, Java, SQL, Microsoft Excel, IBM SPSS Modeller, Hadoop, MapReduce			
ELECTIVES : Corporate Finance, Supply Chain Analytics, Venture Analytics, Game Theory, Pricing and Revenue Optimization			

ACADEMIC QUALIFICATIONS

Year	Degree/Board	University/Institution	%/CGPA
2019*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-
2013	B.Tech (Mechanical Engineering)	IIT Jodhpur	7.81/10
2008	Class XII, CBSE	Gyan Bharati school-Saket, New Delhi	88.00%
2006	Class X, CBSE	Gyan Bharati school-Saket, New Delhi	82.60%

AWARDS & ACTIVITIES

- All India Rank 3037 in **IIT-Joint Entrance Examination** 2009 among 4 lakh entrants (99.2% percentile)
- Secured a Rank **343** in Delhi **Combined Entrance Exam(CEE)** 2009 amongst top **0.35 %** conducted by **NSIT/DCE**
- Qualified for Global semifinals amongst 100 teams out of 600 universities of **Airbus Fly your idea challenge 2013**
- Presented an Idea (amongst top 55) on knowledge sharing dashboard to the Chairman of IOCL at the Idea conclave-2015

POSITION OF RESPONSIBILITY & CERTIFICATIONS

Position of Responsibility	- Selected as Student Guide for the Counseling Service team , IIT Jodhpur- 2010 - Promoted as Assistant Coordinator for the Counseling Service team -IIT Jodhpur - 2011 - Spearheaded the Oil & Gas conservation Fortnight (Govt of India) activities for IOCL in Goa 2014
Certifications	- Certified Marketing Analytics and Customer Valuation, IIM Kashipur - 2017 - Certified in Data Science, Deep Learning, & Machine Learning with Python, Udemy - 2018

WORK EXPERIENCE (45 months)

Key Accounts Manager	Indian Oil Corporation Ltd	(Aug '13 – May '17)
Lubricants Sales(B2B & B2C)	- Managed key accounts such as MRF, Vedanta, JSW in Goa and five districts of Maharashtra - Offered facilities like bulk storage tanks & Vendor managed Inventory (VMI) to 'A' Category customers - Boosted sales for distributors through Bazaar Trade, OEM Service Stations and Retail Outlets - Arranged campaigns at IOCL retail outlets and OEM service stations (Maruti, Hyundai & M&M) - Provided technical support to industries via lube surveys, condition monitoring and seminars	
Inventory planning and forecasting	- Coordinated with plants for Supply and distribution for Servo lubes in the sales area - Liquidated the slow moving and Dead stock of Lubricants (Barrels & Small packs) in Goa - Estimated sales of new products based on customer requirement and market information	
Achievements	- Signed long term contract for 100% business and VMI facility with major Fertilizer Company - Reduced lead time of supplies of lubes to Shipping companies at Mormugoa port (Vasco) - Increased brand visibility of SERVO in Goa through advertisements and sponsorships - Obtained certification of MLT Level-1, International Council of Machine Lubrication -2015	

DATA SCIENCE PROJECTS

Forecasting returns and volatility on stocks	- Predicted 2-days returns, conditional volatility & correlation in five stocks based on 5 year data - Applied ARIMA for returns ,three GARCH variants for volatility & DCC-GARCH for covariance - Tuned model using RMSE,direction accuracy criteria performed Sharpe ratio based portfolio optimization
Investment Recommendation	- Developed a rating dashboard (Shiny R) to present list of potential retail firms based on Financial reports - Analyzed 17 Financial ratios and selected key metrics to cluster companies using K-means - Trained Random Forest to predict company's performance based on the clusters created by K-means
Sales Forecasting	- Created a model to predict sales of a grocery store in Ecuador based on store items & promotion data - Applied ARIMA, ETS,MA for forecasting sales and used Dickey fuller Test to check seasonality - Evaluated the model based on the Normalized Weighted Root Mean Squared Logarithmic Error.
Toxic comment classification (NLP)	- Devised NLP based system to classify comments into six categories for the Google Kaggle competition - Employed word embedding (word2vec & GloVe) to vectorize words without contextual information - Compared traditional methods : SVM,Naïve Bayes vs Deep learning methods: LSTM & GRU - Obtained best ROC-AUC score of 0.9867 by applying deep learning technique-LSTM & GRU
Santanders Product Recommendation	- Built a recommender system serving ~1,000,000 customers by suggesting most relevant product - Recommended few relevant products (7 out of 24) for users, based on their past 15 months data - Applied XGBoost multi class classification , logistic regression techniques, Hierarchical clustering
Network analysis on Wikipedia page	- Created a uni-partite & Bipartite network to understand the Editor interactions on Wikipedia page - Analyzed nine different Wikipedia pages including Amazon, Google, Facebook and Yahoo - Detected communities of Editors using Louvian & K-Clique algorithm on the created network.

TECHNICAL SKILLS & RELEVANT COURSEWORK

- Tools/Languages** – R, Python, Tableau, MS Project, MS Excel, VBA Macros
- Machine Learning methods** – Scikit-learn,Keras,Pandas,Numpy,Scipy,Tensorflow,ShinyR
- Electives** – Computational Finance, Supply Chain analytics, Pricing and Revenue optimization, Econometric



Roles Offered :

- a) Data Scientist

Job Description: On Campus presentation

Internship Stipend Offer : Rs 100000

Shortlist Steps :

- a) Resume Shortlisting
- b) Interview with the chief data scientist of Myntra

Duration : Maximum Duration was 30 minutes.

Technical+HR Round

Technical+HR Round Questions (Shreyas Mangalgi)

Project & Competitions :

- a) Data Science Bowl Silver Medal
- b) Data Science Game AIR 4
- c) Projects on Image classification, Instance segmentation, Fake review detection, Recommender system, Information Retrieval, Pricing optimization, Time series analysis

Questions:

Technical part :

- 1) Questions around image and nlp projects
- 2) Simple Probability question on Bayes theorem
- 3) What kind of projects would you like working on at Myntra ?

Non-technical:

- 1) Why Myntra ?
- 2) What's your motivation in taking up deep learning based projects especially on images ?

Apart from this, I myself told them about an idea combining two research papers I had come across which I feel impressed them.

Technical + HR Round Questions (Preethi V)

a) Mentioned a problem currently solving in MYNTRA:

- a)1. In the search engine, people search for items in different languages for instance, vest vs. baniyan and also brands not offered by myntra. They wanted to display relevant results in case of the former problem and related items in case of the latter. I proposed using word embeddings and in specific glove.
- a)2. Then they asked difference between glove and word2vec.
- a)3. What is global in glove?
- a)4. Construct a neural network model to learn new words.

b) What is p-value? Explain in detail.

c) What is AB testing.

d) Do you know about Gaussian mixture models? I explained in detail about EM algorithm.

e) What is difference between frequentist and Bayesian? When both of them will give the same result?

Other Technical & HR Round Questions

- a) Is it necessary to use word2vec to train an LSTM?
- b) Mention any clustering algorithm? explain k-means clustering?
- c) Why pgdba?
- d) What did you do in workex?
- e) Why myntra?
- f) The project that you like the most? What was the data? What was the business case/impact? What are the techniques used?
- g) What was your job experience and Have you worked on regression in your job ?
- h) Asked about linear regression?
- i) Hypothesis testing
- J) Feature selection and how to validate and check for the best fit.
- k) Application of one of my project (related to image classification) and some concepts of recommendation system?
- l) Mostly technical questions on projects. (Korah T.)
- m) How would you change your approach if you know certain independent variables with only some level of certainty?
- n) (Abhishek Gupta) Interviewed by Pricing and Optimization Guys
- o) CASE: How will you sort the on the basis of profits generated from different catalogue ?
 - o)1. Then asked which algorithm will you use ?
- p) Questions based on Recommendation Algorithms
- q) Explain K means? How will you optimise it? How will you choose the value of k?
- r) Explain other clustering algorithm and which one is better among them and why?

- s) Explain the objective of one of your project and what is it's business impact.
- t) Suppose you have customer data of 1.5 million users and there are 100 advertisements. You have the data about which customer has clicked on which ad. A new user joins the platform, you have his data, how will you find out which ads to show to him?
- u) Suppose you have developed a new cross-sell tool. How will you confirm that it is performing well?
(Answer: A/B Testing)
- v) Describe your work ex.
- w) What was the best project you did in your work ex?
- x) What do you like in linear algebra?
 - x)1.Show how you get the matrix expression of least squares estimate?
 - x)2.First define what the rank of a matrix is and then what happens if the matrix is not full rank?
- y) How do you handle multicollinearity?
 - y)1.Do you always remove variables with high VIF?
 - y)2. Give 2 scenarios where you should not remove variables because it is part of the design of the model (ans: when there are interaction terms in the model, in case of categorical variables)

Courses & Skills Suggested

- a) CDS, FADML and DSL, PRO
- b) IR, Gaussian Mixture models, Clustering, DBMS

Suggestion

Positioning is the single most important thing

1. Your resume should reflect what kind of a role you are seeking. If you want a tech role (data science), position your resume in that way with deep learning projects, data science competitions etc. If you want a business analyst consulting or finance - position yourself accordingly.
2. Attending company PPTs is a key to cracking interviews - In the PPTs, you get to know what kind of a person the company is looking for. Once you attend the PPTs, you can use that knowledge in positioning yourself according to their expectations/requirements during the interview. By being a Placecomm, I was compelled to attend almost all the PPTs which was a blessing in disguise.
3. During the interview, take every opportunity to stress on how you are the most suitable candidate for the role. In Myntra interview, I was asked about my interest in photography (photography contests I had won) and I took that opportunity to explain how deep learning could replace DSLR cameras altogether in future.
4. Have multiple "dream companies" and not hold on to just one - this would reduce anxiety during the interview.
5. Push the placecomms to bring the roles/companies that you want to get into.
6. Be thorough with projects and whatever is mentioned in resume in detail.

Resumes of Selected Candidate Attached.

ACADEMIC QUALIFICATIONS			
Year	Degree / Examination	Board / University	CGPA / %
2016	Dual Degree Electrical Engineering M.Tech Microelectronics	IIT Bombay	7.27 9.50
2011	Class XII	Board of Intermediate Education (A.P)	94.80%
2009	Class X	Karnataka Secondary Education Examination Board	97.12%
RELEVANT CO-CURRICULAR ACHIEVEMENTS			
• Silver Medalist among 3634 teams at the Data Science Bowl 2018, world's top data science for social good challenge by Kaggle • All India Rank 4 in Data Science Game 2018, an international student challenge organized by Kaggle, BNP Paribas & Capgemini • All India Rank 8 in 'Data Hack' organized by Analytics Vidhya & NMIMS Hyderabad for predicting loan default for consumers • World Rank 38 out of 5500+ participants in Machine Learning Challenge held by Hackerearth for earthquake damage prediction • World Rank 213 in iMaterialist Deep Learning challenge organized by Kaggle & 5th Workshop on Fine-Grained Visual Categorization • Finalist among 4900+ participants at 'Brainwaves' Hackathon 2017 organized by Societe Generale's Global Solution Centre • National Finalist 'Data & Beyond 2017' organized by Analytics Vidhya & American Express for credit card recommendation • Selected for Novartis India Biotechnology leadership camp 2017 from 500+ applicants across top Management & Research Institutes			
KEY ANALYTICS PROJECTS			
Trade pattern prediction	• Developed an ensemble model of Light GBM & XGBoost to predict bond trading behavior of customers • Created synthetic transactions to tackle class imbalance & engineered temporal & aggregate features		
Time Series Analysis	• Forecasted sales of 45 Walmart stores across 99 departments over 39 weeks with holiday markdown events • Analyzed different store-department combinations using Time Series Clustering & STL decomposition • Achieved noise reduction using SVD & Kalman Filter followed by sales forecasting using ARIMA & ETS		
Information retrieval	• Synthesized fluent summaries of 25 PDF documents across 5 topics using Text Rank & Degree Centrality • Evaluated summaries on Rouge-N/L scores & used Roccio Algorithm for pseudo relevance feedback		
Fake review detection	• Devised an unsupervised model for group spam detection in Yelp's dataset of 26k+ reviews & 16k+ users • Used Apriori for Frequent Itemset Mining & engineered behavioral/temporal features for candidate groups • Implemented GS Rank Algorithm to obtain an ordered set of candidate groups by their degree of spamicity		
Recommender system	• Built a cross sell model for suggesting philanthropic projects to potential donors of NGO DonorsChoose • Used a combination of word embeddings, graphs & neuralnets to find similar projects for each donor • Developed a hybrid recommendation engine for targeted email campaigns to increase repeat transactions		
Revenue optimization	• Built an ML model to predict renewal rate for 34000+ insurance policies using historical data of customers • Used isotonic regression for transforming scores from ML model to probabilities improving Brier score • Designed an incentive structure for insurance agents to maximize revenue using non-linear optimization		
Image segmentation	• Designed Mask R-CNN & U-Net models for instance segmentation of images using Google Cloud GPU • Used COCO Dataset for transfer learning & OpenCV for post processing image masks for higher precision • Achieved an MAP score of 0.499 (Kaggle Top 5%) by training on just 670 images using Keras Tensorflow		
Image Classification	• Trained a deep neural network on AWS EC2 using pytorch for visual categorization of home goods images • Used ensemble of NASNet-A & Densenet201 for training 2 lakh images (22GB) in 128 product categories		
ELECTIVES & AUDITED COURSES			
• Blockchain Architecture Design - NPTEL (Ongoing) • Introduction to Pyspark (DataCamp) • Deep Learning for NLP			
PROFESSIONAL EXPERIENCE			
Assistant Manager, Tata Communications' Global Campus Program (11 Months) - <i>Selected from top schools in India, Singapore & U.S.A</i>			
Ticket Prediction	• Identified business case for early detection of network failure events leading to customer raising a ticket • Implemented POC model for Machine Learning based ticket prediction for the firm's global IP network		
Project Management	• Evaluated the performance of SDWAN deployments & explored avenues for additional value creation • Published & presented an internal white paper on 'Scalable Internet Traffic Measurement using Hadoop'		
Capacity Planning	• Devised a Network, Service & Application level Inventory for Hosted Contact Center solutions team • Identified items of capacity monitoring & designed a work-flow of portal for inventory management		
RESEARCH PROJECTS & INTERSHIPS			
• Modeled physics of FINFETs & proposed quantum correction method for Monte Carlo simulation [<i>Prof. Udayan Ganguly, IIT Bombay</i>] • Designed Izhikevich model based Spiking Neural Network for musical instument classification [<i>Prof. Bipin Rajendran, NJIT U.S.A</i>] • Built an NFC device for instant transfer of apparel images from interactive screen to cellular devices [<i>Fynd - A Google funded company</i>]			
POSITIONS OF LEADERSHIP			
Placement Coordinator <i>PGDBA 2017-19</i>	• Brought key changes to placement policies of PGDBA resulting in 26.7% increase in final compensation • Coordinated with students, recruiters & authorities of IIM, ISI & IIT for organizing campus placements		
Overall Coordinator <i>NSS IIT Bombay</i>	• Led IIT Bombay's largest student body of 40 member core team & 250+ volunteers across 4 departments • Pioneered 'OLI' - The most followed YouTube channel across all IITs (62,000+ followers, 0.5 crore views) • Organized health camps for 350 construction workers, raised INR 2.15 Lakh for Kashmir flood relief		
EXTRA-CURRICULAR ACHIEVEMENTS			
• Won four 12 hour Swimming Marathons at IIT Bombay between 2012 & 2015 consistently covering a distance of 12+ kilometers • Secured first place in 'ShutterBuzz' photography contest at IIT Kharagpur & second place in 'Click for Change' contest at IIT Bombay			

ACADEMIC QUALIFICATIONS

Year	Degree/Board	University/Institution	%/CGPA
2019*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-
2014	B.E. (Electronics and Communication Engineering)	Sri Venkateswara College of Engineering, Chennai	8.4/10
2010	Class XII, Tamil Nadu State Board	DAV Higher Secondary School, Gopalapuram	96.5%
2008	Class X, CBSE	Kendriya Vidyalaya CLRI	93.6%

WORK EXPERIENCE

35 months

Associate Consultant **ZoomRx Healthcare Technology Solutions Pvt. Ltd.** *Chennai (Mar '15 – May '17)*

Received 'Letter of appreciation' from CEO for achieving high client satisfaction on tight deadlines; Fastest to get promoted in the organization

Physician Target Segmentation (Clustering)	- Led a team of 4 in identifying target MDs by discerning factors influencing prescriptions using regression - Segmented 150 physicians into three clusters by performing K-means clustering using SPSS software - Examined difference in cluster means using ANOVA; Identified 6/9 features to be significant at 0.01 level
MD Profiling and Positioning	- Created robust MD profiles from market research data by probing usage behavior & treatment attitudes - Aided 48.1 billion \$ client in gaining 12% market share by suggesting segment-wise positioning strategy
Statistical Significance Testing	- Devised a methodology to visually interpret significant differences by mapping Z-test to scaled CIs - Enabled signalling alerts to the client on performance deviations; Adopted team-wide as a process standard
Sales Force Effectiveness	- Delivered insights to 30.4 billion \$ client on structuring of sales interactions and messaging with physicians - Gauged delivery impact on perception; Suggested structural changes increasing portfolio impact by 10%
PPT Automation	Saved ~50% man-hrs by leading a team of 5 from concept to execution of VBA-based tool for slide creation

Trainee Decision Scientist **Mu Sigma Business Solutions Pvt. Ltd.** *Bangalore (Jun '14 – Feb '15)*

Sampling behavior	- Healthcare Analytics – Performed critical analysis on sales reps' deviation from sampling planned quantity - Recommended aligning sample plan to physician's response, leading to a 6% improvement in adherence
Store set-up Analysis	- Retail Analytics – Delivered key insights by analyzing factors that drive performance of a display model - Lifted sales by 9% optimizing placement & assortment of SKUs; Simulated model in corporate workspace
Database Mining	Coalesced tables devoid of data dictionaries to create a master database of 350 variables easing retrieval

DATA SCIENCE PROJECTS

Pipe Breakage Prediction (Machine Learning)	- Predicted next breakage of ~78k pipes using Expectation Maximization algorithm and Xgboost Classified the pipes using Random Forest with an accuracy of 80%, based on features selected by Boruta - Applied Box-cox while predicting time to serial breakages; Predicted 70% of breakages with 60 days RMSE
Search Engine (Information Retrieval)	- Built search engine to retrieve relevant documents based on user query using TF-IDF vector representation Pre-processed ~7k documents to build inverted index; Improved F-score by 27% points using Lucene Expanded queries using word embedding; Used Pseudo relevance feedback perking up F-score to 82%
Image Classification (CNN)	- Detected 14 different classes of thoracic diseases from chest X-rays by leveraging transfer learning - Handled high class imbalance using class weights & used VGG-16, ResNet and Inception network models - Achieved a weighted F-score of 35% using Inception network model, against radiologist score of 38.7%
Malicious Software Detection (LSTM & ARIMA)	- Developed robust model for malware detection by modeling normal system behavior & limiting false +ve - Used LSTM based auto-encoder to improve upon boosting (F-score: 86%) by detecting new ransomware - Transformed data to frequency domain & fit ARIMA; Detected ransomware in 5s with 99% confidence
Sales Prediction (LSTM & ARMA)	Predicted the next month sales for about 200k store cross item combinations based on ~3 years of data - Used CatBoost Regressor on engineered features; Checked stationarity & forecasted using ARMA model - Achieved RMSE of 0.92 units using 2-layer LSTM against benchmark model with RMSE of 0.86 units
Document Summarization	Summarized documents using Degree centrality & text rank; Evaluated summaries using rouge measures - Achieved F-score of 45% (Rouge-1) using text-rank by applying a threshold of 0.2 on the weighted graph

CERTIFICATIONS AND ACADEMIC DISTINCTIONS

Certifications	Completed Machine Learning, Deep Learning and CNN courses authorized by Stanford University
Academic Distinctions	- Garnered merit-based Books and Instruments Scholarship; Graduated among top 5% in ECE, SVCE - Conferred Silver in Mini Project on Infrared Counter in Techfest '11 of SVCE; Secured 96 %ile in GATE '14

POSITIONS OF RESPONSIBILITY AND EXTRA-CURRICULAR ACTIVITIES

Positions of Responsibility	- Handled scheduling and assignment for 2 major placement drives in ZoomRx with a footfall of 4000+ - Conducted on boarding sessions for 20+ associates and associate consultants on principles of analysis Organized the Mini project event in Techfest '12 of SVCE; Held the post of class representative in college
Extra-Curricular Activities	- Won the second prize at the International Art and Craft Contest organized by Pidilife Organized blood donation camps & tutored ~200 high school students as a Member of Youth Welfare Trust

Key Skills: Received professional training in **MuSigma University** on SQL, Excel & Statistics; R, Python, SPSS, VBA

Courses/Electives: Corporate Finance, Information Retrieval, Pricing Revenue and Optimization, Supply Chain Management



Roles Offered : a) Analytics Business Consulting Role: Experienced Associate

Job Description :

Interns will work in our Analytics Consulting practice in Mumbai/Bangalore. They will be integral part of business analytics teams in India alongside clients and consultants in the U.S., performing high-end analysis and providing business recommendations to project teams. Interns will have following Roles and Responsibilities:

- Execute on project and analysis plans under the guidance of Project Manager
- Interact with consultants/clients in US to formalize data sources, understand data, acquire and use relevant business knowledge
- Develop analytics path to align with overall project objective
- Validate analysis & process with consultants/clients in US
- Analyze, develop recommendations, and conduct so-what analysis
- Build storylines and make presentations
- Manage large data sets; apply basic statistical techniques to understand data

Internship Stipend Offer : 1,25,000 INR per month .

Shortlist Steps : a) Resume shortlist
b) Technical

Duration : 15-25 minutes.

Technical Round

Technical Round Questions : (Tarun yadav)

- a) predictive maintenance - how would you go about to predict that a machine would fail at some point in future?
- b) supervised learning - a machine can fail if any of its components fail. such a component can either be replaced or repaired. based on images of previous damaged components can you shorten the duration for this activity?
- c) puzzle - you have 2 stacks of 5 and 15 coins respectively in which the 5 coins have heads up. you are blind folded and the coins are mixed. how would you create 2 stacks now such that both have same no of heads up? you can flip the coins

Technical Round Questions : (Palvika Bansal)

- a) What data you need to predict box office revenue of an upcoming movie?
- b) Further questions on the data and approach and what kind of models can be used.
- c) Discussion and cross questioning on 2 projects - NLP and Time series

Technical Round Questions : (Aayush chaurasia)

- a) work ex related questions
- b) case study on stock out prediction at retail outlets, also lost sales (data, methodology)
- c) questions about various sectors and industries, not necessarily as depicted by your work ex or your projects

Technical Round Questions : (Vishakha Bansal)

- a) B.Tech. Internship experience
- b) Problem on Face Recognition

Technical Round

Technical Round Questions : (BSK Teja)

- a) Asked me to explain Work-Ex project (Image Processing)
- b) asked me to explain a project in my cv (on complex networks)
- c) Problems faced while scraping the dataset (twitter)
- d) what is t-SNE, what does it do
- e) case study:

There is going to be a football league in Germany. a German broadcasting company has two options before it

option1(subscription): charging a subscription fee

only this company can telecast matches played by Germany (only Germany matches, other matches can be telecasted by any company)

option2(free): free telecast (all matches, but delayed)

what option should the company opt??

Technical Round Questions : (Burhan Usman)

- a) Was asked about the Time Series Project I had done. How to select the right time-series project.
- b) Case Study : Sales are increasing, but the profits are going down. Why is this happening

Other Technical & HR Round Questions

- a) Case- Croma retail company- If there are different departments in the store, we have customer purchases history. We need to come up with an idea on how to bundle cross products from different departments. There will be plenty of products in each department and the customer purchases are very sparse.
- b) Guesstimate: Number of persons using bus service in Kolkata
- c) Logical Puzzles(2)- Solving the puzzles and explaining the mathematical approach (Eg- Using Optimization to solve- how many minimum number of cheques I need to write to withdraw an unknown amount of money in the range $[0,x]$, will this number change if there is a penalty for each cheque that gets bounced?)

Courses & Skills Suggested

Tarun Yadav :

- a) Emphasis On Your Thought Process, Structure And Application Of What You Have Learned Over The Period.
- b) All Courses At IISc , Complex Network, Regression And Time Series, Projects –IIT , Corporate Finance, Computational Finance, Econometric, Economics

Palvika Bansal :

- a) R, Python, Sql, Time Series, Regression, Aptitude, Logical Reasoning
- b) Regression and time series, CDS, DBMS, Predictive Modeling in time series, Inference, Information Retrieval

Aayush chaurasia :

- a) Business courses, probability and stats applications to business problems

Vishakha Bansal :

- a) Various projects and competitions

BSK Teja :

- a) KGP Projects, statistics, case study, puzzles

Burhan Usman :

- a) CDS, Time Series Analysis ,RTSM ,SSD



ALVAREZ & MARSAL

Roles Offered : a) Consultant

Job Description :

The successful candidate should be able to:

- Quickly develop an understanding of the target business and its drivers
- Manipulate large data sets to provide compelling, value add outputs in quick time.
- Collaborate extensively with A&M engagement teams and clients
- Clearly and effectively communicate requirements, timelines and insights to both internal A&M team members, as well as clients
- Have experience on conceptualizing analytics and driving insights on complex business problems
- Be able to translate business objectives into actionable analysis and own end-to-end analytics.
- Demonstrated communication and presentation skills (both oral and written) in addition to strong project management skills
- Ability to work & manage multiple projects simultaneously and under challenging timelines
- Create insightful automated dashboards and data visualizations to track key business metrics
- Knowledge of R/Python/SAS/Tableau/Qlikview/SQL/Java/D3JS/ any other scripting language
- Experience in consulting or analytics field is a plus

Internship Stipend Offer : INR 100,000 per month.

Shortlist Steps : a) Resume shortlist
b) Technical interview
c) Hr Round

Duration : 30 minutes.

Technical Round

Technical Round Questions : (Vishal Singh)

a) Related to work ex only

Technical Round Questions : (Nishanth Kumar)

a) The offer was for a consultant role, so it was more like a discussion on the skill set which I'll bring to the company and my expectation with the role offered.

b) There was a brief intro, then discussion on the projects I have worked on during my work. Possibility of leveraging the experience on the client side by implementing the solution. Then the interviewer picked the timeseries project and asked to explain it. Discussed case of finding non-performing corporate loans.

HR Round

HR Round Questions (Vishal Singh)

a) Why A&M? Leadership skills, etc.

HR Round Questions (Nishanth Kumar)

a) AnM looks for culture fit than the knowledge. It'll be great if you show that you can handle project individually with a great pace.

b) Be positive and show that you have interest in the profile. The profile doesn't give you to apply ML techniques during internship so don't say that you want ML.

Other Technical & HR Round Questions

- a) AbInBev - Prepare a pitch for improving beer sales in Kolkata (Heavy focus on PRO) and cultural fit.
- b) Introduce yourself
- c) What are your strengths?
- d) Where do you see yourself in 3 years?
- e) Why do you want to join AbinBev?
- f) Asked me something about my past experience
Gave a puzzle
- g) There are 8 identical balls(shape & size), one of them is of different weight. How will you identify this ball using a weighing scale? And in how many steps? (Solve the problem in less than 3 steps)
- h) Explain Neural networks to a layman?
- i) How will you decide the price of a new product to be launched in the market? What is perceived value by customers of a product? How will you estimate this ?

Courses & Skills Suggested

- a) Describe your work ex profile such that they fit to the company's requirement. No technical skills was tested.
- b) Communication skills, clarity of thought, go through resume carefully, any achievements related to work ex.



BNP PARIBAS

Roles Offered : a) Data Scientist, AI Labs

Job Description :

You will participate in the design and development of core Data and AI Lab projects whose ambition is to leverage the huge amount of data we accumulate daily, both structured and unstructured.

We're looking for students in:

Statistics, Mathematics, or Computer Science Demonstrating knowledge of Machine Learning techniques (classification, regression, natural language processing via standard statistical models or neural network...) With a very good understanding of the statistics and mathematics behind the models used And very good programming skills (fluency in Python is preferable) Problem solvers with a delivery-driven mindset)

Internship Stipend Offer : INR 100,000 per month.

Shortlist Steps : a) Online machine learning + coding test + resume
b) Technical interview (Hr questions were also asked)

Duration : 10 minutes.

Technical Round

Technical Round Questions :(TEJA RAM KOTAPROLU)

- a) Expectations of company and candidate.
- b) Why BNP Paribas for me? looking for managerial capabilities who can build and lead AI Labs team in Mumbai similar to the lines of Singapore, Paris & NY for BNP Paribas. Challenges, regulatory constraints, deadlines, timeline, resources, long term commitments etc.

Courses & Skills Suggested

- a) Machine learning Models with Business Application that can bring measurable value to the organization .
- b) CDS, RTSM, Predictive Modelling, Stochastic Processes, Inference (electives are company specific)



Roles Offered : a) Business analyst

Job Description :

The Gig: Business Analytics Intern

Flipkart is making eCommerce happen in India. Flipkart's vision is to make analytics part of its overarching competitive strategy, and push it up to decision makers at every level. The Analytics Team serves data driven decision making along with a diverse set of customers – from Business Teams to Supply Chain Teams to Product Teams. An Analyst is expected to grasp the underlying business/ product needs of these teams and develop reports/ analyses/ models that solves problems of those respective areas/ functions.

Do you have the passion to analyze data and impact the business materially through what you observe?

What this Job Entails?

Being part of Central Analytics Organization at Flipkart, you get an opportunity to work on diverse set of problems that cut across Marketing, Consumer as well as Seller Facing Products on Web & Mobile, Pricing, Customer, various Business Categories, end-to-end Supply Chain, Customer Support/ Call

Centre and so on...At Flipkart, high focus on Impact and Ownership allows for freedom to each to experiment and innovate. The pride is when you see the impact of what you have created. The ability to see what your contribution does to the business is a rare experience, add to that the fact that the impact of it is felt by your friends and family in India - the experience is extremely gratifying.

You're Part:

You will work closely with internal Flipkart Teams to identify, define, collect, and track key business metrics for products or business process. Conduct business analysis (prescriptive and/or predictive), build reports, dashboards and metrics to monitor the performance. Work with the stakeholders to take data driven decisions.

Internship Stipend Offer : INR 1,00,000/- + 2200/- Food Card per month

Shortlist Steps : a) Resume shortlist
b) Technical interview

Duration : 15 minutes.

Technical Round

Technical Round Questions : (Ashish Das)

- a) Mainly from the work experience (As I had experience in analytics I guess) and also describe one of the projects (in recommendation system).
- b) Why do you want to join the company?

Other Technical & HR Round Questions

- a) General Analytics questions like how can I boost my sales, where to open stores, etc. Questions on regression & trees.
- b) Questions on Clustering algorithms
- c) Cross questions on Projects done
- d) Guess estimate on How many Cars in Calcutta

Courses & Skills Suggested

- a) Time series, Regression analysis, Basic statistics .
- b) Regression and time series in IIT and IIM C



Roles Offered : a) Risk Analyst

Shortlist Steps : a) Technical interview

Duration : 40 minutes.

Technical Round

Technical Round Questions : (Abhishek Gupta)

- a) Cross Questions around a competition, related to problem statement, features created and model implemented
- b) Difference between hierarchical and K means clustering? When and how will you use it? How to optimize it? Which is Computationally more efficient?
- c) How will you impute missing values? Cross questions on answers given?
- d) Imbalanced Data Handling questions? What is SMOTE and how does it work?
- e) Rate yourself on a scale of 1-10 at programming in Python and R

Courses & Skills Suggested

- a) CDS, FADML, Course projects and Competitions



ACADEMIC QUALIFICATIONS			
Year	Degree/Board	University/Institution	%/CGPA
2019	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-
2014	B. Tech (Production and Industrial Engineering)	NIT, Allahabad	7.81/10
2009	Class XII, ISC	DBMS English School, Jamshedpur	92.14%
2007	Class X, ICSE	DBMS English School, Jamshedpur	91.16%
WORK EXPERIENCE			24 Months
Assistant Manager, Machine Engineering		Maruti Suzuki India Ltd	Jul '14 – Jul'16
Major Responsibility	▪ Delivered process development and improvement projects for engine component production line		
Process Control and Optimisation	▪ Performed statistical analysis of design parameters to achieve process control and desired quality ▪ Analysed impact of process parameters on production cost to optimise tool life and productivity		
Quality Control	▪ Employed methods like correlation, GRR, process capability index to analyse measurement data ▪ Set up the new process and quality control standards for the production of Alto 800-cylinder head ▪ Responsible for the procurement of metrology equipment for the new 1.5l diesel engine facility		
Initiative	▪ Developed a standardized fixture unit with modular design for measuring 10 different components ▪ Achieved a reduction in average wait time of components by 500 seconds and cost savings of \$10K		
Achievement	▪ Led development of Alto 800 engine cylinder head along with R&D, Production and Supply Chain ▪ Improved the fuel efficiency of the existing engine by 12% with an investment of \$7.5 M ▪ Achieved a cost savings of \$2.9 M by tool localisation and modification of existing production line		
Faculty	T.I.M.E. Pvt Ltd, Jamshedpur		Aug'16 – Apr'17
Responsibilities	▪ Worked as a Faculty of Quantitative Aptitude and also assisted in Course-Material development ▪ Conducted Seminars , in colleges for CAT/GMAT and in schools for IPM/CLAT, in Jamshedpur		
ANALYTICS PROJECTS			
Financial Risk Modelling	▪ Predicted propensity of a bank account to become delinquent based on the customer's credit history ▪ Down-sampled majority class and synthesized minority class instances to handle class imbalance ▪ Applied feature selection algorithm (Boruta) and Logistic Regression to achieve AUC-ROC of 82% ▪ Implemented Hierarchical Clustering to discover clusters of defaulters and non-defaulters		
Fraud Detection (Payment Services)	▪ Detected fraudulent behaviour in a highly imbalanced data of 6 million mobile money transactions ▪ Explored data and engineered features to identify features associated with fraudulent transactions ▪ Tuned Random Forest and XGBoost models for prediction to achieve an AUC-ROC of 90%		
Recommendation System	▪ Estimated propensity of a user to buy an item on an e-commerce website based on clickstream events ▪ Handled sparse data of 33 million clicks and engineered customer and time-based features ▪ Optimised the Factorisation Machine model parameters using SGD to obtain an AUC-ROC of 64% ▪ Used the method of Item-Item based Collaborative Filtering to recommend items to customers		
Portfolio Optimisation	▪ Created an optimal portfolio of stocks by applying the market neutral strategy of Pair Trading ▪ Selected stock pairs having least Sum of Squared Distance between the normalised stock prices ▪ Developed a Hidden Markov Model with simulation to obtain an optimal threshold level to trade ▪ Achieved an average return of 5% by trading the stock portfolio at the threshold level		
Anomaly Detection (Ransomware)	▪ Analyzed multivariate time series sequence data for hardware performance counters of a system ▪ Reconstructed the original input sequence by implementing a LSTM based autoencoder on the data ▪ Applied outlier detection technique on reconstruction error distribution to identify ransomware		
Analysis of Social Temporal Networks	▪ Predicted human interactions in temporal networks by using the data of previous interactions ▪ Leveraged the Expectation-Maximization algorithm for predictions using community detection ▪ Improved predictions by considering higher order interactions and obtained an AUC-ROC of 70%		
AWARDS AND ACTIVITIES			
Academic Awards	▪ Achieved a certificate of merit in (level II) Talent Search Institute Maths Olympiad in 2004 ▪ Winner of the inter-house GK Quiz competition organized during the academic year 2008-09		
Extracurricular Activities	▪ Increased awareness about the PGDBA program through an article in the Analytics India Magazine ▪ Member of the organizing team for the annual Futsal tournament of NIT Allahabad in 2014 ▪ Won event Panorama (represent State through arts) at Culrav for two consecutive years (2012,2013) ▪ Winner in 2012 and 1st runner up in 2013 in the inter-batch football competition of NIT Allahabad		
TECHNICAL SKILLS AND RELEVANT COURSEWORK			
Languages and Tools	▪ R, Python, SQL, MS Excel, IBM SPSS Modeler		
Courses of Interest	▪ Inference, Supply Chain Analytics, Product Analytics, Complex Network, Decision and Games		

ACADEMIC QUALIFICATIONS

Year	Degree	Board/University	CGPA/%/Rank
2019*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-
2014	B.Tech (Instrumentation and Control Engineering)	NIT Jalandhar	9.16/10 (Rank 2)
2010	Class XII	CBSE	86.00%
2008	Class X	CBSE	91.60%

ACADEMIC DISTINCTIONS AND CO-CURRICULAR ACHIEVEMENTS

Institute Rank 5 : PGDBA 2017-2019

Data Science Competitions	Ranked 1 st out of 150 teams in Forecasting challenge 'Data Play' sponsored by KPMG and IIM Calcutta	'18
	1 st Runner-up at national level Consulting event - 'Sumantran' organized at IIT Kharagpur	'18
	World Rank 26 out of 750 teams in 'AO Tennis end point prediction' hackathon by Tennis Australia	'18
	India Rank 7 and World Rank 57 in Data Science Game organized by BNP Paribas and Kaggle	'18
	Ranked 12 th out of 418 students in Statistics and Probability skill test conducted by Analytics Vidhya	'18
	Among Top 20 teams out of ~5000 in Financial Technology Hackathon 'Brainwaves' by Société Générale	'18
	Among Top 5 teams in Customer Targeting challenge 'Data and Beyond' sponsored by American Express	'17
Achievements/Certifications	Awarded Black Belt Certification on Lean Six Sigma Methodology by KPMG	
	Presented a poster paper on 'Power Generation' at the Texas Instruments India Conference (TIEC 2013)	
	Awarded certificate of achievement by CBSE for distinguishing performance in class X; School Rank 1	

WORK EXPERIENCE

31 MONTHS

Instrumentation Engineer, Oil and Gas		Samsung Heavy Industries India (SHI)	Noida (Jul '14 – Feb '17)
Roles and Responsibilities	- Coordinated with cross functional teams to review Procurement and Design specifications for 10+ projects - Assessed vendor proposals on cost, delivery, quality based KPIs to map vendor capabilities to project requirements Led a team of 3 to evaluate vendor packages worth \$5 million for a \$3.3 billion Egina oil field project		
Impact Achieved	- Spearheaded modules to create SOPs through analogous estimation to reduce procurement cycle time by 8% - Deployed Remote and Wireless Instruments in place of wired devices, saved 40% cost & 90% installation time Saved 2k+ man-hours by developing Standard Work Procedures followed by 8k+ employees of SHI		

ANALYTICS PROJECTS

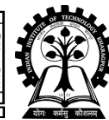
Quora Duplicate Question Pairs (NLP)	- Identified questions with same intent by modelling semantic representation using word embeddings and TF-IDF - Captured similarity patterns with Fuzzy Ratios, Cosine Similarity, Word Movers Distance for 0.4 mn ques pairs - Implemented LSTM and Attention models through Siamese architectures to achieve log loss of 0.17 on test data
KKBOX Music Recommendation	- Built a Recommender System for ~0.33 million users by predicting probability of a user relistening to a song - Applied Collaborative Filtering using Jaccard similarity and dealt sparsity problem by dimensionality reduction - Modelled song-user interaction based on latent features through SVD and used XGBoost to achieve AUC of 0.69
Trade Prediction (Classification)	- Predicted customer interest in a given bond trade using 6.7 mn customer-bond interactions in 119-week period - Generated synthetic transactions to handle class imbalance; Engineered entropy and mean encoded features - Built an Ensemble of LightGBM and CatBoost through rank averaging to achieve an AUC score of 0.66
Sales Forecasting (Time Series Analysis)	- Forecasted 6 weeks of daily sales for 1,115 Rossman stores influenced by factors including holidays, promotions - Identified seasonality by STL Decomposition, stationarity by ADF test and ARMA lags by ACF-PACF plots - Built Time series based ARIMA, TSLM models and achieved best score of RMSPE as 0.13 through TSLM
Image Segmentation (Deep Learning)	- Automated process of detecting nuclei in biomedical images of cells to understand how cells react to treatments - Implemented U-Net, Mask-RCNN architectures for pixel wise classification of images to find mask of nucleus - Improved MAP score from 0.41 to 0.48 by Image Processing, augmented more images from existing train data
Credit Risk Modelling (Classification)	- Predicted loan repayment capability of applicants using previous application, credit history data for 'Home Credit' - Interpreted bivariate plots to analyze trends among features, selected features through Boruta and Random Forest - Imputed missing values through KNN and handled class imbalance using SMOTE to improve model performance
CyberSecurity (Anomaly Detection)	- Detected Wannacry Ransomware attack on system by analyzing deviation in the normal operating performance - Developed an unsupervised model through Auto-Encoders to train on sequence data of performance counters

EXTRA CURRICULARS

Positions of Responsibility	- Contributed as Tutor in 'Samsung Dream Class program' designed to provide academic assistance to schools - Represented department in Young Management Board and ideated team building events for 300+ employees - Served as event organizer, Co-Head for Robotics Club during national level college tech-fest (techNITI)
Sports	- Organized SHI-India Olympics, an annual company sports event involving participation of 200+ employees Bronze Medalist in relay race (4x200) in NIT annual athletics meet, participation of 10+ teams - Award of appreciation for commendable contribution as hostel sports representative at NITJ

Tools and Languages: R, Python (keras, nltk, sklearn), SQL, Tableau, MS Excel, MiniTab

Electives: Information Retrieval, Supply Chain Analytics, Venture Analytics, Corporate Finance, Pricing and Revenue Optimization



ACADEMIC QUALIFICATIONS			
Year	Degree/Board	University/Institution	%/CGPA
2019*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	--
2018	Financial Risk Manager (FRM Level I)	GARP (ID-594319)	Qualified (1st attempt)
2013	Post Graduate Diploma in Industrial Management	NITIE, Mumbai	7.65/10
2010	B.Tech. (Biological Sciences and Bioengineering)	IIT Kanpur	8.3/10
2005	Class XII, CBSE	St. Joseph's Co-Ed School, Bhopal	78%
2003	Class X, CBSE	St. Joseph's Co-Ed School, Bhopal	90%
ACADEMIC & CO-CURRICULAR AWARDS			
Department Rank- 8			
Scholastic Achievements	- Achieved All India Rank of 1764 out of 300000+ students in IITJEE (top 1 percentile) (2006-2007) - Conferred Certificate of High Distinction in Australian National Chemistry Olympiad (2003-2004) - Awarded Certificate of Merit in Social Science subject for 10th CBSE Board exam (2002-2003) - Granted Certificate of Distinctive Performance in 3rd National Science Olympiad (2001-2002)		
Co-curricular Achievements	- Conferred Bronze medal in Toxic Comment Classification challenge (top 7% of Kaggle competition) Finalist among top 5 teams of Nestle Plan-o-Chain - a Supply chain competition Finalist among top 7 teams of GSK Stratedge, West Zone finals- a Strategy competition - Achieved World rank-92 in Capsim Global Challenge- a global business simulation competition - Awarded Certificate in Management Consulting by Consultancy Development Center, Govt. of India		
WORK EXPERIENCE (44 months)			
Deputy Manager		State Bank of India	Hyderabad (Aug '14 – Jun '17)
Corporate Advances: General responsibilities	- Handled high value MSME advances (multiple crores) generating interest of ~12-14% per annum - Generated credit ratings (20+ clients) based on business, management, industry and financial analysis - Carried out loan documentation and disbursals as part of customer service and bank campaigns - Improved client servicing in trade finance business (with guidance of Trade Finance branch) - Ensured timely submission of stipulated performance reports like stock statements and monitoring accounts		
Corporate Advances: Initiatives	Trained new employees in creating credit ratings for various products like CC, term loans, LCs & BGs - Spearheaded adoption of internet and mobile banking at the branch (added 30+ corporate customers) - Focussed on risk management and fulfilled multiple corporate customer requests w.r.t. tax payments etc.		
General Banking	- Handled operations of the RBI Currency Chest (~200 crores), ATMs, branch accounts & cross/up selling - Mitigated complaints w.r.t. operations, internet and mobile banking (added ~10 customers/month)		
Operations Manager		RCI Logistics Pvt Ltd	Indore (June '13 – May '14)
Branch Operations	- Handled branch operations and Logistics by efficiently managing costs, revenue and customer satisfaction level - Coordinated healthy stakeholder's relationship through effective communication & supplier management		
DATA SCIENCE PROJECTS			
Toxic Comment Identification	- To improve online communication used multi label classification models to detect 6 types of toxic comments - Used n-grams, TF-IDF, GLOVE embeddings, Naïve Bayes-Logit on ~160k comments of Wiki's talk page - Performed stacking of GRU & Bidirectional LSTM to capture context of comments improving AUC to 0.987		
Stock Prediction: Time Series analysis	- To predict one day ahead stock prices utilized time series techniques of ARIMA, ARCH & GARCH - Performed analysis on past ~5 years daily data of Infosys stock which resulted in MAPE ~4% of prediction		
Home Credit Default Risk	- To predict probability of customers to repay home loans used classification on data with 200+ variables - Utilized cross section data on ~308k previous applications, bureau data and other socio-economic factors - Handled class imbalance with SMOTE and applied SVM, LightGBM classifiers to improve AUC to 0.8		
Start-up acquisition analysis	- Analysed important factors for start-up acquisition through Random Forest model on data of ~56k start-ups - Performed feature engineering and learnt business patterns related to geographic and funding dependencies		
Stock Prediction: Sentiment analysis	- To predict Stock movement used bag of words based Sentiment analysis on news headlines & tweets - Applied classification on ~8 years news data of NYT (Dow Jones) & ~3 years tweets data (Apple stock) - Achieved accuracy of 68% in predicting direction of movement using Random forest & SVM classifiers		
INDUSTRIAL PROJECT		Madura F&L Ltd	Bangalore (April '12 – June '12)
Order Fulfilment: B2C e-commerce	- Suggested redesign of warehouse layout and distribution for operations suited to new e-commerce channel - Created information flow model design, facilitating in reducing time (~31%) of pareto dominant activities - Evaluated cost/benefits of outsourcing vs in-house e-commerce logistics, created AHP for Logistics supplier		
TOOLS & LANGUAGES: R, Python, SQL, MS Excel, MS project, Hadoop (basics), MapReduce (basics)			
AUDITS & ELECTIVES: Complex networks, Information retrieval, Supply Chain Analytics, Cloud computing (NPTEL ongoing)			



ACADEMIC QUALIFICATIONS

Year	Degree/Board	University/Institution	%/CGPA
2019*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-
2014	B.Tech. (Electrical Engineering)	IIT(BHU) Varanasi	9.02/10
2008	Class XII, CBSE	Dewan Public School, Meerut	94.00%
2006	Class X, CBSE	S. Teja Singh Kandhari Public School, Patiala	91.67%

WORK EXPERIENCE (36 months)

Senior Applications Engineer		Oracle India Private Ltd.	Hyderabad (June'14 – June'17)
Solely executed the projects used by the internal customers that contributed to Build, Release and Integration of Oracle's business applications			
Build Analysis and Failure Detection	- Consolidated data from 4 different sources into Hadoop and used ElasticSearch to provide build success analysis - Built PoC for automatic remediation of real time build failures using build log analysis in Python and Hadoop Predicted build failures using real time analysis of keywords and patterns and alerted the right team for remediation - Ideated web dashboard for build analysis using ElasticSearch and Hadoop utilized by ~200 users across the team		
Conversational Bot	- Developed a bot for +1000 users to handle build related inquiries and perform operations using automated replies - Applied Document Categorizer for intent classification and Named Entity Recognition for finding entities - Utilized Java based Apache Open NLP toolkit and achieved an accuracy of 89% of all the utterances		
Mobile Applications	- Built secured mobile app for ~200 users to provide customized build alerts with notifications and cloud backend - Designed automation and integration framework for mobile app builds used by 6 teams (Team Size:20-100) Led 2 newly hired people, trained and guided 2 other teams for developing demo mobile applications for clients		

DATA SCIENCE COMPETITIONS AND ACADEMIC ACHIEVEMENTS

Data Science Competitions	Silver medalist for ranking 151/3634 in Data Science Bowl 2018-Nuclei Detection hosted on Kaggle Ranked 4th nationally, 40th globally in Data Science Game 2018 on Kaggle to build Bond Recommendation Engine Ranked 6th in Data and Beyond 2017 sponsored by American Express to design model for customer targeting Ranked 64th in Chest X-Rays Image Multiclass Classification Challenge 2018 by National Institute of Health, U.S Ranked 378/2384 in Mercari Price Suggestion Challenge 2018 on Kaggle using Word Embedding and GRU model Ranked 139/2951 in Click Prediction Data Science Hackathon 2017 to predict the user click conversion pay-out
Publication	Submitted paper for IEEE International Conference on Data Mining (ICDM) 2018 on Deep Reinforcement Learning for Derivative Pricing and Trading Strategy
Academic Achievements	Ranked AIR-3100 among +4.5 lac students in IIT-JEE 2010 and ranked 7th among 52 students in PGDBA Secured 90.32 percentile and ranked 934th globally in 2nd International Mathematics Olympiad 2008 Awarded scholarship by Department of Science and Technology India for scoring 100 in Mathematics in 12th - Certified in Java SE-Oracle WDP and Java EE Applications, SQL and PL/SQL-Oracle University Training Class
Technical Skills	R, Python, Java, SQL, MS-Excel ElasticSearch, Hadoop AngularJS, Spring Web MVC Framework

DATA SCIENCE PROJECTS

Predicting Company Acquisition	- Performed feature engineering to predict company acquisition using profile and news articles on Crunchbase - Implemented XG Boost model on basic, financial and managerial factors and achieved recall of 80.5% - Enhanced the model by applying Latent Dirichlet Allocation on news articles and achieved recall of 82%
Anomaly Detection	- Devised an approach to identify presence of Ransomware in computer using Hardware Performance Counters - Applied PCA (Recall:12%) and LSTM Variational Autoencoders (Recall:84%) on multivariate time series data - Achieved recall score of 98% after converting the data to frequency domain using Fourier Transform
Retail Sales Forecasting	- Forecasted sales for 99 departments across 45 Walmart stores in different regions for a horizon of 39 weeks - Performed data imputation using Kalman Filter and used ARIMA on different store-department combinations - Applied SVD for noise reduction and used Exponential Time Smoothing to reduce WMAE by 19.2%
Object Instance Segmentation: Nuclei Detection	- Developed model that can identify and segment range of nuclei across the cell images of varying sizes and modality - Trained Keras based Mask R-CNN with Resnet-101 and U-Net model for dataset of just 670 training images - Utilized transfer learning and performed image augmentation using morphological operations
Derivative Pricing and Trading Strategy	- Devised model to predict European Call option prices and trading strategy on Nifty dataset of 1 min frequency - Labelled actions using EMA Crossover System and applied LSTM (R2 score:0.981, Categorical Accuracy:0.664) - Improved model by using Deep Reinforcement Learning with LSTM (R2 score:0.985, Categorical Accuracy:0.704)
Opinion Spam Detection	- Identified candidate spammer groups from Yelp Reviews Dataset using Frequent Itemset Mining (Apriori) - Performed feature engineering to identify textual, behavioural and temporal indicators of group spam - Implemented Expectation Maximisation algorithm to rank the candidate groups on the basis of spamicity

EXTRA-CURRICULAR ACTIVITIES

Positions of Responsibility	Newsletter Committee Member at Oracle for team and was responsible for releasing monthly online newsletter Creative Head, PRASTUTP14 Electrical Engineering Fest in IIT-BHU and coordinated various creative activities Organizing Team Member at Oracle for one-day informal event for team as part of bootcamp training programme
Extracurricular Activities	Stood 1st in Edifice 2012 Paper Presentation by Emerson, Stood 3rd in Goldberg's Alley, Technex 2011, IIT-BHU - Conducted Tech Talks and organized several Team Building activities at Oracle for the team

ELECTIVES: Information Retrieval, Selected Aspects of Predictive Modelling, Derivatives and Risk Management, Supply Chain Analytics, Corporate Finance, Pricing and Revenue Optimisation

ACADEMIC QUALIFICATIONS

Year	Degree/Board	University/Institution	%/CGPA
2018*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-
2016	B.Tech (Instrumentation Engineering)	IIT Kharagpur	7.67/10
2012	Class XII, BIE-AP	Sri Chaitanya Junior College, Vijayawada	95.4%
2009	Class X, SSC	International School, Hyderabad	86.66%

DATA SCIENCE COMPETITIONS & ACHIEVEMENTS

Academic Achievements	- Secured Department Rank 1 in IIT Kharagpur semester of PGDBA focused on Machine Learning 99.5 percentile in AIEEE'12 among 11L applicants & Top 1.4 percentile in IIT-JEE'12 among 5L applicants
BrainWaves Hackathon (Soc Gén)	- One of Top 6 Finalists, 2nd in final LB among 9600+ in Machine Learning Challenge by Société Générale - Predicted annualized returns of portfolios based on security and market data; achieved R^2 value 0.99 - Predicted the probability of a transaction being fraudulent using Random Forest with an AUC of 0.73 - Built a Credit Approval model for Equipment Finance using customer financials & creditreforms data - Constructed a Data Extraction model to retrieve trade details from world-wide broker transaction PDFs
Mckinsey Hackathon	Top 2 percentile among 5000+ participants in Analytics Hackathon conducted on Analytics Vidhya - Predicted the customer's propensity to renew the policy given demographics, income & renewal history Optimized agent's incentive plan to increase propensity & maximize net revenue collected from policies
ZS Data Science Challenge	Top 1.8 percentile among 5000+ participants in Data Science Challenge conducted on HackerEarth - Forecasted sales for 11 product-country combinations using Exponential Smoothing & Holt's Linear Trend

WORK EXPERIENCE

Software Engineer, R&D Team	ARC Document Solutions	Kolkata(Jul '16 – Jun '17)
R&D Image Processing	- Developed "Skysite", a proprietary tool to digitize Architectural Drawings as part of dedicated research team - Pre-processed parts of Architectural Engineering Drawings to generate clean & "OCR friendly" images Devised a new algorithm resulting in 15% improved accuracy as compared to the conventional approach - Algorithm involved Hough transformation, Contours, Raster Scanning, Coded in Python (OpenCV)	
Open Source Team	- Part of a specialty handpicked four member task-team named "Front-end Open Source" team - Responsible for migrating existing front-end services to a single page web application (AngularJS) - Developed a dynamic end user interface to facilitate document uploads & display task progress - Implemented S3 client side uploader library to handle uploads for faster, secure and reliable data handling	

DATA SCIENCE PROJECTS

Quora Question Pairs	- Identified duplicate question pairs on Quora platform to enhance user experience for writers and seekers - Extracted text features - tf-idf, distance similarities on pre-processed data and built XGBoost model - Attained AUC of 0.876 by using GloVe word embeddings with LSTM - Siameese Recurrent Architecture
Dynamics of Social Movement	- Scraped & cleaned tweets for over 600k users to create a dataset on #blacklivesmatter movement - Constructed a retweet-mention graph to identify authorities and hubs based on Jaccard Similarity metric - Clustered the similarity graph by Newman-Girvan and Louvain methods to find user characteristics - Used t-SNE to find influential bots in user accounts with tf-idf of user description & time based features
Electricity Load Forecasting	- Forecasted hourly load demand for a month using 5-year data to plan power availability in advance - Adjusted for multiple seasonality to reach stationarity; Analyzed ACF/PACF plots to fit ARMA model
Travel Time Prediction	- Modelled intra-city travel time using GPS location data of 300+ car drivers captured over 2 months - Performed feature engineering to remove data inconsistencies using geodesic distance and speed - Represented travel-path by sequence of grids for a stacked model involving RForest, XGBoost & SVR
Credit Risk Default	- Predicted the Default Risk of underserved loan applicants based on their credit history and personal details - Engineered relevant features and implemented LightGBM model to identify loan defaulters

OTHER ACTIVITIES

Extracurricular Achievements	- Awardee of the Best Illumination Cup for outstanding contribution to the Inter Hall event, IIT Kgp - Trained as Cadet for 2 years at (3) Bengal Tech Air Squadron National Cadet Corp(NCC)
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TOOLS & LANGUAGES : Python, R, Open CV, AWS S3, MS Excel, MATLAB, SQL

ACADEMIC COURSES: Supply Chain Management, Corporate Finance, Advanced Predictive Modelling

ACADEMIC QUALIFICATIONS				
Year	Degree/Board	University/Institution	%/CGPA	
2019*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-	
2016	B.Tech (Chemical Engineering)	IIT Guwahati	7.88/10	
2012	Class XII	CBSE	95%	
2010	Class X	CBSE	9.2/10	
AWARDS & ACHIEVEMENTS				
Data Science Game '18	- AIR 4 in Data Science Game - Bond Prediction Challenge sponsored by Microsoft and BNP Paribas - Built ensemble of models to predict the propensity of a customer to buy/sell bonds based on historical data			
Data Science Bowl '18	Silver Medallist (top 5%) among 3634 teams in DSB sponsored by Booz Allen Hamilton and NVIDIA - Implemented state-of-the-art deep-learning techniques to detect and segment nuclei in medical images			
International Competitions	Kaggle Competition Expert – ranked in top 3% among 83547 competing Kagglers around the world			'18
	Bronze Medallist in Kaggle Fraud Click Detection among 4000+ participants organised by TalkingData			'18
	Secured World rank 18, AIR 5 amongst ~450 teams in Capsim Global Business Strategy Challenge			'18
National Data-Science Competitions	1st Rank in Data Tales-National house price prediction competition organised by Great Lakes Institute			'18
	1st Rank among 1018 participants in Data Supremacy product cross-sell challenge organised by AV			'18
	2nd Runner-Up among 1142 participants in Data Identity Challenge organised by Analytics Vidhya			'18
	AIR 7th in Student Hunt 2017 Resource Allocation - Footfall Prediction Challenge organised by AV			'17
	AIR 8th in Data Hackathon for Prediction of Credit Risk organised by NMIMS, Hyderabad			'18
	National Finalist among 9600 participants for Brainwaves hackathon organised by Société Générale			'18
Scholastic Achievements	Top 2.1% in IIT-JEE among 470,000+ entrants & Top 0.6% in AIEEE among 1,100,000+ entrants			'12
	Top 1% in AISSCE Examination, 2012 and was eligible for the Inspire scholarship from CBSE, India			'12
WORK EXPERIENCE		Virtusa Polaris	Aug '16 – June '17	
Software Developer	- Built telecom network optimisation system with a scalable map interface to enable network planning - Developed a GIS application with 25+ functionalities for a major telecom conglomerate in the UK - Delivered a telecom network visualisation tool for the network planners using AngularJS and D3.js - Implemented the Open-Source-First strategy to transform legacy systems leading to reduction in costs - Drove product meetings and gathered business requirements from client in form of HLD & LLD documents			
DATA SCIENCE PROJECTS				
Detection of Click Fraud in Online	- Analysed a dataset of 200 million clicks and built a fraud detection system for pay-per-click online ads - Generated features from the device, IP address and channel details using Target Encoding and aggregation - Obtained an AUC of 0.982 with Word-Batch, Follow Regularized Leader and Factorisation Machines			
Object Instance Segmentation: Nuclei Detection	- Developed a model that can identify and segment range of nuclei across cell images of varying sizes and modality - Trained Keras based Mask R-CNN with Resnet-101 and U-Net model for dataset of just 670 training images - Utilized transfer learning and performed pre-processing using morphological operations			
Time Series Forecasting of Retail Sales	- Forecasted sales for 99 departments across 45 Walmart stores in different regions for a horizon of 39 weeks - Performed data imputation using Kalman Filter & used ARIMA on different store-department combinations - Applied SVD for noise reduction and used Exponential Time Smoothing to reduce WMAE by 19.2%			
Prediction of Customer Interest in Online Ads	- Predicted the customer propensity to click an ad using it's description, images, historical features and price - Extracted text-based features using TF-IDF, Word2Vec and clustered users using Topic Modelling - LDA - Ensembled models built using LightGBM, CatBoost and Ridge Regression using out-of-fold stacking			
Recommender System in Retail	- Built Recommender System for a retail firm to predict the user ratings for products using their past ratings - Implemented Collaborative Filtering, Singular Value Decomposition and Factorisation Machines			
Fake Review Detection	- Fake review groups were obtained using Frequent Itemset Mining–Apriori method from Yelp dataset - Identified behavioural & temporal features and used GS-Rank algorithm to predicted group spamicity scores			
INTERNSHIP		Summer Research Fellow	May '15-July '15	
IISC Bangalore	Modelled high lifetime Lithium-ion batteries using COMSOL and experimentally found the open-loop potential			
POSITIONS OF RESPONSIBILITY				
E-Summit, IIT-G	Team Lead: Led a team to build corporate relations with organisations and industries in North-East			'15
Department Fest	Marketing Executive : Acquired major sponsors for Reflux, Chemical Engineering Fest, IIT Guwahati			'14
E-Cell, IIT-G	Led Member: Conceptualized working plan for the year including club activities, lectures & workshops			'16
EXTRA-CURICULARS AND COURSES				
Extra-Curricular	Awarded the “Best Actor” and “KalaPrathibha” titles for the best performer in Higher Secondary level			
	Semi Finalist among 40+ teams in Red Bull sponsored Neymar Jr. futsal tournament at IIT Kharagpur			
	District Rank 1 in General Quiz Competition organised by Oisca International, South India Chapter			
Courses	Information Retrieval, Supply Chain Analytics, Pattern Recognition and ML, Optimisation Techniques			
Technical Skills :- R, Python, Java, SQL, MS-Excel, Tableau, AngularJS, D3.JS, OpenLayers				

ACADEMIC QUALIFICATIONS

Year	Degree/Board	University/Institution	%/CGPA
2017-19*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-
2009-13	B.Tech (Telecommunication)	PES Institute of Technology, Bangalore	7.5/10
2008	Class XII, CBSE	Chinmaya Vidyalaya, Bokaro	74.00%
2006	Class X, CBSE	DAV Public School, Patna	90.80%

WORK EXPERIENCE (33 months)

Programmer Analyst	Cognizant	Pune (Oct '13 – Jul '16)
Business Development	- Generated empirical insights on Pricing dynamics based on the data collected from Web harvesting - Designed reports using Tableau & Excel for visualisation of Price position across e-commerce websites - Operated on management of applications related to card payments and fraudulent bank transactions Improved business efficiency by 40% by analysing and migrating applications on latest Linux servers Reduced application downtime by 50% by improving quality control metrics & project monitoring tool - Performed business analysis functions incorporating requirement gathering, analysis and validation	
Leadership and Mentoring	Batch Representative of 20 new trainees; Designed and conducted 20+ training sessions for the batch - Conducted 12+ deployments in production system; Supervised team of 4 developers for 5 applications - Facilitated teaching and mentoring of 50 underprivileged children as a part of CSR activities in 2016	
Awards and Achievements	- Awarded Cognizant Rising star for consecutive quarters for delivering total 4 projects before deadline - Received Cognizant Trendsetter for outstanding work and extending peer support to Analytics team - Accorded quarterly Cognizant Pillar of the month award for delivering 3 zero defect projects timely - Received a note of appreciation by client for exemplary personal contribution among team of 10 people	

DATA SCIENCE PROJECTS

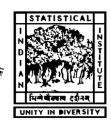
Yelp Photo Classification (Deep Learning)	- Built a Deep learning model using 2,34,000+ Yelp images that tags 2000 restaurants with multiple labels - Classified multi-class, multi-label image dataset which suggests tags to users when uploading images - Extracted features using pre-trained CNN architectures such as Inception V3, VGG16 and ResNet50
Movie Recommender	- Trained the model on Movie Lens dataset of 26 million ratings given by 270000 users to 45000 movies - Designed Hybrid Recommender by combining SVD and Restricted Boltzmann machines algorithm - Handled cold start problem of new users/ movies; used both Content based and collaborative filtering
YouTube Comment Classification (Centrality, LSTM)	Scraped, preprocessed & annotated YouTube comments from 6 different channels into 12 categories - Captured different sub-categories via LSTM, N-gram & other Machine learning classification models - Leveraged centrality measures & community detection algorithm to extract features from word graph
Time Series Modelling	- Analyzed most profitable market segments from the given data & devised ARIMA model for the same - Crafted a robust Linear modelling techniques to smoothen & remove trends from the time series data - Implemented Holt Winters Forecasting and concluded which market segment is the most profitable
Sales Forecasting (Kaggle)	- Modelled Sales forecasting for a large grocery retailer with 125 million observations & 16% NA values - Developed Xgboost, Median Average Algorithm using golden ratio & Exponential time series model - Designed weighted ensemble of Xgboost and glmnet algorithms to reduce Normalized RMSE to 0.11
Spatio-temporal prediction of link travel times	Predicted time taken by a taxi driver to cover a route between given latitude & longitude coordinates - Utilized an ensemble of Random Forest and LightGBM algorithms to build the time forecasting model - Employed travel time data of Cabs from a Cab aggregator app for trips within London from 2014 - 2017

AWARDS & ACTIVITIES

Academic Awards	- Secured 1st position out of 30 participants in Technical Quiz at Annual Inter-college cultural fest, 2012 - Accorded Distinction in a class of 120 students by the Director during Engineering for years 2011-13 - Consistently ranked among the Top 3 students in a class of 60 for six consecutive years from 2000-06 - Received scholarship for 3 consecutive year for scoring highest marks in Mathematics for year 2003-06
Positions of Responsibility	Coordinated convocation ceremony of PGDBA 2015-17 students as a core team member of Alumni cell In charge of administrative work during the final placement of PGDBA 2016-18 batch at ISI Kolkata Core member of Initiative for Community Action club who mentor underprivileged children at IIMC Active member of Corporate Social Responsibility group named OUTREACH in Cognizant (2013-17)
Extracurricular Achievements	Among Top 4% in Data Supremacy, Student Datafest - Analytics Vidhya: Utilized LightGBM model - Received certificate of appreciation from Lions Club International for humanitarian work in year 2010

TOOLS & LANGUAGES : R, Python, SQL, Visualization in Tableau, MS Excel

ELECTIVES : Decision-Games, Complex Network, Pricing & Revenue Optimization, Supply chain Analytics



ACADEMIC QUALIFICATIONS			
Year	Degree/Board	University/Institution	%/CGPA
2019*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-
2012	B.Tech (Electrical and Electronics Engineering)	National Institute of Technology Surathkal	7.79/10
2007	Class XII, CBSE	DAV Public School, Patna	83%
2005	Class X, CBSE	DAV Public School, Patna	90%
AWARDS AND ACHIEVEMNTS			
Analytics Related Achievements	▪ 1st Rank among 9600 teams in predicting Fraudulent transaction by Societe Generale; Offered Internship ▪ AIR 5, World Rank 47 in Data Science Game 2018 on Bond Recommendation by Kaggle & BNP Paribas ▪ 17th Rank out of 877 participants in forecasting Credit Risk of applicants by NMIMS Hyderabad 2018 ▪ 14th Rank among 261 teams in preliminary round & finalist in House Price Prediction Challenge 2018 ▪ 10th Rank among 129 teams, predicting the criticality of accident by IIT Roorkee,Data Science Hackathon		
Examinations	▪ Ranked among top 1 percentile in AIEEE 2007 (6.45 lac candidates) and GATE 2011(EE, 72K candidates) ▪ Awarded Merit Certificate by DAV Management committee for performance in CBSE Board Exams 2005		
WORK EXPERIENCE - 46 Months			
Product Developer		Freshworks, Chennai (Formerly Freshdesk)	22 Months (Aug'15 – May'17)
Product Management	▪ Awarded emerging star for building application to capture Net Promoter Score as SaaS offering ▪ Designed the application, UX flows; did market research & sizing, & competitive analysis of the offering		
Engineering	▪ Developed browser based plugin free Co-Browsing solution for customer support; 1000+ Customers		
Engagement Analyst	▪ Analysed customer data to generate prioritized Product features list; Strategized implementation,pricing ▪ Built automated system with chat data to generate targeted list of customers for Up-sell and Cross-sell		
Software Engineer		1Click.io, Bangalore	24 Months (Aug'13 – July'15)
Core member – played a key role in development and marketing of 1Click.io; Acquihired by Freshworks			
Swiss Knife Go-to-market	▪ Introduced SaaS from API only business model, Customer acquisition time reduced from months to hrs ▪ Analysed chat/call records to design service plans, optimise resources & forecast future requirements ▪ Increased signup rate by 50% through Google Ad-words campaigns of INR 1 million for the startup ▪ Pushed the page rank of company website/blogs to top through in-depth Search Engine Optimization		
Core Engineering	▪ Full Stack; developed dashboard for Customers Support, Agents management; Integrated payment ▪ Developed Chat infrastructure using Tornado websocket with Redis queue system; served 1mn+ msg/hr ▪ Built streaming platform to cover live events; Covered IIM Bangalore startup events, AMA sessions		
DATA SCIENCE PROJECTS			
Restaurants Photo Classification (Deep Learning)	▪ Built an Image classifier by training 2.34lac+ images from 2000 restaurants to label restaurant attribute ▪ Multi-label Multi-class problem – restaurants on Yelp fall into multiple categories from total 9 classes ▪ Extracted features using pre-trained CNN architectures – Inception V3, VGG16 and ResNet 50		
Grocery Sales Forecasting (Time Series)	▪ Forecasted sales for 15 days for a large grocery chain; 125mn observations, 200K products, 54 stores ▪ Implemented ARIMA and Exponential Time series models to achieve Normalized RMSLE of 0.555 ▪ Improved the score to 0.512 after applying Median Average Algorithm using golden ratio		
Movie Recommendation	▪ Built Recommendation engine on MovieLens-26M dataset to suggest unseen movies to the user ▪ Trained the hybrid model based on SVD and RBM on 26 million ratings by 270K users to 45K movies ▪ Handled cold start problem of new users/movies; used both Content based and collaborative filtering		
Youtube Comment Classification (Sentiment Analysis)	▪ Developed a machine learning model to classify YouTube comments into 12 sentiment categories ▪ Scraped, preprocessed and labelled 10,000+ user comments to prepare the dataset & performed EDA ▪ Leveraged graph theory & community detection algo on top of Boosting Algorithms for classification		
Usefulness of Yelp Reviews (NLP)	▪ Built model to rate usefulness of new reviews by training on old user reviews on the Yelp dataset ▪ Used Centrality and Term-Document Matrix to generate features and implemented XGBoost and GBM ▪ Improved results by 12% by leveraging LSTM on vector representation of text via Word2Vec		
Taxi time prediction (ETA Prediction)	▪ Predicted time taken by a taxi driver to cover routes between given latitude & longitude coordinates ▪ Utilized ensemble of Random Forest and LightGBM algorithms to build the model and find ETA		
EXTRACURRICULAR ACTIVITIES			
Positions of Responsibility	▪ Joint Convener, Cultural Festival NIT Surathkal; Organized Concerts and Fashion show; 25000+ footfall ▪ Alumni Cell: Co-ordinated the convocation of 2015-17 Batch of PGDBA at IIM Calcutta - 2018 ▪ HeadBoy, DAV Patna for 2006-07. Focused on safety, sports; Managed 25 member team for Youth Festival		
Extracurricular Achievements	▪ 1st rank in Bihar state among 10000+ students, Horlicks WizKids 2006, All-India quizzing competition ▪ 2nd prize in Inter-college skit, NIT Surathkal Cultural Fest; 1st prize in Inter-DAV skit among 58 Schools ▪ 2nd place in Foxhunt, Built Radio receptor for competition organized in NIT Surathkal Technical Festival ▪ Captain of the school handball team 2004-2005; Won Gold at handball tournament in Intra-School Event		
TOOLS & LANGUAGES : R, Python, SQL, Visualization in Tableau, MS Excel, Javascript, Django, Tornado, Angularjs, HTML, CSS			
ELECTIVES : Supply Chain Analytics, Product Analytics, Complex network, Pricing and Revenue Optimization, Decision & Games			

ACADEMIC QUALIFICATIONS

Year	Degree/Board	University/Institution	%/CGPA
2019*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-
2018*	CFA Level III Candidate	CFA Institute, VA, USA	-
2014	B.Tech (Chemical Engineering)	Indian Institute of Technology Guwahati	7.44/10
2010	Class XII, BIE-AP	Sri Chaitanya Educational Institutions	90.20%
2008	Class X, BSE-AP	Dr. KKR's Gowtham Concept School	88.50%

DATA SCIENCE COMPETITIONS

Data Science Game 2018 India Rank 1 World Rank 9	- Only team to represent India in the World Data Science Game Finals, Paris sponsored by BNP Paribas - Predicted customer interest to buy/sell a bond in the coming week leveraging past trading & market data Clustered Bonds & Users, engineered 200+ features, reduced feature space dimension to 80 using Boruta Out-performed BNP baseline AUC by ensembling multiple base models from LGBM, Xgboost & Random Forest
Brainwaves Challenge 2018 #Finalist	- One of Top 6 Finalists & 2nd in final leaderboard among 9600+ in Brainwaves Challenge by Société Générale - Predicted Annualized Returns of portfolios with diverse instruments using security and market data - Predicted the probability of a transaction being Fraudulent for 1 mil+ transactions using Gradient Boosting - Built a Credit Approval model for Equipment Finance dept using customer financials & Creditreforms data - Built a Data Extraction model to retrieve trade details from transaction PDFs sent by brokers across the globe
Miscellaneous	- World Rank 18 and All India Rank 5 among 460+ teams in CapSim Global Business Strategy Challenge Ranked 8th among 4000+ in ZS Data Scientist Challenge for Event Prediction and Sentiment Analysis Ranked 13th in Data Tales Beyond Infinity ML challenge by Great Lakes for predicting real estate prices

WORK EXPERIENCE

35 Months

Manager, Centre of Excellence	Reliance Industries Ltd.	Mumbai (Jul '14 – May '17)
Part of Central Olefin Technology group providing technical, operations, production & business support to all Petchem BU's across RIL		
Process Optimization	- Continuously Optimized process units using plant sensory data & models to remove production bottlenecks - Developed a BU wide Optimization model to maximize profit based on daily market prices & constraints.	
Risk Assessment	- Developed a dynamic Simulation model of the entire BU and performed stress testing & scenario analysis - Identified and reported potential risks with current standards and mitigated them by updating SOP's and SOC's.	
Business Intelligence	- Identified and modelled KPI's pertaining to operations, production, business and safety for the entire BU - Responsible for deployment of real-time interactive dashboard with over 100+ indicators adopted by all levels.	
CapEx Projects	- Identified various CapEx projects from GAP & Solomon reports and evaluated them by Cost-Benefit Analysis - Part of ~ \$1.5 Billion Ethane project, revamping entire BU; Involved in Planning, Design and Execution	
Highlights	- Achieved revenue increment of Rs. 110 Cr/yr. by increasing production of High Value Chemicals by 11.5% - Awarded Best Team in entire RTG & Tech-CoE dept in Reliance Annual Rewards and Recognition 2016-17	

DATA SCIENCE PROJECTS

Energy Load Forecasting	Forecasted electricity demand for next 7 days of an individual house to ease integration with Smart grids - Fitted an ARIMA model on stationary data after decomposing, removing Trend & Seasonality gave MAPE of 17% - Implementing Standard LSTM and Seq to Seq LSTM with encoder-decoder architecture improved MAPE to 9%
Portfolio Design Using Deep Learning	Demonstrated a robust portfolio construction process using Deep Auto-encoders to learn non-linearity in returns - Created Market-map for each portfolio through a 4-step routine of encode, calibrate, validate and verify - Used regularization with Elastic-net and Dropout to tune out-of-sample error similar to risk-return trade-off - Created an Efficient Deep Frontier with subset of stocks for re-tracking and beating NASDAQ IBB and NIFTY
Hybrid Recommender Systems	- Built Hybrid Recommender models on movielens dataset to give customized movie suggestions to users - Stacked various Latent factor Models based on MF, combining strengths of Collaborative & Content features - Fitted and ensembled Ordinal Logit Models based on similarities and proximities of users and movies
Information Retrieval (IR)	- Designed a Ranked IR system that retrieves documents from corpus for any given query based on TF-IDF Improved accuracy of the IR system by Updating and Expanding query with Rocchio algo and Word2vec Summarized the documents using Text-Rank & Degree Centrality and evaluated them using ROUGE scores

ACHIEVEMENTS & EXTRA-CURRICULAR ACTIVITIES

Academic	Certified in Deep Learning Specialization on Coursera, focusing on application of various Deep Learning models - All India Rank 3048 (99.32 percentile) in IIT-JEE 2010; All India Rank 1636 (99.87 percentile) in AIEEE 2010
Extra - Curricular	- Active member of National Cadet Corps; participated in several camps and trekking expeditions Organized various National level gaming events during Cultural & Technical festivals at IIT Guwahati

TOOLS & LANGUAGES: R, Python, IBM SPSS, MATLAB, C, SQL, Tableau, MS Office, MapReduce, Bloomberg Terminal

ELECTIVES: Corporate Finance, Advanced Predictive Modelling, Pricing & Revenue Optimization, Supply Chain Analytics

ACADEMIC QUALIFICATIONS

Year	Degree/Board	University/Institution	%/CGPA
2019*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-
2014	B. Tech (Electronics and Communication)	NIT Silchar	7.78/10
2010	Class XII, AHSEC(Assam)	Cotton College, Guwahati	69.2%
2008	Class X, SEBA(Assam)	St. Stephen's School, Guwahati	87.33%

ACADEMIC DISTINCTIONS & CO-CURRICULAR ACHIEVEMENTS

International Data Science Competitions	- Certified "Competitions Expert" by Kaggle and ranked in Top 3% among 83k+ competitors globally Won Silver medal among 3967 teams in Kaggle "Talking Data Ad Tracking Fraud Detection" Challenge AIR 7 and World Rank 57 in the Data Science Game sponsored by Microsoft and BNP Paribas	'18 '18 '18
National Data Science Competitions	Won Data Play, a Forecasting challenge among 150 teams and awarded with INR 20k by KPMG at IIM Calcutta Ranked 2nd in 'Credit Card Recommendation' challenge sponsored by American Express at IIM Bangalore Ranked 2nd in a month-long Stock market trading challenge sponsored by ICICI Direct at IIM Calcutta Ranked 2nd in Sumantran - The Hunt for Star Consultant and won prize money of INR 15k at IIT Kharagpur Top 20 among 9600 teams to qualify for the final round of Brain Waves organized by Société Générale Top 2% among 353 teams and awarded with cash prize in Stat Wars statistics competition at ISI Kolkata	'18 '17 '17 '18 '18 '18
Scholarships & Awards	- Certified Lean Six Sigma Black Belt by KPMG for successfully completing trainings and projects - Awarded GOI scholarship worth INR 53.6k for exceptional academic performance in AIEEE under NEC - Awarded with merit-based scholarship of INR 15k for exemplary performance in Class X from Govt. of Assam	'18 '11 '08

WORK EXPERIENCE

35 months

Analytics Consultant	Brillio	Bengaluru (Dec'15 – Jun '17)
"Best Employee Award" for Excellence in work for 2 consecutive Quarters		
Campaign Management	- Conceptualised campaign management framework for Fortune 500 client and achieved 15% QoQ growth in revenue - Achieved 40% QoQ growth in marketing pipeline by performing customer segmentation using RFM analysis - Co-ordinated with 50+ sales representatives and reduced lead follow up time from 1 week to 3 business days	
Sales Dashboard (CPG)	Led a team of 3 analysts & performed SKU level analysis of secondary sales data for 52 area sales managers in India - Developed dashboard in Power BI enabling the sales team to get stock updates and reduced turnaround time by 95%	
HR Analytics	- Scraped Glassdoor reviews for Brillio and its competitors using Selenium and analysed them using NLP techniques - Performed competitor analysis and recommended actionable strategies to address function level attrition rate	
Business Analyst	TEG Analytics	Bengaluru (Jun '14 – Oct '15)
Market Mix Modelling	- Achieved an estimated lift of 40% by optimising marketing budget for multiple brands using Linear Regression - Automated data pre-processing using VBA script, decreasing human efforts from ~24 man-hours to 5 minutes	
Inventory Management	Single-handedly developed inventory management dashboard using Tableau to monitor KPIs to prevent stock-out - Awarded "Employee of the Month" by client and designed POCs to portray core competencies of the firm	

DATA SCIENCE PROJECTS

Bond Trading Prediction	- Built a model to predict probability of customer transacting a bond in the following week using trading and market data - Generated negative samples; engineered features and blended Random Forest, XGBoost to achieve AUC of 0.6 - Extracted latent features of customer-bond interactions using SVD to improve AUC of the model by 15%
Sales Forecasting (Retail)	- Forecasted 2 weeks of daily store sales using promotions, competitor and sales data to develop effective staff schedules - Analyzed seasonality by STL decomposition, stationarity by ADF tests and ARMA orders by ACF-PACF plots - Implemented ARIMA model to achieve RMSPE of 0.15 and reduced RMSPE to 0.13 using TSLM in R
Credit Risk Modelling	- Predicted the propensity of a customer default using previous loan, POS and Credit Bureau data with 220+ features - Used PCA for dimension reduction and handled class imbalance using SMOTE as sampling technique - Ensembled Logistic regression, Random Forest and LightGBM models to improve AUC from 0.72 to 0.80
Comment Classification (Deep Learning)	- Identified negative sentiments from unstructured data using Wikipedia page edits to facilitate healthy conversation - Compared Logistic regression, Naïve Bayes, SVM models using features extracted from TF-IDF and n-grams - Applied LSTM architecture with GloVe word embeddings and achieved AUC of 0.98; won Bronze medal in Kaggle
Social Media Analytics	- Analyzed reasons for loss of followers using 12GB dataset of web crawled user tweets at two different time points - Created user-user network based on topical similarity of contents; applied Louvain algorithm to detect communities - Built SVM model using graph and content-based features to predict early follower loss and achieved F1 score of 0.80

EXTRACURRICULARS & POSITIONS OF RESPONSIBILITY

Social Initiatives	- Volunteered for an NGO - Samarthanam and taught 250+ underprivileged students and recorded audio books - Developed a handbook for underprivileged to help in employment opportunities for an NGO, Dream a Dream	'18 '18
IIM Calcutta	Member, INCA club: Taught underprivileged children and coordinated with NGO's for organizing live projects Member, Marketing club: Responsible for handling marketing events in Experimental marketing association	'18 '18

TOOLS & SKILL SETS: Languages: SAS, Python, R, SQL | Visualization: Tableau, Power BI | Interest: Equity Investing

ACADEMIC QUALIFICATIONS

Course/Examination	Institution/University	Year	Performance
Post Graduate Diploma in Business Analytics	IIM Calcutta/ISI Kolkata/IIT Kharagpur	2017-2019*	-
B.Tech (Electronics and Instrumentation)	Institute of Engineering & Technology (IET), Lucknow	2013-2017	63.56%
Class XII, CBSE	Kendriya Vidyalaya, IIT Kanpur	2012	88.6%
Class X, CBSE	Kendriya Vidyalaya, IIT Kanpur	2010	9.4

AWARDS AND POSITIONS OF RESPONSIBILITY

National Competitions	- Winner of Industrial Design Case competition among 16 finalists by Indian Institute of Chemical Engineers - Finished in Top 3% among 1018 participants in DataHack Student Supremacy organised by Analytics Vidhya - Selected for Yes Bank Future Ready Scholarship as pre-finalist among 3000+ students from top Indian B-schools Top 14% out of 5000+ participants in Store Demand Forecasting Hackathon organized by ZS Associates	'15 '18 '18 '18
President, Alumni Cell	- Organized the silver jubilee edition of Annual Alumni Meet, managed the event within a budget of INR 8 Lakhs - Initiated a mentorship program for 1000+ students, organizing 10+ seminars and workshops by industry experts	'15 '16
Captain, IET Cricket Team	- Led a 16-member team at various inter-college sports events to win 1 Gold & 1 Silver medal at IIM Lucknow - Mentored 50+ students and conducted team selection trials of 480 students across 10 disciplines in the Institute	'15 '16
Institute Events	Doubled the number of participating colleges through promotional events as part of Core Committee, Sports Fest Sponsorship Head; Secured sponsorship of INR 10 lakhs from Coca Cola and KTM for the Cultural Fest - Drafted and conducted an auction-based gaming league at ISI Kolkata with 100% participation from the batch	'16 '16 '17

DATA SCIENCE PROJECTS

Financial Risk Modelling	- Predicted the propensity of a bank customer to default in next two years based on their financial credit history - Handled a class imbalance ratio of 100, leveraging SMOTE and ENN sampling techniques to generate synthetic data - Performed feature selection using Boruta and achieved an AUC of 0.82 by Logistic Regression, SVM and ANN - Implemented Hierarchical Clustering to segment customers and analysed them to understand defaulting patterns	
Opinion/Fact Classification (Microsoft)	- Classified manually annotated phrases into facts and opinions from the news articles in MPQA Opinion Corpus - Extracted features from topic modelling by Latent Dirichlet Allocation and community detection algorithms - Ensembled GRU, LSTM and SVM models with N-gram features to improve the accuracy from 71% to 76%	
Shopper's Cart Prediction	- Predicted the list of items in a user's next order with 3 Million historical transactional data for 200,000 users - Generated 20+ features capturing user, item and user-item buying behaviour to improve the accuracy of the model - Performed Market Basket Analysis using association rules and achieved F1 score of 0.39 by LightGBM classifier	
Recommender System (BNP Paribas)	- Predicted probability of a customer-bond interaction in the next week using past interactions and macro-market data - Synthesised data and ensembled Random Forest, GBM and XGBoost models to achieve an AUC score of 0.62 - Applied Collaborative Filtering and LightFM models to extract additional features and improved AUC score by 9.6%	
Company Performance Prediction	- Designed an interactive dashboard using Shiny and R to present a list of potentially investable retail firms - Applied Factor Analysis and K-means clustering on financial ratios of the firm in the last 5 years to identify KPIs - Utilized Random Forest for predictive modelling of company performance based on each identified indicator	
Stock Price Forecasting	- Predicted the next day closing price of Infosys stock using 5 years of historical data sourced from Bloomberg - Modeled log returns of stock using ARIMA and further improved MAPE (4%) by capturing volatility with GARCH	

INTERNSHIP EXPERIENCE

Summer Intern	Defence Research and Development Organization (DRDO)	Kanpur (Jun' 2016 – Jul' 2016)
Embedded Systems	- Implemented an Automatic temperature-controlled cooling fan with different operating levels using microcontrollers - Incorporated conditional functions using C/C++ to control the operating levels with variations in temperature	
Research Intern	Bhabha Atomic Research Centre (BARC)	Mumbai (Jun' 2015 – Jul' 2015)
Real Time Data Reporting	- Developed SCADA interface which pulled live data from sensors and actuators attached to the Nuclear Recycling Plant - Engineered new live switches and alarms features on the SCADA interface to remotely control the process mechanism	
Start-up Intern	YRC Weblink	Lucknow (Jun' 2014 – Jul' 2014)
Business Development	- Proposed low-cost features such as child protection and on-screen keyboard for Weblink, a touch-screen browser - Increased the downloads of the improved browser to 250,000+ through third-party websites with a rating of 3.9/5	

EXTRACURRICULAR ACHIEVEMENTS

Sports	Active member of IIM Calcutta's cricket team; Won an aggregate of 1 gold and 1 silver at inter college events	
Learning & Development	- Delivered a 4-hour lecture series on Quantitative aptitude at Unacademy, India's largest online learning platform - Selected among 150 applicants, as a member of Design team in MAC, Marketing association of IIM Calcutta	
Social Initiatives	Taught 200+ underprivileged children for entrance tests in government schools under the Social Club of IET - Among 50 students out of 8000 participants to represent IIM Calcutta for Airtel's social initiative "Run for Education"	

TOOLS AND LANGUAGES: R, Python, SQL, IBM SPSS Modeler, Shiny, Tableau, Google Cloud

ELECTIVES: Financial Risk Management, Computational Finance, Corporate Finance, Econometrics, Decision and Game Theory



Roles Offered : a) Machine Learning
b) Operations and Technology Settlements
c) Data Scientist – Intern

Job Description :

a) Machine Learning:

Overall Purpose: Responsible for owning, defining, implementing and operationalizing data products at scale

Key Accountabilities

- Support delivery of one or more data science use cases, leading on data discovery and model building activities
- Conceptualize and quickly build POC on new product ideas - should be willing to work as an individual contributor
- Open to learn, implement newer tools\products
- Experiment & identify best methods\techniques, algorithms for analytical problems
- Operationalize – Work closely with the engineering, infrastructure, service management and business teams to operationalize use cases

Stakeholder Management and Leadership

- Mentor and guide junior machine learning colleagues
- Liaise and work closely with business areas, product owners, project managers, business and data analysts
- Communicate results and key findings to the stakeholders in a clear, consistent and business friendly manner

Decision-making and Problem Solving

- Should be able to look at problems from different dimensions and use judgement based on past experiences, market research, industry specific solutions to choose the right approach for building data products
- Should be able to decide and look for alternatives if the current approach does not work

Risk and Control Objective

All Barclays colleagues have to ensure that all activities and duties are carried out in full compliance with regulatory requirements, Enterprise Risk Management Framework and internal Barclays Policies and Policy Standards

Education

Masters degree or PhD in statistics, mathematics, computer science or related scientific disciplines.

Relevant work experience in some or all of the skills below

Technical skills

- Experience working with large data sets in order to extract business insights or build predictive models
- Broad knowledge of applied mathematics (probability or statistics, linear algebra)
- Proficiency in one or more statistical tools/languages – Python, Scala, R or SAS and related packages like Pandas, SciPy/Scikit-learn, NumPy etc.
- Hands on experience and understanding of Big Data ecosystem with technologies like Spark/PySpark, MapReduce, Kafka, Hive etc. with unix scripting. Knowledge around NoSQL technologies would be an added advantage.

- Hands-on experience running in running various methods like Regression, Random forest, k-NN, k-Means, boosted trees, SVM, Neural Network, text mining, NLP, statistical modelling, data mining, exploratory data analysis, statistics (hypothesis testing, descriptive statistics)
- Hands of experience and knowledge around feature engineering dimension reduction and model optimization using gradient decent, PCA etc.
- Industry experience in building and operationalizing various machine and deep learning models in finance/other domain would be an added advantage

Soft Skills

- Ability to work with a wide range of stakeholders and convert abstract ideas into actionable requirements
- Ability to work with ambiguous situations and uncertainty
- Out of the box thinking to solve real world problems
- Should be able to juggle between multiple priorities efficiently
- Work collaboratively with teams across different geographic locations and contribute to Machine Learning capability.
- Good presentation skills – Should be able to present findings and insights in a manner understandable to business

b) Operations and Technology Settlements(OTS)

- The Operations and Technology Settlements team in the Investment Banking area of Barclays is in the process of developing a lot of strategic components in line with its target state architecture. These components include
- Delphi – An award winning exception management system built using Angular, Tableau, Mongo DB. Need help here to build in some intelligence using Machine Learning.
- UTIF –This component will act as single platform to parse, validate, transform and route the trades/transactions to sub-ledger systems in real time basis. This component is being built using micro service architecture and hosted on cloud using Open shift as container platform.
- RPA/IPA – Automation of various manual repetitive tasks using Robotics platforms like blue prism. Build inhouse components which give OCR, NLP capabilities.
- Operations Dashboard – A single dashboard for the entire Operations area. Use data analytics capabilities to provide different views/reports to the users.
- The candidates will be assigned to one of the above projects based on their strengths. Experience in any of the above areas/technologies which be preferred.

c) Data Scientist – Intern

- We are looking for a data scientist who is able to learn fast on the job, solve complex problem with a pragmatic approach and carefully balance his / her skillsets between science, engineering and security expertise. As a data security engineer within Fusion, you will have the chance to bring innovative ideas live to our users as well as solving more technical challenges in integrating products, tools and APIs, hence directly contributing to an intelligence driven defence.
- You will be responsible for writing production python based APIs to make intelligence actionable
- You will be working with most senior resources to implement complex Machine Learning (ML) workflows in production
- You will be responsible for reverse engineering Python scripts and notebooks
- You will be responsible for extracting re-usable code to generic libraries and Machine Learning (ML) workflows
- You will be responsible for writing Artificial Intelligence (AI) -enabled chatbot consumable by a web application

Essential Skills/Basic Qualifications:

- Strong knowledge in either Python or Scala
- Experience in Continuous integration / unit testing
- Knowledgeable in big data ecosystem and distributed programming
- Understand data science, ML experience a ++ (can explain basic concepts)
- Understand AI and have a pragmatic approach to chatbot implementation

- Internship Stipend Offer : 75000 INR
- Shortlist steps: a)CV shortlisting
b)Technical Round
- Duration : 30- 40 minutes.

Technical Round Questions(Anshul Singhal - OTS):

Questions were mostly on the finance side as it was role related to having some knowledge of finance ,analysing the data and building models using Machine Learning Techniques. Question were :

- 1) What do investment banks do?
- 2) What do you mean by arbitrage and different types of arbitrage ?
- 3)Questions on a case that was on settlements and what should be the predictor variables if a model should be build?
- 4) Should use a simple machine learning model or any deep learning model?
- 5) Drawbacks and advantages of deep learning over simple machine learning model?
- 6)Question on a project of stock market prediction and asked if comfortable with ocr?(Said no knowledge of ocr but did project in which used twitter sentiments to predict stock market movement. Ready to learn stuff related to reading of pdf documents)

Courses and Skills Suggested:

- Knowledge related to finance(Qualified FRM Level 1) , knowledge of basic deep learning (andrew ng) , CDS(ISI)
- Inference, SSD, CDS-ISI
- Time Series and Regression, Complex Network-IIT
- Corporate Finance, Selective aspects of advanced predictive modelling, Game theory-IIM

Suggestions:

Learn the basic stuff that is more important (ISI). This is very important and companies test the person as to how he/she can bring a business impact to the problem at hand. It does not matter as to how you come up to the solution(technique) but at the end of the day in some cases it should be explainable and have business impact.

- So while preparing for all analytics and stuff you should also prepare for how to approach a problem and to look for business impact it creates. They only look for the thought process and how you approach a certain case given.

- Internship Stipend Offer (ML):
- Shortlist steps: a)case study
b)Technical Round
- Duration : 15-20minutes.

Technical Round Questions(Tanmay Choudhary– ML):

1. Questions were mostly from case study and dsg.
2. What is the flip side of e-commerce data.
3. How would you weight data, model and hypothesis in machine learning.
4. How would you react in case of conflict with a senior.
5. Your teammate is about to give ppt in 15-20 mins but has to leave for an emergency and asked you to give his ppt. What will you do

Technical Round Questions(Ravi– ML):

1. Should Barclays use speech and text for sentiment analysis of customer interaction?
2. How would you do that? How will you collect data for that?
3. Give an instance where correlation between variables came out to be different from what you normally expect.`
4. How would you update a model if new data comes in. would you just add the data or build the model from scratch?
5. Should you access your friends laptop if something urgent is required, given you have his permission?
6. Ans is NO, talk to senior management in such situation.
7. Have you ever gone beyond the expected duties during your workex
8. How do you settle issues within your team when there is a conflict of ideas.
9. Have you ever done something that your management was not confident about. How did you convince them to do it?

Technical Round Questions(Munshi Javed Akhtar):

1. How would you build a product recommender system for a bank? What are the features that are most important for this case and why?
2. Dimensionality reduction techniques (PCA, Factor Analysis) and TSNE (How it works?)
3. Variational Auto encoder (In the context of Mule account detection - This was a question from the case given by them)

In the same round (along with tech round), by another panelist(HR QUESTIONS)

1. What do you want to achieve in three years?
2. What is respect in your view and whom do you respect?
3. What are the incentives required for you to work giving your best?

Technical Round Questions(Ankit Jaiswal):

Only one round and questions were a mix of HR and Technical.

1. Why Barclays
2. Explain one of your project
3. Biggest difficulty in implementing it
4. How much were you confident of implementing deep learning into finance prior and post project
5. How much integrity is valuable to you.
6. Any explanations regarding roles offered

Technical Round Questions(Bhagyashree):

1. Why did you chose this problem statement(3 cases were given)?
2. How to represent data visually in 3D containing million rows and hundreds of columns?
3. Explain project where you used deep learning. Things which you did in your previous company is still carry forwarded as legacy?
4. Why finance?
5. Which is your dream company?
6. HR round was not taken separately but my interview was mostly based on general questions from previous work ex.

Other Technical Questions

Questions related to projects and Completions' and HR questions like why Barclays and dream company which you want to work for?

Courses and Skills Suggested:

Munshi Javed Akhtar

Skills : Recommender systems (PCA, SVD, Collaborative, Content Based filtering etc)

NLP with RNN (and other DL techniques)

Image Processing (If you have projects or company is focused towards this)

Reinforcement learning

Courses: ISI - SSD, CDS (ISLR - thoroughly) , IIT - FADML (Deep learning Part)

Ankit Jaiswal

Skills: Technical- Linear Regression, Time Series, R and Python programming ,Puzzles, SQL

Non-Technical- Structured Communication: Able to explain your projects in a logical flow

Courses: Computational Data Science(CDS) ISI

Regression and Time Series Modelling (RTSM) IIT

Selective Aspects of Predictive Modelling IIM

Tanmay Choudhary

Courses : CDS, Inference, FADML

Ravi

Skills: Python, SQL, communication skills

Courses: CDS, Inference, DBMS, RTSM, Information Retrieval

Bhagyashree

- CDS, Probability, Inference, SQL
- NLP, Deep learning though not covered explicitly in any courses should be prepared well in case you have projects on it. Barclays definitely ask question on deep learning

Suggestions:

1. Be strong at coding and SQL queries. Keep in mind the business application of every technique you study. Be clear with concepts and applications.
2. Be thorough with the projects you have done and all the subjects taught in ISI
3. CDS (read ISLR thoroughly) knowledge will help you in FADML, which will in turn help in Deep Learning.
4. Also focus on RTSM (taught in IIT)
5. Prepare your projects well and business applications of it.
6. Study well all techniques mentioned in resume and prepare questions like why did you chose that, what else could be used in place of that etc.

ACADEMIC QUALIFICATIONS

Year	Degree/Board	University/Institution	%/CGPA
2019*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-
2014	B.E. (Electronics and Instrumentation)	SGSITS Indore	7.03/10
2009	Class XII, CBSE	Bhavan's Prominent, Indore	73.80%
2007	Class X, CBSE	Bhavan's Prominent, Indore	83.66%

INTERNATIONAL COMPETITIONS

Data Science Game, 2018 (World Rank 9, AIR 1)	- Only team to represent India in International Data Science Game at Paris, sponsored by BNP Paribas - Recommended bonds to a customer which they are most likely to buy or sell in the next one week - Leveraged transactional, historical price, market, customer and bond data to engineer 80+ useful features Ensembled multiple base models to improve predictions; Out-performed BNP Paribas baseline AUC
Capsim Strategy Challenge	- Secured World rank 18 amongst 450+ teams in Capsim Global Business Strategy Challenge, Spring 2018 - Formulated strategy across 5 functional areas for multi-million dollar company in a simulated environment

WORK EXPERIENCE (36 months)

Senior System Engineer	Infosys Limited	Bangalore (June '14 – June '17)
Roles and Responsibilities	- Provided Production and Development support to one of the world's largest semiconductor chip maker - Handled migration of 1 million documents from an enterprise product to an open source-based solution - Facilitated collaboration among different teams across 5 global data centres and managed queue timelines	
Impact	- Spearheaded a 30-member team on ticket escalation issues and achieved reduction of 26 man-hours daily Saved 60 man-hours weekly by automating technical processes under Infosys 'Zero Distance Program'	
Recognitions	- Received letter of appreciation from client for consistent on-time task planning, execution and delivery - Awarded Team of the Month under Cloud and Infrastructure division for exemplary performance	

DATA SCIENCE PROJECTS

Product Recommendation System	- Predicted the list of items a user will add to the cart in the next order using 3 million orders for 20K users - Engineered 20+ features using collaborative filtering and captured users and item buying behaviour - Generated association rules using Apriori algorithm and achieved F1 score of 0.39 with boosting classifier
Restaurant Visitor Forecasting	- Predicted number of visitors for 829 restaurants using their past one-year reservation and visit data - Modelled seasonal ARIMA, LSTM and generated 300+ features by rolling window to capture seasonality - Achieved better performance using boosted tree model with RMLSE 0.5 in comparison to 0.8 with LSTM
Portfolio Optimization	- Formulated a trading strategy to build an optimal stock portfolio using statistical arbitrage techniques - Identified pairs of co-integrated stocks using Euclidean distance and Augmented Dickey-Fuller test - Optimized trading threshold with Hidden Markov Model and generated portfolio return of 4.7%
Predicting drunk texting using Tweets	- Engineered text-based features on supervised hashtag-based datasets to identify drunk/sober tweets - Analyzed interest-based communities by Topic Modelling and Louvain community detection algorithm - Achieved AUC score of 0.92 with SVM and improved it to 0.95 using model based on LSTM architecture
Start-up funding Prediction	- Predicted propensity of a start-up project being crowd funded on Kickstarter using 300K projects - Carried out geographical and temporal analysis of successful start-ups and compared category wise result - Generated actionable insights for investors and entrepreneurs with 88% accuracy using Random Forest

EXTRA CURRICULAR ACTIVITIES

Sports	- Secured Bronze medal in Inter-State and Silver medal in Inter-District Kick-Boxing tournament - Bagged Silver medal in Table Tennis amongst 50 participants at Inter-Sectional Sports Meet, IIM Calcutta - Semi Finalist among 40+ teams in Red Bull sponsored Neymar Jr. Futsal tournament at IIT Kharagpur - Won 2 Gold and 2 Silver medals in Football, Cricket and Kho-Kho, at Inter-Branch tournament, SGSITS
Positions of Responsibilities	- Organized 'Graffiti-2k12', certified by Limca Book of Records as longest-running (36 hrs) wall painting initiative Core Committee member of 'Aayam 2k13', annual college cultural fest; managed a budget of INR 4 lacs - Organized 2 college level arts and photography themed competitions on creative and innovative concepts

TECHNICAL SKILLS

Languages/Tools	R, Python, SQL, Microsoft Excel, .NET, Tableau
Electives	Pricing and Revenue Optimization, Corporate Finance, Complex Network, Supply Chain Analytics, Selective Aspects of Advanced Predictive Modelling, Venture Analytics

ACADEMICS

Degree/Board	Year	University/Institution	%/CGPA
Post Graduate Diploma in Business Analytics	2017-19*	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-
B.E. (Electronics & Telecommunication Engineering)	2013-17	IEST, Shibpur	7.28/10
Class XII, WBCHSE	2013	Howrah Zilla School	83.80%
Class X, WBBSE	2011	Howrah Zilla School	90.50%

ACHIEVEMENTS

Scholastic Achievements	- Awarded scholarship from Ministry of Human Resource Development, throughout 4 years of bachelors '13-17 - Selected for the prestigious INSPIRE scholarship from MHRD for scoring 99.46 %ile in 12th Board '13 - Awarded scholarship by Chief Minister (WB) for scoring 99.97 %ile in 10th Board among ~1M candidates '11 - Scored 98%ile in 3rd International Mathematics Olympiad (State Rank 32, International rank 826) '09 - Certified star candidate of National Science Society (Science Talent Search Exam) for two consecutive years '08-09
Competitions/ Actuarial	- Winner of Mathemania, a math contest organised by INSTRUO - the technical festival of IEST, Shibpur '16 Student member of Institute of Actuaries of India, Cleared Actuarial Common Entrance Test (ACET) '18

POSITIONS OF RESPONSIBILITY

Public Relations	- Member of the PGDBA PR team; Guided candidates for interviews & assisted with smooth admission process - Published articles for PGDBA blog & press releases; Increased social media presence of PGDBA by 40%
Tutor (Chegg.com)	Assisted 39 candidates on 44 lessons with different subjects of computer sciences and management studies - Delivered 11 sessions on Business Cases, Marketing & Finance and rest on miscellaneous Comp. & Data Sc. topics

DATA SCIENCE PROJECTS

Toxic Comment Classification	- Developed an NLP model to facilitate effective online conversations by minimizing threats of abuse and harassment - Preserved contextual information of comments employing advanced embedding techniques (word2vec & GloVe) - Applied Attention mechanism on Bidirectional-LSTM & GRU layers to capture decisive features in the sentences - Obtained cross-validated accuracy of 98.3% & AUC score of 0.9867 using LightGBM on top of the classifiers
Detecting WannaCry Ransomware	- Explored effects of ransomware on normal system behaviors with dataset of infected machine's CPU parameters - Built Intrusion Detection System with LSTM based Variational Autoencoder to identify attack within 4 seconds - Applied Fast Fourier Transform on time domain values of different hardware events to reduce false positive rate
Derivative Pricing & Trading Strategy	- Devised model to predict European Call option prices & trading strategies on 1 minute frequency Nifty dataset - Used Exponential Moving Average Crossover technique to derive the optimal trading strategies for training - Improved overall payoff by 10% using Deep Reinforcement Learning with LSTM; Submitted for ICDM 2018
Time series modeling of Nifty stocks	- Forecasted conditional returns, volatility & correlations of 5 Nifty stocks using historical data of 1 day frequency - Leveraged techniques: ARIMA for returns, three GARCH variants for volatility and DCC-GARCH for covariance - Tuned model using RMSE, direction & sign accuracy criteria; Performed portfolio Sharpe ratio maximization
Stock Price Prediction using Complex Network Theories	- Demonstrated a network theoretical approach to forecast S&P500 stock prices using 1 hour interval market data - Calculated Relative Strength using Mutual Information Matrix of all stock pairs to obtain strong interactions - Represented the market as weighted graph & extracted communities of similar stocks using Louvain algorithm - Improved MSE of best fitted ARIMA model by about 20%, adding several network measurement parameters
Investment Recommendation of Indian Retail Firms	- Developed an interactive company ratings dashboard (ShinyR) to present list of potential firms for investment - Employed factor analysis to identify KPIs and k-means clustering to bundle companies based on their financial ratios - Compiled a list of prospective Indian retail firms as investment options based on the investors' risk appetite
Image Background Segmentation	- Developed a CNN based Encoder-Decoder architecture (U-Net) to automatically remove image background - Applied Otsu thresholding for foreground detection & trained models at different resolutions of augmented images - Post-Processed images with Erosion, Dilation & Binary Hole Filling to obtain F1 score of 0.995 on test data
Santander Product Recommendation	- Built a recommender system serving 1 million customers by suggesting the most relevant products (7 out of 24) - Methods Implemented: Content based filtering, Collaborative filtering & Context Aware Recommender Systems - Applied Hierarchical Clustering, Logistic Regression, Naïve Bayes (OneVsRest) classifiers with XGBoost

INTERNSHIP/UG PROJECT

Video Inpainting	- Interned at 2nd Summer School on Computer Vision, Graphics and Image Processing, ISI Kolkata '15 - Employed different techniques of video processing to remove an unwanted moving object from a short video
Metamaterial absorber Design	- Designed an electromagnetic absorber in micrometer range using High Frequency Structure Simulation '17 - Demonstrated an ultra-broadband, polarization-insensitive structure that could absorb frequencies of THz range

PROFICIENCIES

- Languages & Tools:** R, Python, C, SQL, Visualization in Tableau, IBM SPSS Modeler, Minitab, Excel, MS Project, MapReduce, MATLAB
- Relevant Coursework:** Computing in Data Science, Regression & Time Series Modelling, Strategic Management, Decisions & Games, Corporate Finance, Computational Finance, Categorical Data Analysis, Sentiment Analysis, Advanced Predictive Modelling



ACADEMIC QUALIFICATIONS			
Year	Degree/Board	University/Institution	%/CGPA
2017-19*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-
2018	Financial Risk Manager (FRM) Level 1	GARP ID: 673248	Qualified (1st attempt)
2013-17	B. Tech (Electronics and Comm. Engineering)	Jamia Millia Islamia, Delhi	8.52/10
2012-13	Class XII, CBSE	Apeejay School, Delhi	90%
2009-10	Class X, CBSE	Apeejay School, Delhi	9.4/10
ACADEMIC ACHIEVEMENTS			
Analytics Competitions	- Placed among Top 3% of 4700+ participants in Kaggle’s House Prices: Advanced Regression challenge - Finished among Top 3% out of 1018 participants in Data Supremacy, 2018 on Analytics Vidhya - Ranked among Top 9% out of 1142 participants in Data Identity competition, a month-long hackathon - Ranked 1st out of 608 participants in Python for Data Science skill test, 2018 on Analytics Vidhya - Secured a rank in Top 3% out of 418 participants in Statistics and Probability skill test, 2018		
Scholastic Achievements	- Scored in 1st quartile in 2 sections of FRM: i) Quantitative Analysis ii) Financial Markets and Products - Placed among Top 2.8% in the Joint Entrance Engineering (JEE) mains exam conducted in 2013 - Scored a perfect 10 CGPA in Mathematics and Science in Class X Boards Examination conducted in 2010		
DATA SCIENCE COMPETITIONS			
Housing Price Prediction	- Engineered features related to plot size, neighbourhood, amenities to augment 79 exploratory variables - Imputed missing values and applied Box-Cox Transformation for highly skewed predictor variables - Improved RMSLE by stacking base models Lasso, Ridge, XGBoost with meta regressor LightGBM		
Employee Attrition Rate	- Predicted interest of employee in switching job based on qualification, experience and company size - Performed feature engineering using univariate, bivariate analysis and mean-encoded variables - Applied 5-fold Cross Validation on training data, tuned LightGBM parameters to get test AUC of 68.5%		
DATA SCIENCE PROJECTS			
Product Recommendation (E-commerce)	- Predicted propensity of a customer buying an item based on sequence of click and buy events - Handled data of 30M clicks, 9M users, 53K items of an e-commerce website spanning over 6 months - Performed Exploratory Data Analysis and engineered 14 features on user, item and time attributes - Implemented Factorisation Machines and used Collaborative Filtering to recommend similar items		
Portfolio Optimization	- Applied Market Neutral Pair Trading Strategy to come up with an optimal portfolio of stocks - Obtained top five pairs of stocks based on Sum of Square Distances and Co-integration approach - Employed Hidden Markov Model to obtain sequence of hidden states, corresponding transition matrix - Used transition matrix to simulate time series and maximised returns based on optimal threshold		
Stock Market Prediction	- Predicted bi-directional movement of stock price using Apple’s historical stock price and twitter data - Performed Sentiment analysis on Apple’s tweets on features extracted from Bag of Words and Bi-gram - Augmented ARIMA and GARCH with sentiment models, boosting accuracy from 67.3% to 75.2%		
Crowd-Funding Prediction on Kickstarter	- Predicted probability of a start-up being successfully crowd funded on Kickstarter using 300K projects - Identified important variables related to geographical and amount funded using Lasso Regularization - Implemented Random Forest and ensembled with LightGBM to improve accuracy from 87% to 91.3%		
Ransomware Detection	- Implemented an anomaly detection algorithm for time series data obtained from registers in processors - Employed LSTM architecture based Autoencoder on sequenced data to analyse reconstruction error - Selected threshold value for identification of ransomware by setting confidence interval of 99.7%		
SUMMER TRAINING AND B.TECH PROJECT			
Delhi Metro Rail Corporation	- Evaluated the security and induced redundancy issues of data being transmitted using Fibre Optics - Acquired thorough understanding of TVM, AFC gate technologies used to increase operational efficiency		
Wireless Weather Station	- Developed prototype for low-cost wireless weather station to gauge temperature, humidity, luminosity - Equipped with a safety alert feature activating buzzer on breach of sensor’s critical threshold		
EXTRA-CURRICULAR ACTIVITIES			
Content Writing	- Drafted proposal for PGDBA’s first analytics summit including Data Science related hackathon - Featured in letters to editor column in editorial section of renowned newspapers The Statesman, The Hindu and Telegraph on variety of topics ‘Obama-Sharif meet’, ‘Fighters of the sky’		
MOOC	Deeplearning.ai by Andrew Ng, A to Z in Deep learning and Machine learning with hands on training		
TOOLS/LANGUAGES: R, Python, SQL, Microsoft Office, Tableau, IBM-SPSS Modeler, Google Cloud Platform			
ELECTIVES: Corporate Finance, Decision & Games, Complex Network, Supply Chain Analytics, Venture Analytics			

ACADEMIC QUALIFICATIONS			
Year	Degree/Board	University/Institution	%/CGPA
2019*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-
2014	B.Tech. (Electrical Engineering)	NIT Allahabad	8.38/10
2010	Class XII, CBSE	G.N. National Public School	80.4%
2008	Class X, ICSE	Little Flower School	85.0%
AWARDS & POSITIONS OF RESPONSIBILITY			
Publications & Distinctions	- Published a research paper on ‘Model Order Reduction’ in IFAC collated by Science Direct [link] ‘14 - Presented a research paper on Control Systems in an International Conference (ACODS) at IITK ‘14 - Awarded merit certificate for 393rd rank in final round of International Mathematics Olympiad, SOF ‘10 - Awarded for securing 24th state rank in IOM; Currently ranked 18th among 52 students of PGDBA 17-19		
Extra-Curricular Achievements	- Won an aggregate of 2 Gold and 2 Silver medals in inter-college and inter school football competitions ‘14 - Bagged silver medals in department event at Avishkar (Annual tech fest) and in FUTSAL tournament ‘12 - Semi Finalist among 40+ teams in Neymar Jr. Futsal tournament sponsored by Red Bull at IIT KGP ‘18		
Positions of Responsibility	- Member of Sports Committee: Organized various sports and cultural activities in HMEL township ‘16 - Organized the cricket championship Techno with a budget of 30k attracting participation of ~60 teams ‘14 - Event Organizer: Organized the department event Connections at Avishkar (Annual Tech Summit) ‘13 - Led the college football team as Vice-Captain, guiding the team to victory in annual college sports fest ‘12		
WORK EXPERIENCE (35 months)			
Engineer	HPCL Mittal Energy Ltd.		Bhatinda (June ‘14 – June ‘17)
Roles and Responsibilities	- Spearheaded a team of 7 members to monitor plant functioning at Utilities and Captive power plant - Formulated inventory control with indigenous spare development in SAP with follow-up on logistics - Executed projects to improve plant efficiency and evaluated technical feasibility of MOCs in Water Block - Conducted illumination survey of refinery; co-ordinated various audits as a part of risk assurance team - Initiated training modules on power systems and cable fault detection for colleagues and was later awarded		
Impact Achieved	- Led a team for INR 30 million project on cable routing of CT motors in major refinery turn-around - Officiated assembling of bus coupler in RODM which led to an aggregate cost saving of ~INR 4 million		
Professional Achievements	- Awarded thrice in Individual Category for Reliability Improvement and once in Quarterly Conclave - Selected for training on ‘Reliability improvement’ conducted by National Productivity Council, Delhi - Won Poster making competition, ranked 2nd in Safety slogan competition for 2 consecutive years in HMEL		
DATA SCIENCE PROJECTS			
Portfolio Design (Deep Learning)	- Demonstrated a robust portfolio construction process using Deep Autoencoders to unravel nonlinearities - Devised a four-step routine of encode, calibrate, validate & verify to pick bonds based on communality - Applied elastic-net regularization with dropout to tune error similar to Markovitz’s risk return trade-off - Built an efficient frontier with subset of stocks for re-tracking and beating NASDAQ, IBB and NIFTY		
Stock Movement Prediction	- Predicted the movement of the Apple stock’s closing price based on historic daily data using sentiments - Extracted salient features using bag of words to score sentiments of Tweets associated with Apple Stock - Ensembled Logistic Regression, XGBoost, Random Forest & SVM models to achieve accuracy 65.85%		
Anomaly Detection (WannaCry)	- Devised a methodology to detect WannaCry ransomware using 5 hardware performance statistics (HPCs) - Implemented an auto-encoder using LSTM architecture to detect anomaly within 5s with 99% confidence - Transformed data of an anomaly into frequency domain using FFT to classify SPEC from Ransomware		
Venture Analytics	- Predicted propensity of a start-up being crowd funded using data from Kickstarter repository at Kaggle - Leveraged Boruta to identify important predictors and obtained an accuracy of 87.2% using Random Forest - Recommended actionable insights for entrepreneurs and investors based on Exploratory Data Analysis		
Energy Load Forecasting	- Forecasted electricity load demand using individual house hold data to ease integration with smart grids - Fitted an ARIMA model on decomposed stationary data using STL; Applied DF test to confirm stationarity - Implemented Standard LSTM and Seq to Seq LSTM improving earlier MAPE of 17% (ARIMA) to 9%		
Credit Risk Modelling	- Predicted the tendency of a customer default using previous loan and Credit Bureau data with 220+ features - Applied PCA to reduce dimensions & used over sampling methods like SMOTE to deal class imbalance - Ensembled Logistic regression, Random Forest and Light GBM model to improve AUC from 0.72 to 0.80		
Document Summarization	- Summarized articles in 250 words using graph-based approaches such as Text Rank & Degree Centrality - Designed a Ranked IR system that retrieves documents from corpus for a given query based on TF-IDF		
TOOLS AND LANGUAGES: R, Python, SQL, IBM SPSS Modeler, Microsoft Office, SAP, Tableau			
ELECTIVES: Supply Chain Analytics, Corporate Finance, Information Retrieval, Pricing and Revenue Optimization			

ACADEMIC QUALIFICATIONS

Year	Degree/Board	University/Institution	%/CGPA
2019*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	
2014	B. Tech in Electrical Engineering	Veermata Jijabai Technological Institute, Mumbai	8.90/10
2010	Class XII, Maharashtra Board	Amolakchand Mahavidyalaya, Yavatmal	95.33 %
2008	Class X, Maharashtra Board	Rani Laxmibai Vidyalaya, Yavatmal	96.76 %

AWARDS AND ACHIEVEMENTS

Achievements	- Ranked 3rd among 77 students of Electrical Engineering, 9+ SGPA in 5 consecutive B.Tech. Semesters '14 - Ranked 2nd in HSC among 98871 candidates with cash reward of Rs.50000 and 'National Merit Award' '10 - Ranked 3rd in SSC Amaravati Board and accoladed with 'Dhirubhai Ambani SSC Merit Award' '08 - Bestowed with 'Amul Vidyashri Award' for securing 96.76% in SSC Board examination '08
Scholarships	- Conferred 'Dhirubhai Ambani Undergraduate Scholarship' of Rs.12750 annually for four years '10-14 - Awarded 'Voltas Scholarship' of Rs.12500 for scoring CGPA of 8.9/10 in B. Tech Electrical at VJTI '14
Competitions	Ranked in top 20% (4179 teams) in Deng-AI competition hosted by Driven data to predict dengue '17

WORK EXPERIENCE

34 Months

Electrical Site Engineer	Lodha Group, Mumbai	July 2014-May 2017
Received ' Certificate of Appreciation ' from EHS head for ensuring safe place to work at Lodha with 0 fatality & 0 hospitalization rate '16		
Project Management and Reporting	Handed over 9 storey Commercial tower to Customers pertaining to Schedule, Cost & Time constraint - Saved 3 months of Workhours with Effective Reporting, Quick Responsiveness & Resources Availability - Reduced 3 manhours per week by providing consolidated daily progress report for 18 towers, 27 storey	
Stakeholders Management	Institutionalized formats of Project Progress Reporting for Top Management decision making and review - Coordinated amongst 8 inter departments for timely deliverables related to material, services and design - Handled 14 vendors effectively as a client representative on daily basis through schedule trackers, MOM's Identified risk upfront & made contingency plans for labour shortage and late shipments using past record	
Material Management	Reduced manhours of 2 intermediaries by initiating process of issuing material online through SAP Estimated Resources required and availed them through timely Purchase or Service Requisitions on SAP	
Quality Control	Standardized Work Inspection Methods through check lists, change over requests & discrepancy memos	

DATA SCIENCE PROJECTS

August 2017-August 2018

Dengue Cases Prediction	- Developed a model to predict dengue cases in week with training cases of 34k+ spanning over 20 years - Trained model using Random Forest and Support Vector Regression on 24 environmental predictors - Achieved MAE of 25.9 using Random Forest on 5 years data; Among top 20% globally on leaderboard
Stock Movement Prediction	- Designed a model to predict DJIA Movement of 178 stocks using Financial News article & Stock data - Summarized 5 lakh Financial text news (Bloomberg) using Text Rank algorithm to generate Sentiments Boosted accuracy to 68% over econometric model like ARIMA by embedding sentiments with LSTM
Sales Prediction	- Predicted weekly sales at Walmart's 45 stores using non-tree based approach of Multivariate Regression Validated assumptions with regression analysis plots; Identified interaction effects and removed outliers - Evaluated model performance on validation set with MSPE of 0.059 explaining 85% of variance in data
Node Similarity using Graph Theory	Identified Friendship between 34 members of Zachary's Karate Club Network with Node Embeddings - Analysed node embeddings with Cosine Similarity measures to detect friendship link between members Scaled model from homogeneous to synthesized heterogeneous graph with 500+ nodes to find similarity
Spam Detection	- Built a robust Classifier to Distinguish spam messages by training 5572 raw text messages with labels - Exercised NLP techniques for Pre-processing along with Bag of Words and TF-IDF for feature extraction - Obtained accuracy of 97% on test dataset using Naive Bayes classifier & Support Vector Machine classifier
Usefulness of User Rating	- Devised a model to Predict Usefulness of User Rating by training Yelp Dataset with Tree-based methods - Garnered approaches of Graph Theory (Degree Centrality) and NLP to impute features in XGBoost model - Model to be utilized for enhancing User Experiences by accounting for critical reviews given by customers
Recommendation System*	Building a model to recommend Products with Highest Propensity to be purchased by 13.6MM customers of Santander Bank; Analysing possible models (like XGBoost) with different metrics like MAP@7, NDCG

POSITIONS OF RESPONSIBILITY

Public Relations PGDBA	- Contributed to enhance brand value of PGDBA programme by increasing followers of FB page to 2261 '18 Assisted candidates with their queries related to programme, exam structure & interview preparation '18
Sports & Cultural	Organized and Coordinated 2 Events at Sports and Cultural festival at VJTI for around 50 participants '13
CERTIFICATIONS	Python for Data Science and Machine Learning Bootcamp, Probability & Statistics authorized by Udemy
TOOLS	Languages-R, Python, SQL, C, C++; Tools- MS Excel, SAP, MATLAB, SPSS
INTERESTS	Supply Chain Analytics, Project Management, Predictive Modelling, Corporate Finance and PRO

ACADEMIC QUALIFICATIONS

Year	Degree/Board	University/Institution	% / CGPA
2019*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-
2016	B. Tech (Chemical Engineering)	IIT Bombay	6.99
2011	Class XII	CBSE	85.6%
2009	Class X	CBSE	94.2%

ACADEMIC AND EXTRA CURRICULAR ACHIEVEMENTS

Academic Achievements	- Awarded Bronze medal out of 3967 teams in Kaggle's 'Toxic Comment Classification Challenge' '18 - Ranked 10th out of 179 in Credit Card Recommendation Challenge organized by IIM Bangalore '18 - Among 162 out of ~1.7 lakh applicants to qualify the UPSC conducted SCRA written exam '12
Sports	- Awarded Roll of Honor for winning 7 medals in cricket, basketball, volleyball, hockey in IIT B '16 Captain, IIT Bombay Football Team: Led a team of 16 players in inter-college tournaments '15 1st Runner-up out of 160 teams in 3rd division league of Mumbai District Football Association '14 - Represented IIT Bombay in 50th Inter IIT Sports Meet as a member of institute football team '14

WORK EXPERIENCE (10 months)

Business Analyst	Deloitte TTI LLP	Mumbai (Jun '16 – Apr '17)
Lead Management System for an Automobile Manufacturer	- Generated monthly scorecards to rank car dealerships sales performance for a fortune 500 company - Extracted and transformed Salesforce data, created relevant variables, and analysed sales KPIs - Automated business report generation using VBA script, delivered more than 100 reports in a week - Optimized resource allocation by decreasing report development time from 1 man- hour to 1 min - Prepared business reports with analytical insights and actionable recommendations on sales practices	
Report Automation for Retail Chain	- Developed VBA script to create automated inventory reports for more than 700 SKUs sold by the client - Balanced resource allocation and project budget by on time generation of client ready reports	

DATA SCIENCE PROJECTS

Toxic Comment Classification	- Developed a multi- label classifier to detect negative sentiments using comments on online platform - Created features using Naïve Bayes log-count ratio, TF-IDF, word2vec, and GloVe word embeddings - Ensembled Logistic Regression, Naïve Bayes, and Bi-Directional LSTM and achieved an AUC of 0.98
Credit Card Recommendation	- Developed a model to target right customers with the right cards to improve lead conversion rate - Conceptualized a marketing campaign framework to maximize the return on \$1M marketing budget - Achieved 40% increase in expected return using ensemble of Logistic Regression and XGBoost models
Time Series Analysis	- Devised a time series model to forecast Infosys's next day stock price using historical data of 5-years - Performed ADF test for stationarity and analysed ACF-PACF plots to identify ARMA orders - Implemented ARIMA for modelling mean and ARCH for volatility, and achieved MAPE score of 6%
Home Credit Default Risk	- Predicted loan repayment ability of applicants using previous applications, POS, and Credit Bureau data - Used PCA for dimension reduction and handled class imbalance using SMOTE as sampling technique - Ensembled Logistic regression, Random Forest, and LightGBM to improve AUC from 0.72 to 0.80
Information Retrieval	- Summarized 125 articles in 250 words using Text Rank, Degree Centrality, and cosine similarity - Evaluated summary against reference summary using Precision, Recall, F1 score & Rouge N metrics - Designed a ranked IR system that retrieves documents from corpus for a set of queries based on TF-IDF
Start-up Acquisition Prediction	- Predicted acquisition status of U.S. based start-ups using Random Forest and Gradient Boosted Trees - Performed exploratory data analysis and selected features using Boruta and achieved acc. of 81.97%

POSITIONS OF RESPONSIBILITY

Institute Football Secretary	- Organized 50th Inter IIT sports meet with the participation of 16 IITs; managed budget of INR 0.5 MM - Proposed and consulted the construction of futsal arena in IITB Gymkhana worth INR 3 MM
Coordinator IITB Football League	- Led a two- tier team of 10 managers to organize IITB Football League for two consecutive years - Conceptualized use of online payment portal for registration which increased registration by 20%

TOOLS AND ELECTIVES

Tools / Languages	R, Python, VBA, SQL, Tableau, Google Cloud Platform, and Microsoft Office Suite
Electives	Supply Chain Analytics, Information Retrieval, Price and Revenue Optimization



Roles Offered: The intern will be attached to senior management at Corporate Centre, New Delhi

Job Description:

Qualification:

- Graduate degree in Engineering with Post Graduation in Business Analytics (full-time) with at least 60% marks
- Upper Age limit: 29 years

Job Profile:

- Mine, validate and document large sets of data of power business, competitors, energy/power sector and national economy
- Analyze data sets and draw insights, inferences and correlations as decision support system.
- Develop energy/power sector and competitor benchmark reports
- Design surveys and analyze survey data
- Design and monitor key performance indicators
- Regular generation of business analytics dashboards, visualizations and reports.

Location: New Delhi

- Internship Stipend Offer : 1,15,000 INR
- Shortlist steps: a)CV shortlisting
b)Technical Round
- Duration : 15- 20minutes

Technical Round Questions(V Adishesu Naik):

1. A case was given about the natural disaster occurs and the data given about the economic losses because of it how would you monetize the data?

Technical Round Questions(Arjun):

1. You have the credit card data transaction data of all the people in India. How can you create an Investment Opportunity out of it?
2. How to do Missing value imputation?

Technical Round Questions(Darshan Dhodapkar):

1. Mostly questions were asked on data preprocessing,
2. how can we use the machine data generated in NTPC for forecasting and what other inferences can be generated from this data

Courses and Skills Suggested

- Machine learning, NLP and a bit time series
- Data Preprocessing, Understanding of the ML algorithms used in projects and have some business sense of the projects in CV .

ACADEMIC QUALIFICATIONS			
Year	Degree/Board	University/Institution	%/CGPA
2018*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	--
2015	Dual Degree in Information Technology & MBA in HR Ops	IIITM Gwalior	6.25/10
2010	Class XII, BIE-AP	Sri Narayana Jr. College, Hyderabad	76.4%
2008	Class X, BSE-AP	Montessori High School, Kurnool, AP	79%
WORK EXPERIENCE			
Analyst, Transaction Management		The Royal Bank of Scotland Group	Chennai (Jan 25 – Dec 3, 2016)
Trade Compliance	- Part of compliance team that handled verification of trades with ticket size greater than \$10million - Performed initial segregation of 20K+ trades to detect potential money-laundering acts each month - Acted as an interface between client’s Investment broker and Anti Money Laundering/KYC analysts - Provided timely alerts to Investment brokers of loyal clients to renew AML/KYC nearing expiry date - Majorly involved in compliance check to park Foreign Exchange and Money Market transactions - Leveraged ODBC on the central database to facilitate data pre-processing of clientele classification		
Certifications	Certified in Foreign Account Tax Compliance Act & Basic Banking by The Royal Bank of Scotland		
Management Trainee Intern		EISAI Pharma technology Pvt Ltd,	Visakhapatnam (May ’12- June ’12)
Marketing Intern	- Strategized the process architecture to minimize transit time between post manufacturing operations - Built a Shortest Expiry Date First scheduler that established real-time consignment load-time tracking		
POSITIONS OF RESPONSIBILITY			
Founder & President, Rotaract IIITM (2013-15)	- Supervised a team to organize the health camp in Adivasi area and treat the diseases they are facing - Organized motivational and personal skill development workshops for juveniles in Borstel school - Hired capable members from college; provided training sessions and conducted education camps in remote areas as a part of Literacy Development Program under the guidance of Rotary club, Gwalior - Organized Health check-up & Blood donation camps, awareness on alcohol addiction in remote areas		
Hospitality Head	- Lead the Hospitality team of National techno fest in IIITM to facilitate all the needs of participants ’13 - Strategized and coordinated events during College Fests with around 5000+ of students participation		
DATA SCIENCE PROJECTS			
Walmart Sales Forecasting (kaggle)	- Devised a model to predict the weekly sales for each department across 45 different Walmart stores - Pre-processed the data by transforming it to remove heteroscedasticity for a better prediction - Implemented multivariate regression model and predicted the weekly sales with an accuracy of 85%		
DengAI – Driven Data Competition	- Predicted weekly Dengue cases for a city based on environmental variables, vegetation, humidity etc.. - Devised a Regression analysis to predict total Dengue cases considering after imputing missing value - Improved accuracy by Implementing SVR and Random forest & performing Hyper-parameter tuning		
Deep Portfolio	- Crafted a robust portfolio construction using Deep Auto-encoders to learn non-linearity in returns - Created Market-map for each portfolio through a 4-step routine of encode, calibrate, validate & verify - Regularization with elastic-net & dropout to tune out-of-sample error similar to risk-return trade-off - Created efficient deep frontier with subset of stocks for re-tracking & beating NASDAQ IBB & NIFTY		
Ratings Prediction for Twitter Review	Objective: To assign probable ratings (1-5) to live tweets & posts during electronic product launch - Leveraged 1 Million Amazon electronic product review-ratings to train a FM model & a LSTM model - Demonstrated both models using 50 sample tweets on electronics with 80% & 92% 1-step accuracies		
Predict Future Sales (Kaggle)	- Predicted the weekly sales of 200k shop-item combinations by ensembling LGB & Catboost Regressor - Improved RMSE by implementing LSTM after taking each store-item as independent time series		
Stock Price Prediction (Network Theories)	- Demonstrated an approach to forecast future price changes of S&P500 stocks using networks science - Calculated relative strength of interactions between stock pairs using Mutual Information Matrix - Used Louvain algorithm to extract communities of similar stocks in each timestamp, from stock data - Improved MSE of best fitted ARIMA model by about 20%, adding network measurement parameters		
IT Related Projects		B.Tech and MBA Thesis	IITM, Gwalior
Parkinson Gait Recognition	- Built an Android Application for Parkinson Gait Recognition using Soft Computing Tech. - Used Artificial Neural Network (Back Propagation Algorithm), Results using Genetic Algorithm		
MBA Thesis	- Worked on Organizational Citizenship Behavior & Knowledge Sharing Behavior - Used SPSS to work on the data collected by the students that contains behavioral questionnaire		
TOOLS & LANGUAGES : R, Python, SQL, MS Excel			
ELECTIVES : Computational Finance, Corporate Finance, Pricing and Revenue Optimization			

ACADEMIC QUALIFICATIONS

Year	Degree/Board	University/Institution	%/CGPA
2019*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-
2013	B.Tech (Electrical and Electronics Engineering)	National Institute of Technology, Hamirpur	7.48/10
2009	Class XII, BIE-AP	Srichaitanya Junior Kalasala, Gajuwaka	94.20%
2007	Class X, BSE-AP	Sri Saraswathi Vidya Vihar, Gajuwaka	90.16%

DATA SCIENCE PROJECTS

Stock Price Prediction	- Demonstrated a network theory approach to forecast S&P500 stock prices using 1 hour interval data - Calculated Relative Strength of interactions between stock pairs using Mutual Information Matrix - Represented market as weighted graph & extracted communities of similar stocks using Louvain algorithm - Improved MSE of best fitted ARIMA model by 20%, adding network measurement parameters
Electricity Demand Forecast	- Forecasted hourly load demand for next one month using 5-year data to plan power availability in advance - Adjusted for multiple seasonality to reach stationarity; Analyzed ACF/PACF plots to fit ARMA model
Home Credit Default Risk	- Predicted the Default Risk of underserved loan applicants based on their credit history & personal details - Engineered relevant features and implemented LightGBM model to identify loan defaulters
Quora Question Pairs	- Identified duplicate question pairs on Quora platform to enhance user experience for writers and seekers - Extracted text features: TF-IDF, distance similarities on pre-processed data and built XGBoost model - Attained AUC of 0.876 by using Glove word embeddings with LSTM - Siamese Recurrent Architecture
Amazon Review to Rating Prediction	Objective: To assign probable ratings (1-5) to live tweets and posts during electronic product launch - Leveraged 1 Million Amazon electronic product review-ratings to train FM model & LSTM model - Demonstrated both models using 50 sample tweets on electronics with 80% & 92% 1-step accuracies
DengAI : Predicting Dengue cases	- Predicted weekly Dengue cases for a city, based on environmental variables like vegetation & humidity - Devised a Regression model to predict the weekly Dengue cases after imputing the missing values - Compared SVR & Random Forest models in terms of accuracy and the latter explained higher variance
Predict Future Sales (Kaggle)	- Predicted next month sales of 200k shop item combinations of 59 stores using sales data of last 33 months - Implemented an ensemble of LightGBM and CatBoost Regressor to predict weekly sales for a month - Modelled each store-item as independent time series and improved cross-validated RMSE using LSTM

WORK EXPERIENCE (41 months)

Business Development Associate	Washapp	Hyderabad (Jun'16 – Mar'17)
T-Hub	- Washapp is a Laundry service start-up generating yearly revenue of 6M with 1200 active monthly orders - Worked under T-Hub, the largest incubator for start-ups in India, as a member of Washapp's team	
Cross Marketing Program	- Coordinated with multiple B2C start-ups for mutual co-operation by encouraging Cross-Marketing - Generated more than 150 leads by partnering with local startups - Leanspoon, Tinmen & Smartlivein	
QA Engineer	Infosys Limited	Hyderabad (Sep'13 – May'16)
Root Cause Analysis	Managed and supervised Business applications working on accounts of U.S based brokerage firms Processed Business requirements from clients into test cases and progressed them to production Analyzed issues in client's production environment and identified the root causes to resolve issues	
Enterprise Security Master (ESM)	- Recommended testing procedures and ensured smooth transition and release of ESM into production Coordinated with the team at Wipro to fix critical issues and improve performance of applications - Deployment of ESM generated \$7.5 billion annual revenue with a growth of 17.2% in 2016 for the client and made it possible to achieve the mark of 44 million transactions published in a day - Received a note of appreciation from the Project lead for effectively screening defects in the client UI	

AWARDS & ACTIVITIES

Academic Awards	- Recipient of MCM Scholarship, awarded by HRD Ministry throughout 4 years of B.Tech 2009-13 - Among top 2%ile in GATE (Electrical Engineering) among 1.5 lakh aspirants (AIR-3004) 2013 - Developed an Online Library Management System during the training period in Infosys (Mysore) 2013 - Awarded for successfully completing training in Infosys with a GPA of 4.4 out of 5 2014
Position of Responsibilities	- Coordinated events for EEE Dept's TEAM OJAS at the annual Technical Fest - NIMBUS 2012 Led School Sports team in numerous District level Kabaddi, KoKo, & 1600m Sprint, 4*400 relay races
Achievements	Finalist in the Pool(Billiards) competition held at the Intersection games conducted by IIM Calcutta - Two times Bronze winner of Sparring event in State Karate Championship, Andhra Pradesh 2005

TOOLS & LANGUAGES : R, Python, Java, C, C++, Perl, SQL, PL/SQL, HTML, MS Excel

SUBJECTS : Corporate Finance, Game Theory, Games and Information, Complex Networks, Predictive Modelling

ACADEMIC QUALIFICATIONS			
Year	Degree/Board	University/Institution	%/CGPA
2019*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-
2015	B.Tech, Civil Engineering	Dr. Babsaheb Ambedkar Technological University	6.83/10
2010	Class XII, CBSE	Jawahar Navodaya Vidyalaya, Aurangabad (M.S)	69.80%
2008	Class X, CBSE	Jawahar Navodaya Vidyalaya, Aurangabad (M.S)	77.00%
AWARDS AND POSITIONS OF RESPONSIBILITY			
Academic Achievements	- Secured AIR-19 and World Rank-84 in Capsim’s Business Strategy Simulation Game - Awarded full-ride scholarship by Govt. of Maharashtra for selection into PGDBA 2017-19 - Among ~2% students to be selected on the basis of merit for ‘Jawahar Navodaya Vidyalaya’		‘18 ‘17 ‘06
Company Sergeant Major, NCC	- Conferred with 3rd highest rank of ‘Company Sergeant Major’ and the highest NCC certificate - Represented the university in three annual training camps organized by the Army Wing of NCC - Led a unit of Discipline Committee during Maharashtra’s Inter-University Youth Festival		
DATA SCIENCE PROJECTS			
Credit Risk Analysis	- Predicted propensity of a customer to default in the next 2 years using data of 1.5 lakh individuals - Synthesized training data using SMOTE to tackle class imbalance & identified outliers using t-SNE - Compared the performance of Logistic Regression, Random Forest and SVM models using AUC - Obtained best AUC score of 0.85 among all models with debt ratio as the most significant predictor		
Detection of Identical Question Pairs on Quora	- Identified duplicate questions on Quora using word embeddings to capture semantic similarity - Trained multiple Random Forest models using TF-IDF and distance-based similarity features - Captured context of the questions leveraging Siamese Recurrent Neural Network architecture - Improved Random Forest’s performance with LSTM-based models to obtain AUC score of 0.876		
Product Classification	- Built models on 90+ features to classify ~60,000 products into 9 anonymous product categories - Compared the performance of Artificial Neural Networks, XG Boost and Random Forest models - Achieved a cross-validation accuracy score of 82% and multi-class logarithmic loss of 0.544		
Retail Store Sales Prediction	- Developed a sales prediction framework for 45 stores to facilitate inventory planning and control - Pre-processed data by applying log-transformation to remove heteroskedasticity and skewness - Implemented Multiple Linear Regression to predict daily sales and achieved MAPE of 15%		
Knowledge Graph for Legal Domain	- Built a knowledge network for legal domain by extracting entity relationships from various sources - Created a Web Scraper to extract data from Wikipedia and Westlaw by traversing related pages - Applied Entity Resolution to match the nodes using exact & longest common subsequences		
ECONOMETRIC PROJECTS			
Economic Quantification of Accidents	- Examined the impact of weather conditions on road traffic deaths due to deterioration of roads - Analyzed cross-sectional data of accidents in 2012 for 20 Indian States and Union Territories - Used Link & Ramsay test for model specifications; handled heteroskedasticity by log-log models Devised a mathematical formula to estimate the total economic cost due to deaths in accidents		
Social Impact of Rule of Law	- Studied the impact of Rule of law on property crimes, inequality and education across countries - Used cross sectional data for 76 countries from reports published by UNODC, UNDP and CIA - Computed elasticity of number of thefts, inequality loss and Gini coefficient w.r.t rule of law index - Identified high correlation between number of years of education and rule of law index		
INTERNSHIP AND B.TECH PROJECT			
D.S.Construction (Internship)	- Coordinated with the Site Engineers to achieve planned targets and maintain Quality Control - Assisted the Procurement team in acquiring high-quality materials for use in the construction site		
No-Fines Concrete (B.Tech Project)	- Studied the impact of sand-removal while making concrete-mix, on the properties of concrete - Carried out sensitivity analysis of size of coarse aggregates vs. tensile and compressive strength		
EXTRA-CURRICULAR ACTIVITIES			
Certifications	- Certified in Introduction to Statistics: Inference by UCLA, Berkeley course on EDX - Certified in Neural Networks and Deep Learning course on Coursera by Andrew Ng - Qualified ‘B’ & ‘C’ certificate exams conducted by Army Wing of National Cadet Corp.		(Sept ‘13) (May ‘18) (May ‘14)
Dramatics & Arts	- Performed in variety of cultural events as a member of dramatics club during graduation - Secured 2nd rank in Essay writing competition conducted by NSS in Dr. B.A. Tech. University		
TOOLS & LANGUAGES : R, Python, SQL, IBM SPSS Modeler, STATA, Microsoft Office			
COURSES: Econometrics, Business Economics, Pricing and Revenue Optimization, Decision and Game Theory			

Roles Offered: Analytics

Job Description:

Roles & Responsibilities:

Identify business critical issues in the organization that could be potentially solved leveraging the tools and techniques of Big data analytics, Machine learning etc. Design, Develop, Implement and Standardize Analytical models to solve business critical issues. Effectively communicate derived analytical results and provide actionable insights to various business stakeholders. Design and implement course corrective actions basis analytical results, in co-ordination with the business stakeholders. Standardize various structured and unstructured data reports across company and across. Implement sophisticated Business Intelligence tools and systems for databases, dashboards, systems etc. Set up methodologies & best practices for Analytics wing. Design future strategies, road maps and action plans to be undertaken by Analytics team. Efficiently manage and lead a team of Associates and Executives.

Competency:

Experience in working with large sets of structured and unstructured data sets. Relevant knowledge of Machine learning algorithms and Statistical modelling techniques. Proficiency in Statistical tools such as R and Programming languages such as Python is a must. Experience with relational databases and SQL is an added advantage. Relevant experience in using BI tools such as Tableau, Qlikview etc. Ability to understand company specific business issues and derive data points to run Analytics solutions. Ability to translate analytical results into actionable solutions. Should possess excellent Problem solving and Critical thinking skills. Should possess strong leadership skills and ability to motivate and inspire team members.

Location: Bangalore

- Internship Stipend Offer : 1,00,000 INR
- Shortlist steps: a)CV shortlisting
b)Technical Round
c)HR round
- Duration : 40-45 minutes

Technical Round Questions(Nisant Manepalli):

- Mostly business application of ML (they assume we have sufficient tech background of ML)
- Challenges faced in the project and how ML could have been applied in my previous job(my previous job was pure engineering-no ML scope)
- why take a risk in joining this course

Technical Round Questions(Praneeth Ponnekanti):

- Given the business problem of Churn rate reduction, what data would you require and how would you model this problem
- I was also asked about what data I would gather for third party vendors if possible & about their pricing and revenue strategies
- Also asked them about how they evaluate customer satisfaction, proposed a GIS based idea about creating a satisfaction index and check for variations every month in the areas they operate, proposed an idea for segmentation based on GIS approach. They did tell me that things are being analyzed using excel and might need GIS approach once they operate all over India.

HR Questions(Praneeth Ponnekanti)

- How good listener are you ?Why was your engg CGPA low? to which I had to substantiate the facts behind why I was chilling in the initial years.How confident and intimidated can you get ?
- Any complicated scenarios where you had to take a firm decision and how did the decision pay off? - I mentioned about my brother's marriage the day before placements and how I attended PPT's and interviews and I hope that it would pay off good !
- Was also questioned upon the effort required to take the above decision , to which I swiftly replied saying that it was an easy decision to which I arrived at, due to the application of pay-offs and Nash Equilibria discussed in Games & Decisions class to which they started smiling.
- Questions on Team Dynamics, how good a team player are you and How authoritatively can you be at the same time.

Courses and Skills Suggested

- Business Data Mining, COS (Nisant Manepalli)
- Basic ML (CDS), Marketing & Operations, Business Sense, Strategic Management concepts, PRO (To an extent where you can discuss things related to Bundle pricing, extracting consumer surplus)-Praneeth Ponnekanti

Suggestions:

- Attend as many PPTs as you can and actively participant in it, it makes an impact when they find students they interact with in the ppt are on the other side desk during the interview
- Atria was not just looking for technical expertise but also business minded people who can handle cross functional responsibilities. Be Honest in the interview and the panel are very good listeners and it was very healthy conversation. Attend the PPT to get a grasp of their current problems and how you would go about solving it.
- They look for this fit and they would interview all the possible candidates they want in the slot provided to them. They rejected all the candidates who they find a fit for.
- Substantiate your proposals with business/tech knowledge as why things would work out. Be honest and make sure you are not cooking up things in the psychometric/HR questionnaire round. The interviews on an average went for around 35 mins. Be patient.

ACADEMIC QUALIFICATIONS

Year	Degree/Board	University/Institution	%/CGPA
2019*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-
2015	B. Tech - Chemical Engineering	IIT Madras	6.23/10
2010	Class XII, BIE-AP	ST. Patricks Jr. College, Hyderabad	88.60%
2008	Class X, CBSE	Kendriya Vidyalaya, ONGC - Panvel, Mumbai	90.40%

DATA SCIENCE PROJECTS

Time Series Analysis of Cryptocurrencies	- Analysed the time series aspect of major cryptocurrencies based on their market value as of Nov 31, 2017 - Verified stationarity and variance of Bitcoin, Ethereum, Litecoin and Ripple after data transformation - Developed ARIMA and volatility clustering models (GARCH) and cross-compared cryptocurrencies
Forecasting returns & volatility	- Built time series model to forecast 2-day conditional returns, volatility & correlation for Indian stocks - Used forecasting models: ARIMA for returns, GARCH for volatility and DCC-GARCH for covariance - Tuned model using RMSE, direction & sign accuracy criteria; performed Sharpe ratio maximization
Investment Recommendations	- Developed interactive dashboard (ShinyR) listing potential investment options amongst retail firms - Employed factor analysis and k-means clustering to bundle companies based on their financial ratios - Compiled catalogue of retail firms as prospective investment options based on investors risk appetite
Toxic Comment Classification (Kaggle)	- Designed NLP based models for detecting 6 different categories of toxic comments on Wikipedia - Vectorised words using word embedding's - word2vec and GloVe to capture contextual information - Applied Attention models on Bi-LSTM & GRU layers to capture decisive features in the comments - Employed LGBM on top of classifiers; Obtained cross-validation accuracy of 98.3%, AUC score of 0.9867
Santander Product RS (Kaggle)	- Built a recommender system to suggest most relevant financial products for 1 million customers - Recommended few relevant products (7 out of 24) for users, based on their past 15 months' data - Methods implemented: One-vs-Rest Classifier, Content Based filtering, Collaborative Filtering
Wikipedia Network Analysis	- Analysed online vandalism on 9 different Wikipedia pages based on hyperlinked editor comments - Created uni-partite and bipartite networks to understand editor interactions on 9 different Wiki pages - Detected communities and overlapping communities using Louvian and k-clique algorithms

WORK EXPERIENCE

23 MONTHS

TECHNICAL CONSULTANT

Wood Group PLC

Chennai (June '15 - May '17)

Roles & Responsibilities	- Designed Oil & Gas Flow Assurance solutions for major middle eastern upstream operator companies - Used OLGA (industry exclusive multiphase flow software) for case analysis of clients operating problems - Generated project quality reports to achieve product excellence through quality control techniques Authored several executive level technical reports outlining engineering recommendations - Analysed client data to formulate optimal Cost Time Resource (CTR) sheets for project bidding process
Results & Achievements	- Developed extensive VBA macros for pre/post processing data leading to faster project completion - Tailored windows & python utilities to generate speedy system graphs from large data files (~5 GB) - Awarded project phase-II for valuable recommendations in phase-I, solidifying client relationship
Recognition	- Received Pre Placement Offer (PPO) for exemplary performance during Bachelors internship 2014 - Recognised for delivering consistent performance and promoted to Flow Assurance Consultant

POR & EXTRA-CURRICULARS

Sponsorship & PR Core ChemClave '14	- Led a team of 6 people as the Sponsorship & PR Core for ChemClave 2014, with a footfall of 1,000 - Expanded sponsor pool leading to 70% increase in sponsorship budget for the fest over previous year Public Relations: Expanded the outreach by spearheading the first ever publicity event of the fest - Foresaw Walk-on-Water experience for the fest attendees, using non-Newtonian fluid based pool setup
Extra-Curricular Activities	Sponsored a child through ActionAid India (NGO) which works for the upliftment of rural children Directed and acted a stage play of 25 people during department and institute fresher's night 2011, IITM - Part time content writer with Rosen Pictures, a YouTube channel and was part of four short films - Awarded Karate Green-II belt certification which is the 4th highest ranking belt in Karate - Successfully completed Trekking Expedition at Chanderkhani Pass(13000 ft. above sea level), Kullu-Manali

TOOLS & LANGUAGES : R, Python, SQL, Visualization in Tableau, MapReduce, MS Excel, MS Project

COURSES : Computations in Data Sciences, Complex Networks, Categorical Data Analysis, Regression & Time Series Modelling, Econometrics, Strategic Management, Decisions and Games



ACADEMIC QUALIFICATIONS

Year	Degree/Board	University/Institution	%/CGPA
2019*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-
2016	B.Tech in Computer Engineering	IIITDM Kancheepuram	6.60/10
2011	Class XII, BIE - AP	NRI Junior College, Gajuwaka	89.40%
2009	Class X, BSE - AP	Chaitanya Public School, Ukkunagaram	91.33%

WORK EXPERIENCE & INTERNSHIP

Assistant Systems Engineer Trainee **Tata Consultancy Services Ltd.** *Chennai (July'16 – June'17)*

Location Intelligence & Retail Analytics	- Represented one half of the offshore GIS consulting team that serviced a major American retail giant - Determined optimal locations for new stores using demographics, competitors, sales geospatial data - Visualized local community trends to proactively evaluate demand, reduce shortage at all~13k stores Innovated health check reporting to fix sporadically surfacing issues in the client LI web-app modules - Leveraged Python, SQL in providing production and development support for client LI web-app
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UG Summer Intern **IIM Lucknow** *Lucknow (May'15 – June'15)*

Marketing Management	- Analysed the dynamics in consumer markets that influence buying decisions, campaign responses - Presented a comprehensive analysis of Harvard Business School case -“Dove : Evolution of a brand” - Received certificate & LinkedIn recommendation from faculty towards remarkable intern performance
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POSITIONS OF RESPONSIBILITY

Member of Institute Web-ops team (IIITDM, '13 –'16)	Co-Engineered '4pi' - a successful premier social networking student portal that catered to ~750 users Designed timestamp based algorithm that scored real-time activity of posts on basis of likes, comments - Augmented the algorithm with activity-based sorting of posts and a novel save to mail functionality - Co-Developed the front-end of IIITDM websites - Samgatha'14 & 4pi using HTML, CSS, Bootstrap
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Extracurricular POR's	PGDBA PR team: Established a centralized FAQ answer forum for PGDBA shortlisted candidates ['18] HackerRank CA: Established the online platform coding culture within IIITDM community ['15,'16] Event Core, Samgatha: Led a team of 15 and bolstered Gaming events to the top2 profitable events['15,'16] Co-Organizer, Samgatha: Co-strategized operational workflow; Negotiated event sponsorships ['16] Mentored the annual trophy winning team of 15 enthusiastic singers at the IIITDM music club ['16]
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DATA SCIENCE PROJECTS

Movie Recommender System	- Built Hybrid Recommender models on MovieLens data to give customized movie suggestions to users - Stacked various Latent Factor models based on Matrix Factorization, by combining strengths of Collaborative and Content based filtering techniques; trained using Stochastic Gradient Descent - Implemented an ensemble of Ordinal Logit Models based on similarities & proximities of users and movies
Abnormality detection in X-Ray images	- Developed a tool to detect one of 14 different thorax diseases given an X-ray as an image (jpeg format) Preprocessed images, handled high class imbalance by data augmentation & assigning class weights - Implemented transfer learning using pre-trained weights of InceptionNet and achieved F1-score of 0.35
Stock Price Forecasting	- Conceptualized a time series model to forecast Infosys stock price a day ahead using past 5 year data - Leveraged ADF test to confirm stationarity; Created ACF and PACF plots to determine ARIMA order - Modelled volatility of logprices using ARCH & GARCH; Achieved the lowest AIC of -2424 on ARCH
Product Classification	- Built a predictive model to classify 190K e-commerce products into 9 categories based on 93 features - Trained RandomForest, XGBoost and Artificial NN models; Achieved best accuracy of 0.82 on RF;
Network Community Analysis	- Conceptualized a framework to accelerate new community detection by identifying k-core subgraphs - Leveraged NCP plots to analyze the dynamics of communities in real world & synthetic networks - Achieved modularity of 0.59 on 16-core subgraph of email-eu network in contrast to 0.61 by Louvain
Credit Default Evaluation*	- Predicted the Default Risk of underserved loan applicants based on credit history and personal details - Engineered relevant features and implemented LightGBM model to identify loan defaulters

ACHIEVEMENTS AND EXTRACURRICULARS

Scholastic Achievements	- One among the top 20 applicants to be selected for GNOME FOSS Hacknight at Shaastra, IIT M ['14] Certified in Neuro Linguistic Program skills towards completion of the NLP Leadership camp ['07]
Extracurricular Achievements	Awarded 1st prize in solo singing event “VOX”, Clustervents'15, techno-cultural fest of IIITDM ['15] - Received accolades for consistent inter-school quizzing achievements at District & State level ['07-'09]

TOOLS & LANGUAGES : R, Python, SQL, Visualization in Tableau, MS Office, C++, Shell Scripting, HTML, PHP, Bootstrap

ELECTIVES : Corporate Finance, Pricing & Revenue Optimization, Product & Supply chain Analytics, Decisions & Games



(Estee Advisors)

Roles Offered: Algorithm Trading Strategy

Job description:

- Design and implement mathematical models for fundamental valuation of securities. The person will need to understand latest research in quantitative finance and implement the same.
- Design, back-testing and implementation of high-frequency trading strategies on international exchanges. Work as part of the market-making team to determine the signals and trading strategies to go live with.
- Conduct performance attribution of live portfolios.
- Location: Gurgaon
- Internship Stipend Offer : 125,000 INR

Shortlist steps: a)Quant Aptitude Test

b)Coding

c) Technical Round

Duration : Max 45 minutes

Technical Round Questions(ABHIRUP SEN)

- 3 Probability based logical questions,
- 2 basic questions on Econometrics (time-series ARIMA, cointegration)
- 1-2 questions on projects related to financial analytics
- Why desire to pursue finance trading as career after MBA

Courses and Skills Suggested

- Strong conceptual understanding of Probability and distributions; Basic knowledge of Indian financial markets
- ISI-Stochastic Processes; IIT-KGP-RTSM/IIM-Advanced Predictive Modeling;

ACADEMICS			
Year	Degree/Board	University/Institution	%/CGPA
2016-18	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-
2014-16	MBA (PGP)- Finance & Information Systems	IIM Lucknow	6.44/10
2006-10	B.E. (Electronics & Communications Engineering)	Jadavpur University	9.11/10
2006	Class XII, ICSE	St. Thomas Boys’ School, Kolkata	95.2%
2004	Class X, ICSE	St. Thomas Boys’ School, Kolkata	92.5%
ACADEMIC AND CO-CURRICULAR ACHIEVEMENTS			
▪ Cleared two levels of Chartered Financial Analyst (CFA) and two levels of Financial Risk Manager (FRM) in first attempt ▪ Scored 99.49 percentile in CAT 2013 among approximately 1.73L candidates and 750/800 in GMAT (98 percentile at global level) ▪ Published IEEE Paper: Inertia Adaptive Particle Swarm Optimizer algorithm for improved Global Search, ISDA, Taiwan, 2008 ▪ Achieved top 1 percentile of 5.25L candidates in AIEEE and 99.95 percentile in West Bengal Joint Engineering Entrance Exam ▪ Selected in final round of KPMG International Case Competition (KICC) for analysis of Indian PE versus non-PE backed firms ▪ Received award for topping ICSE Class XII and X board exams in school and scoring 100% in Class-XII Mathematics examination			
DATA SCIENCE PROJECTS			
Investment Recommendation using Data Mining	▪ Compiled an interactive ratings dashboard (ShinyR) to list potential firms for investment based on investor risk appetite ▪ Applied factor analysis to identify key performance indicators & K-means clustering for grouping the companies ▪ Used Random Forest for predictive modeling of company performance based on each key factor identified		
Time-series Forecasting of Nifty stocks	▪ Built univariate time series models to forecast 2-day conditional returns, volatility & correlations of five stocks ▪ Used forecasting techniques: ARIMA for returns, three GARCH variants for volatility and DCC-GARCH for covariance ▪ Tuned model using RMSE, direction & sign change accuracy criteria, performed Sharpe ratio based portfolio optimization		
Toxic Comments Classification (Kaggle Competition)	▪ Designed an NLP system to classify toxic comments from social networks into six toxicity severity classes ▪ Used word embedding techniques-Word2Vec & GloVe to vectorize words without losing important context ▪ Achieved best ROC score 0.9867 using advanced deep learning techniques-Bidirectional LSTM and GRU		
Product Backorder Prediction (Kaggle Competition)	▪ Performed product backorder risk prediction modeling on a retail store using historical data of 1.6M SKUs ▪ Used bootstrap technique ROSE to handle imbalanced data classes & Boruta to extract relevant features ▪ Scored 63% recall & 64% precision by logistic regression and boosted to 86% & 95% respectively using RF ▪ Performed Monte-Carlo Simulation to locate the ROC threshold point to maximize expected profit		
Santander Product Recommendation System	▪ Built a bank recommender system serving 1 million customers suggesting 7 most relevant out of 24 products ▪ Techniques used: Content based filtering, collaborative filtering & context-aware recommender system		
Econometrics Application in Indian Capital Markets	▪ Designed multivariate time series model to predict Nifty index returns using top-down approach ▪ Performed statistical tests to verify validity of structural regression model based on the macroeconomic indicators ▪ Used Johansen Co-integration & Vector-Error-Correction methods to model the equilibrium relationships		
Derivatives Pricing using Reinforcement Learning (ICDM Paper, 2018)	▪ Devised model to predict European call option prices & trading policies on 1 min. frequency Nifty data ▪ Labeled actions using EMA Crossovers & applied LSTM (R2 score:0.981, categorical accuracy:0.664) ▪ Improved model using LSTM with reinforcement learning (R2 score 0.985, categorical accuracy:0.704)		
Selected MBA Projects (IIM Lucknow)			
Yield Curve Analysis& Hedging	▪ Performed term-structure modeling of spot and forward market interest rates of China bond market ▪ Used piecewise Cubic-Spline & Nelson-Siegel (NS) as yield curve bootstrapping techniques ▪ Implemented “tailing-the-hedge” portfolio mechanism for trading during dynamic market conditions		
TV Advertisements Analysis	▪ Carried out statistical analysis to determine key factors used to decide timing, frequency and duration of advertisements in Indian TV channels		
WORK EXPERIENCE			
Business Consultant	Cognizant Business Consulting	May ’16-Mar ’17	
Roles &Responsibilities	▪ Performed GAP analysis and strategized analytics based user behavior tracking enhancements on a corporate cards webpage to improve campaign management and boost ROI ▪ Conducted exhaustive market research on existing business landscape as part of RFP for new client bids ▪ Published research whitepaper on IT & business models on card payment tokenization systems ▪ Proposed an innovative cloud based approach to improve a card payment acquirer’s POS system		
Technical Analyst-CRM	Cognizant Technology Solutions	Jul ’10-Mar ’14	
Roles & Responsibilities	▪ Proposed BPM/CRM coding solutions for credit-underwriting, POS payments and SIPP trading ▪ Organized knowledge sessions on macro-level or functional aspects of project for technical team ▪ Recipient of BPM Bravo Awards for timely delivery of projects as part of team of 8-10 members		
Internship (MBA)	CRISIL Risk Solutions, Mumbai	April ’15-Jun ’15	
▪ Created library of Early Warning Signals to help infrastructure lenders like banks & NBFCs detect potential defaulters in advance ▪ Classified signals on basis of sector & stage of project financing so that the tool can be used on ongoing basis across entire project lifecycle			
TECHNICAL SKILLS& RELEVANT COURSES			
▪ Languages:C, C++, Java (Oracle Certified), Matlab, SQL, PL/SQL, R, Python ▪ Online certifications (DataCamp): Sentiment Analysis, Credit risk modeling in R, Fundamentals of Bayesian Data Analysis in R, Marketing & Product Demand Forecasting Analytics			



Roles Offered: Data Scientist (Supply Chain)

Job description:

- Internship Stipend Offer :
- Shortlist steps: a)CV shortlisting
 - b)Group Discussion
 - c) Technical Round
- Duration : Max 25 minutes

Technical Round Questions(Ranadheer):

- Tell me about yourself. Questions were manly related to my projects.
- What is encoding and decoding, draw the architecture, where will you use, advantages.
- Explain market basket analysis, Apriori algorithm, definitions of support, lift, confidence.
- How Boruta works? what is the shadow variable.

Technical Round Questions(Pratyusha A)

- How would you construct a neural network for a logistic regression model?
- Briefly describe your projects mentioned in the CV?
- Explain the loss function of an Auto-encoder network (used in one of my projects)?
- Please let us know where in your preferences J&J stands (since this was my 3rd interview)?

Courses and Skills Suggested

- One should be very thorough with the things written in CV. Deep Learning, Random Forest, Boosting, Case based questions are important (ISI is the foundation for everything we learn)
- Computational Data Science (ISI) and Algorithms and Machine Learning (IIT)

Suggestions:

- You master on the things that you have written in CV. Mock interview helps(Ranadheer)
- Behavioral Aspects (such as rejection after previous interview) play a major role in firms like PWC(Pratyusha)

ACADEMICS

Degree/Examination	Year	Board/University	%/CGPA
Post Graduate Diploma in Business Analytics	2019*	ISI Kolkata, IIM Calcutta, IIT Kharagpur	-
M.Tech, Electronics and Communication	2016	National Institute of Technology, Rourkela	8.05/10
B.Tech, Electronics and Communication	2015	National Institute of Technology, Rourkela	7.96/10
Class XII BIE-AP	2011	Vijaya Ratna College-Ramayah, Hyderabad	92.3
Class X BSE-AP	2009	St.Anns' High School, Hyderabad	95.3

ACHIEVEMENTS AND COMPETITIONS

- 99.06 percentile in CAT-2016; 99.98 percentile in IITJEE-2011; among the **top 10% coders** on hackerearth.com for the year 2017
- Ranked 117 in Data Science Hiring Challenge for **HackerEarth, Bangalore** and shortlisted for **Data Scientist** role in Jan'18
- Selected for **Business Analyst** roles at **Capgemini** and **ZS Associates** in the campus recruitment drive at NITR in Sep'15
- Ranked **20th** (~535) in 3 hour **SQL Challenge** Feb'18 and **79th** (~1229) in 6 hour **SQL Challenge** Sep'17 on hackerearth.com
- Secured **7th rank** in **ZS Data Scientist Challenge** on hackerearth.com for event prediction and sentiment analysis, Jul'18
- Secured **13th rank** in Beyond Infinity **house price prediction challenge** by **Great Lakes** on AnalyticsVidya during Jan'18
- Among top **0.02** percentile in several data science problems involving **tabular, text & image data** on **hackerearth.com**

AWARDS

- Project Winner** at **Business Consultant Boot Camp** held for 30 new recruits at **ZS Associates**, Pune during July 2016
- Winner** in Inter hall women's **Chess** tournament at NIT Rourkela for years 2012 & 13; **School Topper** in Class-X BSE-AP '09

WORK EXPERIENCE

Business Analyst	ZS Associates, Pune	Jul'16 - Jun'17
- Updated and managed US diabetes and obesity database, covering 24 million patients with over 1 billion data points - Streamlined the data quality management (DQM) processes by automation and reduced the turn-around time by ~70% - Built patient pathways for 85 current diabetes drugs to map treatment patterns for Fortune 500 global pharma clients - Generated a set of industry standard business rules that were used across the organization for projects in this market - Identified and recommended insights that helped bust several treatment myths and bag follow-on projects for ZS - Designed metrics to periodically measure the efficiency of HCP marketing campaign and the need for target refinement		

ANALYTICS PROJECTS

Analyzing the Dynamics of a Social Movement with #blacklivesmatter
- Created a dataset by scraping and cleaning tweets related to the #blacklivesmatter movement for over 600k active users - Constructed a retweet-mention graph to identify authorities and hubs based on Jaccard Similarity between different users - Clustered the similarity graph of authorities & hubs by Newman-Girvan and Louvian methods to find user characteristics - Measured influence of bot accounts by clustering them with TSNE based on tf-idf of user description & time gap features
Portfolio Selection using Deep Learning
- Demonstrated a robust portfolio construction process using Deep Auto-Encoders to learn non-linearity in security returns - Applied elastic net regularization and dropout techniques to tune out-of-sample error similar to risk-return trade-off - Created an efficient deep frontier with subset of stocks for re-tracking and beating NASDAQ IBB and NIFTY indices
Predicting Next Breakage of Pipeline
- Predicted next breakage by clustering the mixed Gaussian response variable by Expectation Maximization and R-Forest - Performed feature selection for early first breakage and late first breakage clusters using Boruta for 23k pipes in the data - Predicted the next breakage for 70% of the 55k pipes in the data with a 60 day error using XGboost and Gradient Boosting
Suspicious Lesion Detection in Medical Mammograms using mini-MIAS database
- Classified 287 images into cancerous and non-cancerous and identified the location of cancer using SVM with RBF kernel - Used region seed growing, largest area detection, adaptive thresholding and achieved an acc. of 91% with 0.71 FP/Image - Improved acc. to 95% by Transfer Learning technique which uses pre-trained weights of CNN to create bottleneck files
Predict Future Sales for a Gaming Products Firm
- Predicted the weekly sales of 200k shop-item combinations by determining the order of ARMA process using ACF & PACF - Achieved 30% lower RMSE by taking each store-item as independent time series with appropriate lag and using LSTM

POSITIONS OF RESPONSIBILITY

- Worked as **Teaching Assistant** for Analog Electronics Lab in the autumn semester of year 2015 at **NIT Rourkela**
- Co-Editor** of 2016 Institute Year Book; **Member** of Mathematics Club; Debating Society during 2011-16 at **NIT Rourkela**
- Event Coordinator during 2012 and Central Coordinator during 2013, for the **annual techno-management fest** at **NITR**

Languages and Software: R, Python, SQL, SAS, Oracle, Teradata, MS Office, PySpark, MapReduce, Google Cloud, Bloomberg, IBM SPSS
Certifications and Electives: Deep Learning Specialization, BMC, Stanford SQL & Stat. Learning, Game Theory, Corporate Finance



ACADEMIC QUALIFICATIONS			
Year	Degree/Examination	University/Board	%/CGPA
2019*	Post Graduate Diploma in Business Analytics	IIM Calcutta, ISI Kolkata, IIT Kharagpur	-
2014	B.Tech (Naval Architecture)	IIT Kharagpur	7.17/10
2010	Class XII	Board of Intermediate Education, AP	92.4%
2008	Class X	State Board, Andhra Pradesh	87.5%
ACADEMIC ACHIEVEMENTS			
Ranks & Achievements	- AIR 3228 (top 0.7%) in IIT-JEE among 4.5L+ entrants; AIR 2436 (top 0.2%) in AIEEE among 11L+ entrants ‘10 - Selected among the batch of 60 (top 0.01%) for ISI Kolkata, B.Stat. program through two toughest exams & PI ‘10 - Ranked one amongst the top 20 students during academic year 2017-18 in the IIT Kharagpur semester, PGDBA		
WORK EXPERIENCE			23 Months
ASST.MANAGER (Design)		GOA SHIPYARD LIMITED	Aug’14 – Jun’16 (23 Months)
Project for Mauritius Government	- Led a division unit of the design team on a defence project worth ₹200 crore for the Mauritian Government Proposed a robust design to reduce the ship’s weight by 4%, improving fuel efficiency by 1% and speed by 0.8% - Served as liaison between departments to facilitate cross-functional collaboration for smooth execution of project Received an appreciation from the HOD for bolstering the design strength and improving operational efficiency		
INTERNSHIP		MISKA PROJECTS (Startup)	Oct’16 – Mar’17 (5 Months)
Connecting Market	- Pioneered a fixed subscription revenue generation model in the market to attract and create a base of 1L+ vendors - Launched a logo design competition in different colleges with around 200+ entries of the student participants - Raised a seed fund of ₹4 lakhs & strategized a roadmap for phase-1 product launch in 10L+ populated urban area		
DATA SCIENCE PROJECTS			
Product Backorder (Kaggle)	- Predicted product backorder on historical retail data containing 1.6M products to manage its inventory efficiently - Imputed missing values, selected features using Boruta and implemented Logistic regression & Random Forest - Performed Monte-Carlo Simulation to determine the threshold in ROC for achieving maximum expected profit		
Stock Movement Prediction	- Forecasted daily stock price movement of DJIA index by applying the deep learning techniques on historical data - Summarized the data using text rank algorithms, performed sentiment analysis and fitted the model using ARIMA - Achieved an accuracy of 58% by implementing sequence Encoder-Decoder on the historical data of stock price - Embedded Bloomberg news data with the historical data of stock prices & boosted accuracy of the model to 74%		
Consumer Market Basket Analysis	- Designed a recommendation system to predict the products that a user will purchase for the 1st time or buy again - Analyzed dataset containing 3M orders of 50K different products giving information about 2L customer profiles Engineered new features from user-product interactions & implemented Market Basket using Apriori algorithm		
Community Detection	- Identified communities in Amazon network where nodes(3L) are products & edges are co-purchase relationships - Applied a framework that can accelerates community detection by identifying k-core subgraphs within a network - Detected the target communities from the initial seeding set & achieved an F-1 score of 0.910 in less than 15 sec		
Forecasting Electric Demand	- Forecasted household electric load demand using 5-year historical data to reduce wastage & promote sustainability - Removed weekly and yearly seasonality present in the data & confirmed weak-stationarity using Dicky Fuller test - Fitted an ARIMA model on stationary data, & forecasted mean of load demand for next 7 days with MAPE of 13%		
ROLES AND RESPONSIBILITIES			
General Secretary	- Organized events in the hall of 1600 students & negotiated terms in the favour of hall during Gymkhana meetings ‘11 - Facilitated to increase hall budget by 50%, and increased event participation rate to 100% from the previous 66%		
Social Initiatives	- Strategized & coordinated personality development events for VR1(NGO) at IIT KGP impacting 200+students ‘14 - Instrumental in publicizing about “Child Adoption” at the Asia’s largest tech fest that impacted 1k+ students ‘12		
Other Responsibilities	- Spearheaded the hall team to achieve first time ever 4th position in the interhall hardware modelling, IIT KGP ‘11 - Supervised and mentored 10 interns and elucidated them with the procedures and daily tasks in the industry ‘15		
AWARDS & ACTIVITIES			
Distinctive Achievements	Winner in the national level robotics event organized by IIT Kharagpur and received a cash prize of ₹12000 ‘12 Winner of management games at Goa Shipyard for displaying exemplary management and leadership skills ‘15		
Sports Achievements	- First runner-up in badminton at IIM Calcutta; secured 3rd in national level badminton event at school level ‘17 - Achieved an excellent grade in the athletics (NSO) as a part of the extra academic activity at IIT Kharagpur ‘12		
TECHNICAL SKILLS / RELEVANT COURSES			
Certifications	Neural Networks + Sequence Models of Deep Learning, NLP fundamentals and Data Visualization using ggplot2		
Software Tools	R, Python, SQL, C, MS Excel, MATLAB, MS Project, AutoCAD, IBM-SPSS, Bloomberg, Prezi and PowerPoint		
Relevant Courses	Pricing & Revenue Optimization, Product and Supply Chain Analytics, Corporate and Computational Finance		